

Prague University of Economics and Business Faculty of Finance and Accounting



Five Essays on Asset Pricing

Dissertation

Field of Study: Finance

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Declaration of Authorship

I declare that the dissertation on the topic “Five Essays on Asset Pricing” and articles have been composed by the candidate, and any included publications are the candidate’s own work. All referred literature and other sources are correctly cited and listed in the reference list.

Martin Časta

Prague, March 31, 2025

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Motivation

The interplay between macroeconomic and financial variables and financial markets has long captivated scholars, policymakers, and practitioners alike. The dissertation thesis delves into the domain of asset pricing, focusing mainly on how changes in the pricing kernel, also known as the stochastic discount factor (SDF), impact financial markets and forecasting performance. More specifically, the dissertation studies the dynamics between several macroeconomic and financial factors, for example, dividends, inflation, interest rates, and supply and demand shocks—and their impact on equity risk premia, bond yields, exchange rates, stock returns, and yield curve behaviour. Through a collection of five distinct yet thematically interconnected articles, the dissertation thesis sheds some light on the various mechanisms through which macroeconomic and financial variables shape financial market outcomes and explores the predictive capabilities of these variables in forecasting financial returns. In other words, at the heart of this dissertation lies the investigation of how macroeconomic and financial shocks and trends influence the behaviour of the financial markets and if these variables allow for the predictability of financial returns.

The dissertation thesis is, therefore, anchored by a unifying theme: understanding the structural implications of macroeconomic and financial variables on market dynamics and possibly other financial variables. In more detail, the articles presented in the thesis examine the response of financial markets to multiple financial and macroeconomic variables by employing a variety of econometric methodologies to analyze their effects and predict market movements. The dissertation contains five articles in the field of asset pricing. In more detail, the articles in the dissertation are the following: *Recovery Theorem and the Risk Aversion. Evidence from the Czech Republic* (Časta, 2024b); *Deriving Equity Risk Premium Using Dividend Futures* (Časta, 2022a); *Inflation, Interest Rates and the Predictability of Stock Returns* (Časta, 2023); *Supply Shocks, Demand Shocks and Yield Curve Dynamics* (Časta, 2022b); *Forecasting Nominal Exchange Rates Using a Dynamic Model Averaging Framework* (Časta, 2024a). All articles were published in academic journals such as *Finance Research Letters*, *The North American Journal of Economics and Finance*, *International Journal of Monetary Economics and Finance*, and *Heliyon*. The following table shows the names of the articles (chapters in the thesis) and journals in which the articles have been published:

Table 1: Summary Table of Articles

Name	Journal	Status
Recovery Theorem and the Risk Aversion. Evidence from the Czech Republic.	International Journal of Monetary Economics and Finance	Published
Deriving Equity Risk Premium Using Dividend Futures	The North American Journal of Economics and Finance	Published
Inflation, Interest Rates and the Predictability of Stock Returns	Finance Research Letters	Published
Supply Shocks, Demand Shocks and Yield Curve Dynamics	Finance Research Letters	Published
Forecasting Nominal Exchange Rates Using a Dynamic Model Averaging Framework	Heliyon	Published

Each article within the dissertation thesis contributes to an understanding of the relationship between macroeconomic and financial factors and market dynamics. Collectively, they offer insights into how various asset classes and their components evolve over time and under varying economic conditions, highlighting, for example, the importance of considering both cyclical components and long-term trends as well as the time-varying risk aversion. The primary research question of this thesis is, therefore, an empirical assessment of various financial and macroeconomic variables on financial markets and to uncover insights into the predictive capabilities of these variables and their implications for understanding market dynamics.

The first article examines option-implied probabilities of exchange rate movements, aiming to derive expectations about future exchange rate developments and assess investors' risk perceptions. Specifically, it applies the Ross recovery theorem to the CZK/EUR exchange rate in the FOREX market. The study introduces a novel approach to calculating the implied risk premium and provides a theoretical expression of the Ross recovery theorem using different numeraires. The findings reveal that both subjective and risk-neutral densities deviate from unbiased probability estimates of future exchange rates. However, the analysis also highlights a significant rise in implied risk aversion during the COVID-19 pandemic.

The second article studies stock market development. More specifically, I present a simple stock price decomposition model using the dividend discount model and dividend futures. The main contribution of this paper is the use of dividend futures, which represent the risk-adjusted expectations of future dividends. This allows for the calculation of the implied equity risk premium and the decomposition of stock price movements into individual components. Due to the use of daily market data, this method can take into account the structural changes associated with falling interest rates and the COVID-19 pandemic. I empirically show the risk premium development of the S&P 500 Index and Euro Stoxx 50 Index in the last decade.

The third article focuses on the predictive role of inflation and interest rates in forecasting stock returns. I report economically significant in-sample evidence that stock returns are predictable using cyclical components of inflation and interest rates. The out-of-sample analysis results show that these cyclical components have reasonable out-of-sample predictability for the equity risk premium when the comparison is made against a historical average benchmark. I believe this is due to the fact that both inflation and interest rates contain stochastic trends.

The fourth article presents a framework for extracting the supply and demand shocks using the revisions of expectations obtained from a Survey of Professional Forecasters. The main contribution of this article is the identification of these shocks and the subsequent assessment of their impact on yield curve dynamics. This allows for the asymmetry in the bond yield movements, where I show that yields react differently to supply and demand shocks.

The fifth article presents a dynamic model averaging approach for forecasting nominal exchange rates. This framework encompasses most of the approaches commonly used in the forecasting literature and also allows us to study parameters and model uncertainty in exchange rate forecasting. We empirically show statistically and economically significant exchange rate predictability in the medium and long run, and we also present some findings on predictability even in the short run.

To summarise, the dissertation employs a diverse set of econometric and statistical tools to explore the predictive power of macroeconomic and financial variables on financial markets. Dynamic Model Averaging, structural VAR models with sign restrictions, predictive regression models, dividend discount models, and detrending techniques such as exponentially weighted moving averages (EWMA) are among the methodologies utilized across the articles. These approaches allow for an examination of the complex interplay between financial and macroeconomic variables and financial market dynamics. The empirical analysis presented in these articles underscores the variability in the impact of macroeconomic and financial factors on

financial markets. The findings reveal how shifts in inflation, interest rates, and dividends, alongside supply and demand dynamics and other factors, influence equity risk premia, bond yields, exchange rates and the yield curve.

From a theoretical point of view, the common theme of all these articles is that they build upon the asset pricing field. Asset pricing serves as the cornerstone of financial economics, providing a framework for understanding how securities are priced in markets. In other words, asset pricing, at its core, seeks to determine the price of financial assets, which inherently involves understanding the risk and return characteristics that investors demand. Asset pricing theory, therefore, tries to understand the prices or values of claims to uncertain payoffs. To put it even more simply at the heart of financial markets lie questions about how assets are priced and what drives their returns. However, all the theories explaining these questions stem from one simple concept that the price equals the expected discounted payoff (see Cochrane (2009) among others).

Central to this field is the concept of the stochastic discount factor (SDF), or pricing kernel, which bridges the gap between economic theory and financial market reality, providing a framework for understanding the relationship between risk and return. It encapsulates investors' risk aversion and time preference, and discounting future payoffs to present value. Variations in the pricing kernel can be induced by changes in economic conditions, which in turn affect risk premia, which is the additional returns investors require to hold risky assets over risk-free assets reflecting the risk tolerance and expectations about future economic states. The major advantages of understanding asset pricing through this approach (stochastic discount factor/pricing kernel) are its simplicity and universality. Where once there were different theories for stocks, bonds, options, exchange rates, and others now we see each as a special case of the same theory (Cochrane, 2009).

Let us first briefly introduce the asset pricing field in more detail, emphasizing the foundations that underpin the valuation of financial assets and the role of the pricing kernel in these models. At its essence, the asset pricing field is rooted, as in most economics and finance, in the simple idea of expected utility maximization and optimization. This concept represents the process through which economic agents make decisions aimed at maximizing their utility, subject to various constraints. Optimization in finance is a central concept that ties together individual decision-making with market dynamics, including the determination of asset prices through the pricing kernel. Economic agents face decisions regarding allocation between saving and consumption, as well as determining the optimal portfolio of assets to hold. At the core of these decisions lies a fundamental pricing equation derived from the first-order condition. According to this condition, the marginal utility loss incurred by consuming slightly less today and buying a little more of the asset should equal the marginal utility gain of consuming a little more of the asset's payoff in the future. If the price and payoff fail to align with this relation, the agents adjust their asset allocation accordingly, either increasing or decreasing their holdings of the asset. Consequently, it follows that the price of the asset should correspond to the expected discounted value of its payoff, computed using the investor's marginal utility to discount the payoff (Cochrane, 2009). The optimization process is governed by agents' preferences, which are typically characterized by risk aversion and intertemporal substitution. Risk aversion describes agents' tendency to prefer less risky outcomes, while intertemporal substitution reflects their willingness to trade off consumption at different points in time. These preferences shape agents' decisions regarding savings, investment, and consumption, influencing the dynamics of financial markets.

This decision-making process (first-order condition) is encapsulated in the Euler equation, which is a fundamental result derived from the principle of agent optimization, expressing the conditions under which an agent's intertemporal consumption plan is optimal. The Euler equa-

tion reflects the optimal consumption choice, balancing present consumption against future consumption, given the rates of return on assets. It implies that the marginal utility of consuming an additional unit of wealth today must equal the expected marginal utility of consuming that unit in the future, discounted at the risk-free rate, adjusted for risk. This equation lays the groundwork for understanding how economic agents' consumption choices under uncertainty translate into demand for financial assets, thereby influencing asset prices.

The pricing kernel, or stochastic discount factor (SDF), emerges as a natural extension of the agent optimization framework, serving as a bridge between individual preferences and market prices. It represents the rate at which future payoffs are discounted to determine their present value. From the perspective of the Euler equation, the pricing kernel is instrumental in equating the marginal utility of consumption across different states and times. In essence, the terms pricing kernel and stochastic discount factor (SDF) extends concepts from economics and finance to include adjustments for risk (Hansen & Renault, 2010).

To provide further insight into the stochastic discount factor (SDF), we briefly examine several variations of the Lucas Jr (1978) asset pricing framework, incorporating different investor preferences. Specifically, we investigate the stochastic discount factor (SDF) in the context of the following exchange economies: (i) a representative agent with time-separable utility, (ii) the external habit model of Campbell & Cochrane (1999), and (iii) a representative agent with Epstein & Zin (2013) preferences. The examples are borrowed from Aspris *et al.* (2024).

First, we consider an exchange economy where the representative agent is endowed with a time-separable concave utility function. Without loss of generality, we normalise time such that $T - t = 1$. In equilibrium, the aggregate dividends/fruits will be fully consumed each period and hence for all time periods $C_t = D_t$, where aggregate dividends are denoted by D_t and consumption by C_t . This implies that the stochastic discount factor (SDF) in this setting is given as follows:

$$M_{t,T} = \beta \frac{U'(D_T)}{U'(D_t)} \quad (1)$$

where β denotes the rate of time preference. This basic framework serves as the foundation for many classic asset pricing models. For example, if we assume a single-period setting with either quadratic utility or normally distributed returns on the aggregate market portfolio, it results in the standard Capital Asset Pricing Model (CAPM).

The external habit model of Campbell & Cochrane (1999) assumes a representative agent who maximizes expected lifetime utility, expressed as:

$$\max E_t \sum_{\tau=t}^{\infty} \beta^{\tau} \frac{(C_{\tau} - X_{\tau})^{1-\gamma} - 1}{1-\gamma} \quad (2)$$

where γ is the coefficient of relative risk aversion, and it considered to be positive. X_t denotes a habit level. The representative agent derives utility from the difference between current consumption and the habit level, with the latter evolving based on past consumption patterns, introducing time-varying risk aversion into the model. The stochastic discount factor (SDF) can be written as follows:

$$M_{t,T} = \beta \left(\frac{C_T - X_T}{C_t - X_t} \right)^{-\gamma} \quad (3)$$

For the last example, I also examine an exchange economy with a representative agent endowed with Epstein & Zin (2013) recursive utility. In this framework, the market portfolio is

modelled as the claim on aggregate consumption in a pure exchange economy. The stochastic discount factor (SDF) in this setting is given by:

$$M_{t,T} = \prod_{\tau=t+1}^T \beta \left(\frac{C_{t+\tau}}{C_{t+\tau-1}} \right)^{-\theta} \left(\frac{V_{t+\tau}}{\left(E_{t+\tau-1} \left[V_{t+\tau}^{1-\gamma} \right] \right)^{1/(1-\gamma)}} \right)^{\theta-\gamma} \quad (4)$$

where V_t represents the continuation value of future utility, β is the time discount factor, γ denote the coefficient of relative risk aversion, and θ represents the elasticity of intertemporal substitution (EIS). This recursive utility formulation allows for a separation between risk aversion and the elasticity of intertemporal substitution, providing a more flexible structure for modeling investor preferences and asset pricing dynamics.

As explained above, the Euler equation helps us understand how assets are priced and what drives their returns. With some additional work, this relationship can also shed light on other phenomena. Specifically, the Euler equation helps explain exchange rate movements, as demonstrated by Backus *et al.* (2001a). In an arbitrage-free environment, the existence of a pricing kernel implies the following fundamental relationship relating the expected (real) return on an asset to the pricing kernel (stochastic discount factor):

$$1 = E_t(M_{t,T} R_{t,T}) \quad (5)$$

where $R_{t,T}$ represents the gross return of a domestic asset. For the foreign country, an analogous relationship holds:

$$1 = E_t(M_{t,T}^* R_{t,T}^*) \quad (6)$$

where $M_{t,T}^*$ and $R_{t,T}^*$ denote the foreign pricing kernel and gross asset return, respectively. Using these two relationships, Backus *et al.* (2001a) derive an equation that connects exchange rate movements to differences in the pricing kernels of the home and foreign countries:

$$E_t s_T - s_t = E_t \log(M_{t,T}) - E_t \log(M_{t,T}^*) \quad (7)$$

where s_t represents the logarithm of the exchange rate at time t . This equation highlights how differences in pricing kernels between the home and foreign countries drive changes in exchange rates. Notably, this expression also underpins the familiar (un)covered interest rate parity condition, connecting exchange rate expectations to interest rate differentials.

Therefore, understanding the relationship between agent optimization and the pricing kernel has profound implications for financial markets. It provides a theoretical framework that explains the dynamics of asset prices based on changes in economic states and investors' risk aversion. It offers insights into the determinants of asset prices, highlighting the role of risk aversion and temporal preferences in shaping the risk premia embedded in asset returns. This framework allows us to explain why asset prices fluctuate in response to changes in economic conditions and investors' risk perception. In forecasting, a well-specified pricing kernel model can help forecast the movements of asset prices by capturing the relation between economic and financial variables and market sentiment. Furthermore, since the pricing kernel directly influences risk premia across different assets, understanding its variations allows forecasters to better predict changes in asset returns in response to shifting economic conditions. Risk premia, the excess return that investors demand for bearing additional risk, is a direct manifestation of the pricing kernel in financial markets. As such, risk premia are central to the forecasting of asset returns because they embody the market's expectations regarding future risk and its compensation.

In conclusion, asset pricing provides a framework for understanding economic agents' decision-making and its impact on asset prices through the pricing kernel. By elucidating the interconnections between agents' preferences, the Euler equation, and the pricing kernel, this approach offers insights into the dynamics of financial markets. It underscores the significance of risk aversion, temporal preferences in determining the pricing and performance of financial assets.

Chapter 1

Recovery Theorem and the Risk Aversion. Evidence from the Czech Republic

Abstract: The main goal of this study is to obtain expectations regarding the future development of the exchange rate using derived option probabilities and based on them to calculate the perception of risk by investors. More specifically, this study deals with the application of the Ross recovery theorem in the FOREX market using the CZK/EUR exchange rate. From a theoretical point of view, I offer an expression of the Ross recovery theorem using different Numeraire and I also apply a novel approach to the calculation of the implied risk premium. The results show that both subjective and risk-neutral densities are not unbiased probability estimates of a future exchange rate. However, the results show a significant increase in the implied risk aversion during the Covid19 pandemic.

1.1 Introduction

Financial markets can provide useful information about the perception of risk by individual market participants. Options contracts are particularly useful tools, where the reason for their usefulness is due to the fact that one of the key factors determining the prices of these derivative contracts is the probability that certain events will occur. However, these contracts also include information on investors' risk aversion, as investors demand a premium for their uncertainty.

The approach presented in this study makes it possible to derive the implied risk-neutral and subjective densities of exchange rate developments using option prices. The obtained densities can theoretically allow us to extract market beliefs regarding the future. This is because subjective probability, in this case, refers to the market's estimate of the true probability density, i.e., the market's (subjective) expectation of returns, volatility, etc... It should be noted that the risk-neutral probabilities are different from subjective (and real-world) densities. Hypothetically, if the investors were risk-neutral, the risk-neutral probabilities denote what the expectations about future developments would need to be so that the present prices were identical to those observed.

Regarding subjective densities, for a long time, it was considered to be an unsolvable problem to separate risk aversion and the subjective probability that we obtain from the prices of derivative contracts from each other. However, a novel solution was offered by Ross (2015), who uses specific restrictions on the functional form of the utility function of a representative investor together with other limitations, to get the desired solution. Due to the knowledge of

these subjective densities, we can focus, for example, on the development of the risk premium over time and whether there is an improvement in predictive ability, i.e., whether these obtained densities reflect real market developments.

This study presents an application of the Ross recovery theorem in the FOREX market. Based on his work, I extract the subjective densities of individual participants from previously acquired risk-neutral densities. Furthermore, I propose a novel approach using a combination of these results to obtain a risk premium. Finally, I compare statistics constructed from the obtained densities and investigate their predictive abilities.

1.2 Literature Review

The use of the Breeden & Litzenberger (1978) method for estimating risk-neutral density has become especially popular for determining market expectations. For example, the work of Jackwerth & Rubinstein (1996) studies the S&P 500 index, and the authors are interested in the predictive power and probability of tail events implied by the obtained densities. Alonso *et al.* (2005) then applies the same analysis to the Spanish IBER35 index. Furthermore, the work of Figlewski (2010) deals with the impact of monetary policy decisions of the Fed on the development of the S&P 500 index through an event study regarding risk-neutral density quantiles. Other publications focus on subjective probabilities and try to extract them. These are papers by Ait-Sahalia & Lo (1998) and Jackwerth (2000), Liu *et al.* (2007), and Rosenberg & Engle (2000), where all these works assume that the subjective density distribution is equal to the realized historical distribution. The opposite approach is characterized by the follow-up work of Bliss & Panigirtzoglou (2004), where the authors use risk-neutral densities to derive the risk aversion coefficient and test the predictive power. Most similar to this study is Hanke *et al.* (2020), which deals with quantiles of risk-neutral densities of the world stock markets and their response to the COVID-19 pandemic.

Application of the Breeden and Litzenberger method to non-equity markets is less common, mainly due to the lower number of option quotations. In the case of currency markets, the most well-known publication is Malz (1997) and Malz (2014), where the author uses additional assumptions to interpolate the volatility surface. The first study by Malz (1997) assumes a second-order polynomial form of implied volatility, whereas the second work uses an interpolation by the clamped spline method.

It is also possible to list other papers that all use the Malz methodology in different periods and on other currencies. For example, the study of Cuaresma *et al.* (2010) deals with risk-neutral densities in the case of Central European countries. Similar analyses are provided by the Turkish Central Bank's working paper Korkmaz *et al.* (2019) and Bank of New Zealand working papers, Gereben (2002) and Lewis (2012). There is also the work of Bańbuła (2008) on this topic regarding Polish zloty and the euro. Finally, Castrén (2005) examines the risk-neutral density of the exchange rates of the three new EU member states, Poland, Czech Republic, and Hungary.

The literature on the extraction of subjective densities is quite limited, although the Ross Recovery Theorem has attracted some other authors who have built upon Ross's research ever since his publication. Regarding the theoretical foundations, Ross's work was followed by Carr & Yu (2012), who have shown that the recovery theorem can be extended to the continuous state-space setting. Another paper by Qin & Linetsky (2014) shows how to express Ross recovery in the more general state-space of Borel right processes. Backwell (2015) formalized the Recovery Theorem and added the extension, where he relaxed the assumption of state-independent preferences. In contrast, Dubynskiy & Goldstein (2013) and Borovička *et al.* (2016) present a

critique of the recovery theorem, which focuses on the decomposition of the stochastic discount factor.

Empirical work literature applying the recovery theorem is also relatively sparse, and the results vary considerably. Martin & Ross (2019) applies the recovery theorem to analyze long-dated bonds and the yield curve movements. Audrino *et al.* (2015) examines the S&P 500 option prices to investigate whether the recovery yields predictive information and found that market timing strategies based on recovered moments significantly outperform their counterparts based on their risk-neutral densities. Bakshi *et al.* (2018) apply the recovery theorem using Treasury bond futures and reject the recovery theorem. Jackwerth & Menner (2018) show that recovered physical distributions based on the S&P 500 index fail to predict future returns. These disappointing results are even more potent when authors add, as they claim, economically reasonable constraints. Lan (2018) builds on the method of Backwell (2015) using an improved strategy for combining option data with the Recovery Theorem to estimate the subjective densities; authors then investigated whether the Recovery Theorem yields predictive information and rejected the hypothesis.

On the other hand, Kiriū & Hibiki (2019) propose a robust method for the Ross Recovery Theorem that uses prior information and ensures numerical stability. Gagnon *et al.* (2022) investigate whether moments recovered using recovery theorem perform better to forecast realized volatility. The authors found evidence that these factors significantly improved realized volatility forecasts. Flint & Maré (2018) implement the Ross Recovery Theorem using option prices of the stock market in South Africa. Sanford (2018) shows that more robust results can be achieved using a multivariate Markov chain. Appel & Mare (2019) follow up on their work with specific functional forms of pricing kernel and simulated data for determining the robustness of the given method.

1.3 Options and Risk neutral densities

Options are contracts that allow one party to sell/buy an asset at a pre-agreed price in the future. Since options are quoted for several different strike prices at the same maturity, they allow observing the market's perceived probability distribution.

The basic model for determining an FX option's price is the generalized Black Scholes model. (Garman & Kohlhagen, 1983) This model assumes a lognormal movement of the underlying asset and constant volatility over time. As an input for determining the price using this model, six parameters are used $S_t, \tau, X, r_t, q_t, \sigma(t, X)$. These are the spot price, the time to maturity, the exercise price, and the risk-free domestic and foreign interest rate. The last necessary parameter is the volatility $\sigma(t, X)$. However, the value of this parameter does not correspond to the observed historical volatility of the underlying asset. Therefore, the reverse engineering method is used to determine the value of implied volatility. Knowledge of implied volatility makes it possible to assess how market participants perceive the market risks associated with individual options. For this reason, it is typical for the options market that prices are quoted in implied volatility terms.

The currency options market is further specific because there is only a limited number of options with different strike prices. However, the standard of most markets is the existence of at least three options with different exercise prices at the same maturity. In the Czech environment, three quotations of European options are used. These are at the money (*atm_t*), i.e., delta neutral call/put option, 25 delta put option, 25 delta call option.

The issue is more complicated by the fact that instead of quoting the three options, related so-called option strategies are quoted instead. The most often quoted as Risk reversal (*rr_t*) and

butterfly strategy (bf_t). Risk reversal (rr_t) is an options strategy where the investor simultaneously buys an out-of-the-money call option and sells an out-of-the-money put option. Butterfly (bf_t) is a strategy that usually consists of 4 options. The portfolio is created by buying a call option with a relatively low strike price, buying a call option with a relatively high strike price, and selling two call options with a strike price that is midway between these two strike prices. The relationship between individual quotations is expressed as follows (Reiswich & Wystup, 2010):

$$\sigma_t^{call}(\Delta) = atm_t + bf_t(\Delta) + rr_t(\Delta)/2 \quad (1.1)$$

$$\sigma_t^{put}(\Delta) = atm_t + bf_t(\Delta) - rr_t(\Delta)/2 \quad (1.2)$$

The approach presented in Breeden & Litzenberger (1978) is used in this study to determine the risk-neutral density. The approach is based on the simple idea that price always expresses the expected discounted return on a given asset. In the case of a European call option, its price is expressed as follows:

$$C(X) = e^{-rt} E^q \max(S_t - X, 0) = e^{-rt} \int_X^\infty (x - X) f_q dx \quad (1.3)$$

E^q denotes the conditional expectation at time t according to the risk-neutral probability measure q , S_t denotes the price of the asset at maturity, X denotes the strike price, e^{-rt} represents the discount factor and f_q is the risk-neutral density. Assuming continuity and differentiability, a probability density can be obtained using the second derivative with respect to the strike price.

$$f_q = e^{rt} \frac{\partial^2 C(X)}{\partial X^2} \quad (1.4)$$

1.4 Ross recovery theorem

Let us now focus on the decomposition of the risk-neutral density, which includes both the real expectation of future movements and the risk premium. For a better understanding, we would ideally be interested in the subjective probability density f_p of investors; although it occurs in the valuation, it is modified through the pricing kernel, denoted by m_{t+1} . In the simple tree model, the pricing kernel expresses the ratio of marginal utilities of a representative investor Lucas (1978). Using simple algebraic manipulations, it can be shown that only if the ratio m_t is equal to one, thus the investors are risk-neutral, and then the option price would reflect subjective market expectation. (Formally E^p denotes the conditional expectation at time t according to the probability measure p .)

$$C(X) = e^{-rt} \int_X^\infty (x - X) \frac{f_q}{f_p} f_p dx = E^p(m_{t+1} \max(S_t - X, 0)) \quad (1.5)$$

Under strong assumptions, the subjective probabilities of market participants can be extracted from the risk-neutral densities as is noted by Ross (2015). The first assumption of this method is the existence of a discrete and bounded state space Θ , consisting of θ_i , where $i \in (1, 2 \dots n)$, the other assumptions are the absence of arbitrage and a complete market. The next assumption is that the previously defined pricing kernel is m_t transition independent, and thus there exists a representative investor with a time additive inter-temporal utility. Provided that all conditions are met, the subjective probabilities can be obtained by the Ross recovery theorem.

The solution of the Ross recovery theorem can be shown in the following way. Assume a representative investor who solves the following two-period inter-temporal optimization problem

given that that investor is in a generic θ_i initial state of the world (Backwell, 2015):

$$\max u(c_{t,i}, c_{t+1,i}) = u(c_{t,i}) + \delta E^p u(c_{t+1}) = c_{t,i} + \delta \sum_{j=1}^n u(c_{t+1,ij}) f_{i,j} \quad (1.6)$$

Given a budget constraint:

$$c_{t,i} + \sum_{j=1}^n c_{t+1,ij} p_{i,j} = w \quad (1.7)$$

$u(c_{t,i}, c_{t+1,i})$ denotes the two-period utility function. $f_{i,j}$ denotes the real probability of transition from the generic initial state θ_i to the generic final state θ_j . $p_{i,j}$ represents the price of the Arrow Debreu security contingent on the initial state θ_i , which pays 1 unit of Numeraire¹ if the next state is θ_j , and zero otherwise. $c_{t,i}$ denotes initial consumption at time t and final consumption at time $t+1$ contingent on the future state of the world is then denoted by $c_{t+1,ij}$, δ denotes a discount factor and w is positive constant representing the wealth. The first order conditions then lead to the following solution for the price of state-contingent claim for every state i, j

$$p_{i,j} = \delta \frac{u(c_{t+1,ij})'}{u(c_{t,i})'} f_{i,j} = m_{i,j,t+1} f_{i,j} \quad (1.8)$$

Finally, according to Backwell (2015), we assume an equilibrium condition on consumption, that ensures that the consumption is assumed to be dictated only by the state. In other words, for every i, j , the following holds:

$$c_{t,j} = c_{t+1,ji} = c_i \quad (1.9)$$

The above optimal solution for all i, j can be further simplified using the equilibrium condition of time homogeneity of consumption. It should be noted that this significant simplification and the assumption of time homogeneity is one of the main weaknesses of the Recovery theorem. This is due to the fact that it rules out some useful pricing kernel properties, such as possible path-dependence in expectations and different transition state prices in different periods, i.e., we assume that all time periods have identical structures for prices and probabilities. Therefore, the pricing kernel is the transition independent, which means that it is independent of any calendar time.

$$p_{i,j} = \delta \frac{u(c_j)'}{u(c_i)'} f_{i,j} = m_{i,j} f_{i,j} \quad (1.10)$$

Since this applies to all i, j in θ , the result can be rewritten in matrix form as:

$$P = \delta D^{-1} F D \quad (1.11)$$

P is a $n \times n$ matrix of Arrow Debreu prices, F is a $n \times n$ matrix of subjective probabilities of investors and D is an $n \times n$ diagonal matrix of marginal utilities, which consists of d_i on the diagonal and zero outside the diagonal. Using matrix algebra and the knowledge that the matrix F is a stochastic matrix, i.e., the sum of individual rows must be one or $F\mathbb{1} = \mathbb{1}$, where $\mathbb{1}$ is the vector of the ones. Due to the Perron Frobenius theorem, which guarantees the existence of a unique real eigenvalue, the problem is reduced to finding the largest eigenvalue and its corresponding eigenvector. The calculation is as follows, where we defined $D^{-1}\mathbb{1} = z$.

$$Pz = \delta z \quad (1.12)$$

¹Unit of account, for formal treatment see: Geman *et al.* (1995)

1.5 Stochastic Discount Factor in FOREX market

So far, we have dealt with the general valuation of a risky financial instrument. Let us now turn to the FOREX market and the risk premium associated with it. We begin with a definition of the gross return of a risky asset denominated in foreign currency as in Backus *et al.* (2001b):

$$1 = E^p(m_{i,t+1}(1 + r_{r,t+1})) \quad (1.13)$$

Next, we consider an asset denominated in foreign currency. We adopt the same approach as before for pricing this foreign-denominated asset. By $*$ we denote foreign variables, i.e., $m_{i,j}^*$ denotes foreign pricing kernel and $r_{r,t+1}^*$ foreign gross return:

$$1 = E^p(m_{i,t+1}^*(1 + r_{r,t+1}^*)) \quad (1.14)$$

By using the no-arbitrage argument we could alternatively convert foreign instrument returns into domestic returns and value them using domestic pricing kernel, where FX_t denotes the exchange rate at a time t and $E^p FX_{t+1}$ denotes the conditional expectation at a time t of the exchange rate at a time $t + 1$.

$$1 = E^p(m_{i,t+1}(1 + r_{r,t+1}^*) \frac{E^p FX_{t+1}}{FX_t}) \quad (1.15)$$

This leads to the following equality between pricing kernels and exchange rate development:

$$m_{i,t+1} \frac{E^p FX_{t+1}}{FX_t} = m_{i,t+1}^* \quad (1.16)$$

This reason, as was stated before, is that if equality is violated and both currencies are traded, there is an arbitrage opportunity, as there would exist expected excess return by converting to another currency. In conclusion, using the no-arbitrage argument, we have derived the equivalence of these two procedures for the valuation of foreign assets. Combining this result with the Ross recovery theorem stated above gives us the following relation:

$$p_{i,j} = m_{i,t+1} \frac{E^p FX_{t+1}}{FX_t} f_{i,j} = m_{i,t+1}^* f_{i,j} \quad (1.17)$$

By combining this result with the assumption that the exchange rate development can be described by different Markov transition matrix², we obtain an alternative definition of the Ross recovery theorem.

$$p_{i,j} = \delta \frac{u(c_j)'}{u(c_i)'} f_{i,j} f_{x_{i,j}} \quad (1.18)$$

I use $f_{x_{i,j}}$ to denote the ratio of the exchange rate (FX in matrix form). This can be rewritten in matrix form using the fact, that the product of two stochastic matrices is a stochastic matrix, i.e., $F(FX) = (FX)F = Q$ and the Perron-Frobenius theorem thus can be applied:

$$P = \delta D^{-1}(FX)ED = \delta D^{-1}QD \quad (1.19)$$

²Formally, we must also assume the same condition as in the case of consumption regarding equilibrium value.

1.6 Methodology

To determine the price of the option for non-quoted values, it is necessary to use interpolation and extrapolation methods, which lead to the need to make certain assumptions about the form of implied volatility. The calculation method used in this study is the Vanna Volga method from Castagna & Mercurio (2007). Vanna Volga method is based on the assumption of dynamic hedging and the creation of a locally risk-free portfolio. Using the knowledge of implied volatility, we have defined all possible option prices at different exercise prices, which is also equivalent to the knowledge of risk-neutral probability. (Blake & Rule, 2015)

More formally, Castagna & Mercurio (2007) assume that a risk-free portfolio can be created consisting of a long position in a call option, a short position in the underlying instrument, and three short positions in other options. Due to this, the portfolio is not only risk-free in the delta but also Vega, Vanna, and Volga hedged, i.e., the portfolio then is hedged not only against shifts in the underlying asset up to the second order but also against movements in implied volatility also up to second order. In other words, there is a perfect local hedge, provided we have a sufficient number of other options. The great advantage of the Vanna Volga method is its relative simplicity. Therefore, unlike other methods such as SABR, it can be easily applied to less liquid markets with fewer quotations.

Even more formally, let's assume three different options available, each with different strike prices denoted by X_i , and x_i representing the weight of the given option in the portfolio. The portfolio has the following form:

$$\pi = C(X) - \Delta X - \sum_{i=1}^3 x_i C(X_i) \quad (1.20)$$

Using Ito's lemma, we get the following expression for portfolio development:

$$d\pi = dC(X) - \Delta dX - \sum_{i=1}^3 x_i dC(X_i) \quad (1.21)$$

To erase the stochastic parts, it is necessary to determine the individual weights (x_i) and the size of an underlying asset (Δ). Using the well-known relations between the Greeks, we obtain the following conditions for Vega, Vanna, and Volga:

$$v = \frac{\partial C(X)}{\partial \sigma_t} = \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial \sigma_t} = \sum_{i=1}^3 x_i v_i \quad (1.22)$$

$$vd2 = \frac{\partial^2 C(X)}{\partial \sigma_t^2} = \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial \sigma_t} \frac{d2_i}{X_i \sigma_t \sqrt{t}} = \sum_{i=1}^3 x_i v_i d2_i \quad (1.23)$$

$$vd2d1 = \frac{\partial^2 C(X)}{\partial \sigma \partial X} = \sum_{i=1}^3 x_i \frac{\partial C(X)}{\partial \sigma_t} \frac{d2_i d1_i}{\sigma_t} = \sum_{i=1}^3 x_i v_i d2_i d1_i \quad (1.24)$$

Vega (v) denotes a partial derivative with respect to the change in implied volatility of the underlying currency pair. Vanna and Volga are expressed using Vega. This is a system of three equations with three unknowns which leads to the following weights for Vanna Volga's secured portfolio:

$$x_1 = \frac{v \ln(X_2/X) \ln(X_3/X)}{v_1 \ln(X_2/X_1) \ln(X_3/X_1)} \quad (1.25)$$

$$x_2 = \frac{v \ln(X/X_1) \ln(X_3/X)}{v_2 \ln(X_2/X_1) \ln(X_3/X_2)} \quad (1.26)$$

$$x_3 = \frac{v \ln(X/X_1) \ln(X/X_2)}{v_3 \ln(X_3/X_1) \ln(X_3/X_2)} \quad (1.27)$$

To determine the size of the delta, it is necessary that the portfolio is delta-secured. (Since this approach is only an approximation, we assume constant volatility.) The result is the following position in the underlying asset:

$$\Delta = \frac{\partial(C(X))}{\partial X} - \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial X_i} \quad (1.28)$$

The result is a locally risk-free portfolio, the price of which is the sum of the delta hedge and Vanna Volga hedge. To determine the state prices based on the approach presented in Breeden & Litzenberger (1978), we can use the second derivative of the portfolio price with respect to the strike price.

The main practical problem of the empirical implementation of the Ross recovery theorem is, therefore, the estimation of the matrix P containing the Arrow Debreu prices $p_{i,j}$. Ross (2015) proposes a solution using the current state prices vector s_t . Using these prices, we obtain the state space, which consists of n realization prices and m maturities. However, this is not the required matrix P . The difference is that the vector s_t consists of prices (prices of the contingent claims), which give one unit of Numeraire if state j occurs due to the fact that we are in state i . In contrast, the matrix P determines the prices of Arrow Debreu securities for every state i and j . However, there is one row of the matrix P , which is already known. It is the vector s_t that corresponds to the transition time given by the transition matrix P .

For each period, I am able to obtain state prices s_t using the (Breeden & Litzenberger, 1978) method with different maturities up to time t . Using the assumption of time homogeneity of matrix P , I obtain the following relation between state prices and matrix P , which can be written in vector form as follows:

$$s_t = \frac{\partial^2 C(X)}{\partial X^2} \quad (1.29)$$

$$s_t = s_{t+1} P \quad (1.30)$$

Due to the assumption of time homogeneity, the relation can be aggregated for all maturities as follows: $A = BP$ under the condition $P > 0$. Where $A = (s_2, \dots, s_m)$ is a matrix $n \times m$ and $B = (s_1, \dots, s_{m-1})$ is a matrix $n \times m$, provided that $n \leq m$. Due to the numerical instability of the matrix P , the method of generalized ridge regression with a prior probability matrix $\tilde{P}$ is used. The estimation of the matrix P can, therefore, be written as the following optimization problem ³ as in Kiriu & Hibiki (2019):

$$P = \operatorname{argmin}_{P>0} (AP - B)^t (AP - B) + \lambda \left(P - \tilde{P} \right)^t (P - \tilde{P}) \quad (1.31)$$

$$P = \operatorname{argmin}_{P>0} y^{fit}(\lambda) + \lambda y^{reg}(\lambda) \quad (1.32)$$

$\tilde{P}$ is a $n \times n$ matrix containing a prior information. The matrix form is based on the assumptions that $p_{i,j}$ is similar to $p_{i+k,j+k}$ for all $k < \min(n-i, n-j)$. This basically guarantees the unimodality of transition probabilities within this transition matrix and thus the highest

³Practically the optimization problem can be rewritten as n generalized ridge regressions. The non-negative least squares algorithm is then used for the calculation itself due to the need to maintain positive transition probabilities.

values are located on the diagonal of the matrix. This formulation effectively restricts the state space over which probabilities must be estimated, which is necessary given limited data. On the other hand, it also eliminates states that are large outliers, this also means disaster events that are the main reason for people to buy put options. For this reason, it can significantly distort distributions with long tails or significant skewness. This means that if we are in state i at time t , we are most likely to move near it. In general, the matrix $\tilde{P}$ has the following form:

$$\tilde{P} = i_0 \begin{pmatrix} \sum_{k=1}^{i_0} s_{k,1} & \dots & s_{n,1} & 0 & 0 \\ \sum_{k=1}^{i_0+1} s_{k,1} & \dots & s_{n-1,1} & s_{n,1} & 0 \\ s_{i_0+1,1} & s_{i_0+1,1} & \dots & s_{i_0+1,n-1,1} & s_{i_0,n} \\ 0 & s_{1,1} & s_{2,1} & \dots & \sum_{k=n-1}^n s_{k,1} \\ 0 & 0 & s_{1,1} & \dots & \sum_{k=i_0}^n s_{k,1} \end{pmatrix} \quad (1.33)$$

For simplicity, the estimation is performed with the assumption that there is an odd number of states θ_i and the middle column of the matrix $\tilde{P}$ is denoted $i_0 = (n + 1)/2$. It should be noted here, that the values are defined as spot moneyness (X/S). The optimal lambda is then determined through problem-specific function $h(\lambda)$:

$$\text{argmin } h(\lambda) = \frac{y^{fit}(\lambda) - y^{fit}(0)}{y^{reg}(\infty) - y^{reg}(0)} + \frac{y^{reg}(\lambda) - y^{reg}(\infty)}{y^{reg}(0) - y^{reg}(\infty)} \quad (1.34)$$

y^{fit} denotes residuals and y^{reg} denotes the difference between the estimated and the prior matrix. In the denominator, there is the maximum possible range, and in the numerator, there is a range for the given value of λ . Kiriu & Hibiki (2019) showed λ monotonically increases y^{reg} and monotonically decreases y^{fit} .

The actual implementation was done in the following way. The implied state price surface obtained above is discretized in $m = 54$ evenly spaced maturities from 1 month to one year, which roughly approximates the weekly maturities. This is due to the fact that if we used only six non-overlapping periods (with different lengths) we would obtain only six transitions which allow for at most six states. This would result in inaccurate measurement due to the coarse grid. Concerning the strike dimension, we create a moneyness grid from 0.85 to 1.25 ($n = 29$), which seems to be sufficient given the high persistence of the exchange rate. The moneyness grid size is also approximately equal to the half size of Jackwerth & Menner (2018), which deals with much more volatile stock prices. Finally, the state price matrix S consisting of $(s, \dots, s_m)$ is then split into two submatrices with a lag of 3 months, meaning that the recovered physical transition probabilities represent expectations for 3 months ahead.

1.7 Empirical estimation

1.7.1 Data

The Refinitiv database was used as a data source. Bid and ask option end of the day quotations from the Butterfly strategy, Risk reversal strategy and at the money option with maturities of 1,2,3,6,9 and 12 months on each business day since the beginning of the year 2018 were obtained.⁴ 25 delta put option, 25 delta call option quotations are derived as in equations 1

⁴Specifically following tickers were used: EURCZK1MO=, EUCZ1MRR=, EUCZ1MBF=, EURCZK2MO=, EUCZ2MRR=, EUCZ2MBF=, EURCZK3MO=, EUCZ3MRR=, EUCZ3MBF=, EURCZK6MO=, EUCZ6MRR=, EUCZ6MBF=, EURCZK9MO=, EUCZ9MRR=, EUCZ9MBF=, EURCZK1YO=, EUCZ1YRR=, EUCZ1YBF=.

and 2. Strike prices from the OTC quotations are computed as in Reiswich & Wystup (2010). The strike price for at the money delta is computed as follows:

$$X = F_t e^{\frac{1}{2}\sigma_t^2 t} \quad (1.35)$$

Strike price for other values of (unadjusted spot) delta is obtained as follows:

$$X = F_t e^{\frac{1}{2}\sigma_t^2 t - \sigma_t \sqrt{t} N^{-1}(e^{qt} \Delta_s)} \quad (1.36)$$

The last day obtained is October 7, 2020. The resulting value was determined as the average of the current bid and ask prices. IBOR quotes with the corresponding maturity were used as a proxy for the risk-free rate. The mid-price quotation was used for the spot rate. For overall statistics on the FX market, the Czech National Bank survey data were used.

1.7.2 Risk Neutral Densities

The study empirically deals with the foreign exchange market in the Czech Republic, where it focuses mainly on the period after the end of the foreign exchange commitment, emphasizing the period of the COVID-19 pandemic. To begin the description of the foreign exchange market in the Czech Republic, the following Figure 1.1 shows the structure of the Czech foreign exchange market from domestic banks' perspective. It can be seen that over 90% of foreign exchange transactions with the Czech koruna carried out by domestic banks are with the Euro and the US dollar. It can be observed that forward, and FX swaps occupy the largest share in notional volume, and there is a much smaller share of options.

Figure 1.1: FX transactions of Czech banks (daily volume)

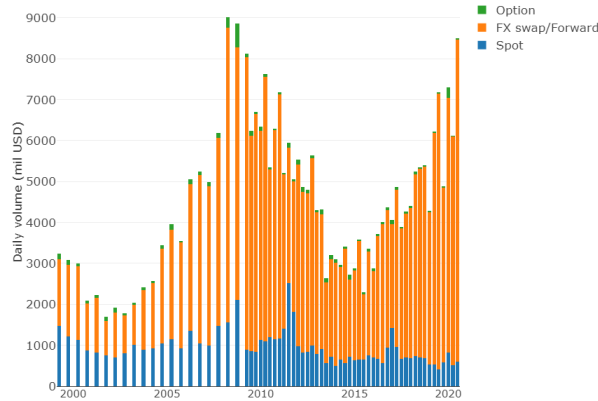
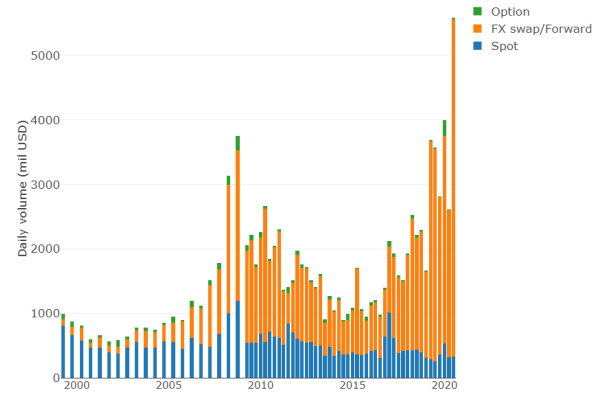


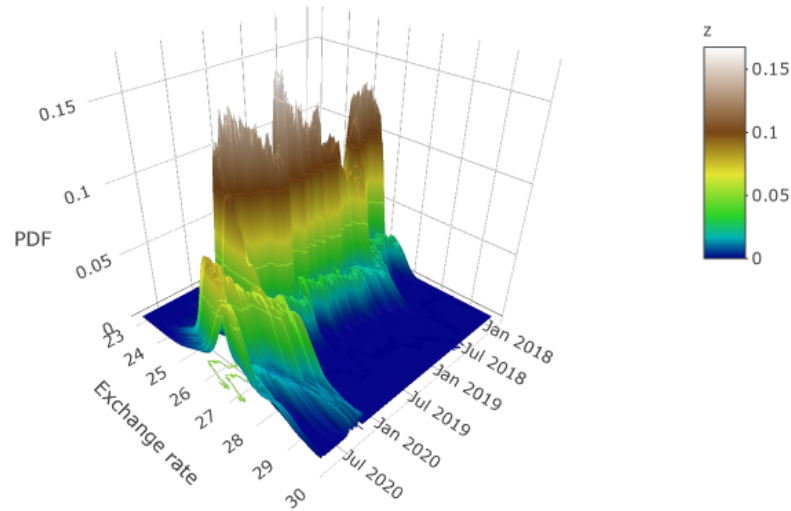
Figure 1.2: CZK/EUR statistics



Source: Czech National Bank

I obtained the risk-neutral density using the Breeden and Litzenberger method. Let us focus on the time series of the risk-neutral probability density of the three-month CZK/EUR exchange rate from 2018 to 2020. From its development, as is shown in Figure 1.3, we can see a stable and almost symmetrical probability distribution. The mean and median of the given distribution oscillates between 25 and 26 CZK/EUR. Furthermore, the 10% and 90% quantiles of the implied risk-neutral distribution consistently fall within the range of 25 to 27 CZK/EUR, with the distance between individual quantiles remaining approximately constant until 2020, or even slightly decreasing in specific periods.

Figure 1.3: Risk-neutral density



Source: Own calculation, Refinitiv

However, already in the first quarter of 2020, we have observed a significant worldwide increase in the uncertainty associated with the COVID-19 pandemic. The Czech koruna weakened to the level of 28 CZK/EUR, and there was a significant change in the implied density of the CZK/EUR exchange rate. This means that the obtained implied densities showed an increase in the mean of the distribution and an increase in the standard deviation, interquartile range, and Pearson and Kelly skewness, as shown in Figure 1.4. These statistical moments peaked in the first half of 2020. In particular, the month of April indicates the highest increase in uncertainty about exchange rate developments, which is manifested by the highest standard deviation in the observed period.

Another statistical moment examined is skewness, which is shown in Figure 1.5. The COVID-19 pandemic results show an increase in skewness, indicating a greater probability of depreciation of the Czech koruna. However, the results probably include a significant increase in investors' risk aversion, as will be illustrated later. This can be intuitively explained by the investor's fear of further depreciation of the Czech Koruna, which can be explained by the Czech Republic's perception as an emerging market and CEE country.

Figure 1.4: RND: Standard Deviation and Skewness

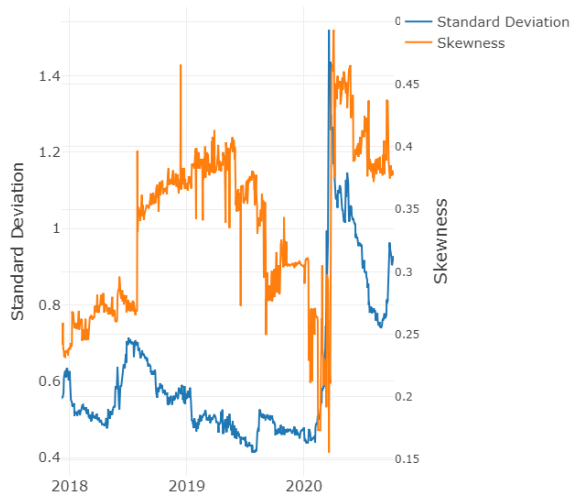
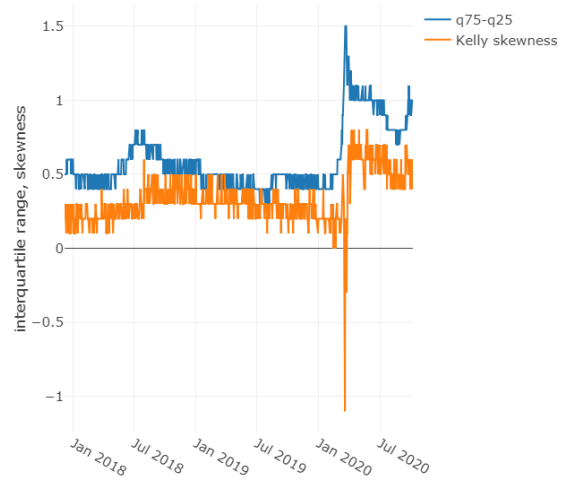


Figure 1.5: RND: Kelly Skewness and interquartile range



Source: Own calculation, Refinitiv

Next, let's focus on the predictive power of derivative contracts, namely whether the market was able to predict the COVID-19 pandemic. Firstly, we can observe that the increase in the variance and individual quantiles of the risk-neutral density occurred essentially simultaneously with the sudden movement in the spot exchange rate, as is shown in Figure 1.6. However, this can also be interpreted as a sign that the market did not expect such a sharp movement, and due to this, the reaction of investors was probably even more pronounced. Since May, the situation has gradually calmed down, but the risk-neutral density remains highly dispersed and skewed to the right.

I also tested the predictive ability of the entire risk-neutral density by using the probability integral transform (Diebold *et al.*, 1998). The results indicate limited predictive ability; the values on the left side of the histogram (Figure 1.7) prevail, which means a more frequent depreciation of the Czech koruna than expected. This was formally confirmed by the Kolmogorov-Smirnov statistic, where the null hypothesis about the uniformity of the obtained distribution was rejected.

Figure 1.6: Risk-neutral density at the time of settlement

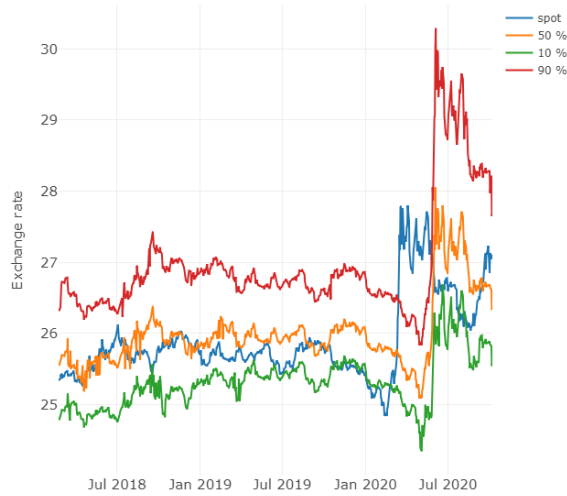
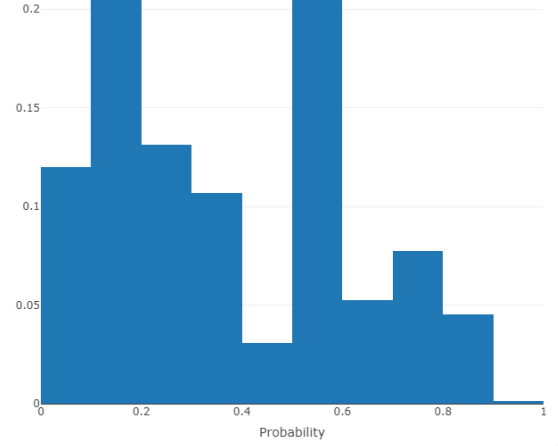


Figure 1.7: RND-Probability integral transform



Source: Own calculation, Refinitiv

1.7.3 Subjective Densities

Using the methodology described above, I was able to obtain subjective densities. The development of the implied subjective and risk-neutral probability densities shows their very similar movements at the horizon of three months. However, the quantiles of subjective densities show a smaller possible range of exchange rate movements in general. In particular, sudden depreciation of the exchange rate becomes less likely; on the contrary, the probability of appreciation is mostly maintained. A significant difference between the development of subjective and risk-neutral distribution does not occur until the period of the COVID-19 pandemic in the year 2020.

Figure 1.9 shows that the increased range of risk-neutral probability quantiles did not fully manifest in the subjective probability quantiles. This can be illustrated by the fact that in the second quarter of 2020, 90% quantile of the risk-neutral probability quantiles almost reached 30 CZK/EUR, while at the same time, the 90% quantile of subjective probability was almost a unit less. These results are in line with the intuition about the global increase in the investors' risk aversion.

Regarding the moments of obtained densities, in the case of the subjective probability density, Figure 1.8 shows that the mean of the distribution, standard deviation increased, i.e., the same development as in the case of the risk-neutral distribution. Skewness slightly increased but did not reach the values in 2019.

Figure 1.8: Standard Deviation and Skewness

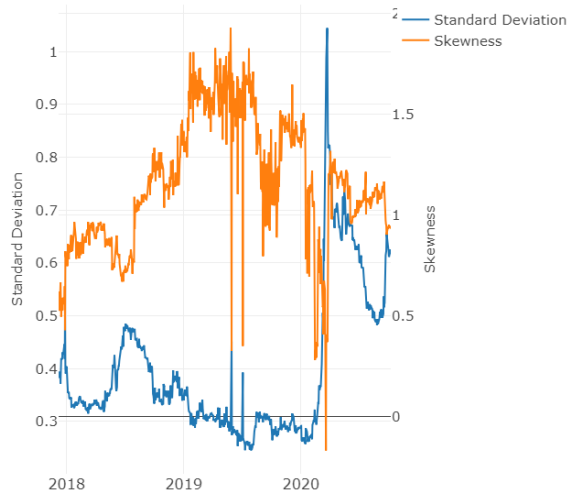
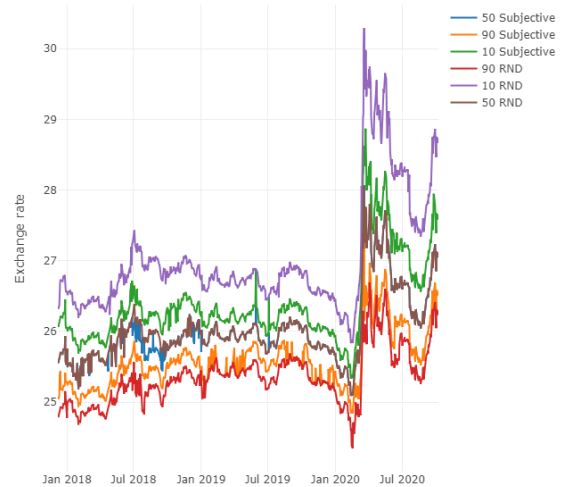


Figure 1.9: Comparison of risk neutral and subjective densities



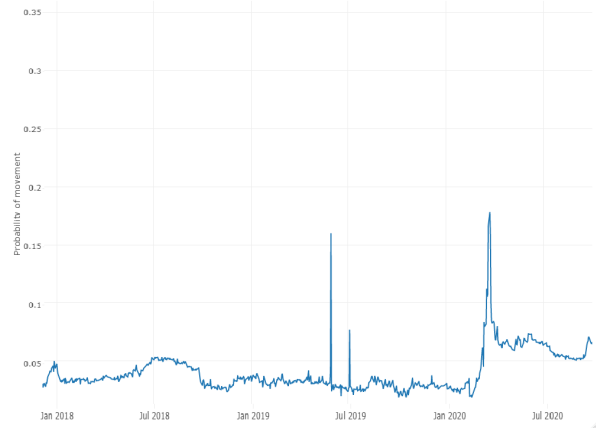
Source: Own calculation, Refinitiv

Regarding uncertainty during the COVID-19 pandemic, I calculated the probabilities of extreme movement in exchange rate implied by the risk-neutral and subjective densities, as shown in Figure 1.10 and Figure 1.11. I focus on the probability of the exchange rate movement by more than 10 % in every direction compared to the spot rate. As in the previous case, I can observe that until the COVID-19 pandemic, this indicator does not show any significant shifts, and for most of the observed period, this probability is not higher than 10 %. On the contrary, this indicator reached its peak in April 2020, where the option-implied probability of a sudden movement is almost 30 %. In the case of subjective density, we can observe that for the most time during the observed period, the probability is not higher than 5 % and also reaches its peak in April 2020, where it implies a probability of sudden movement of almost 20 %. Overall, the development of the two indicators is very similar, with the difference that subjective probability densities for most of the observed period show a lower probability of a sudden and extreme movement.

Figure 1.10: Risk-neutral probability of movement more than 10 %



Figure 1.11: Subjective probability of movement more than 10 %



Source: Own calculation, Refinitiv

However, it should be noted that the results need to be interpreted very carefully, as it is possible that rather than the real subjective probability of market participants, we have extracted the probability of a representative investor with a time separable utility function using Vanna Volga hedging.

However, if we assume that we have obtained real subjective probabilities of market participants, then due to the knowledge of both probability densities, we can calculate the implied risk premium rate. If we start from the equation of the price of the FX forward using risk-neutral and subjective probability measures, equating these two equations, then we are able to extract the risk premium coefficient from them. Due to the interest rate parity, the FX forward price is equal to the expected value of the exchange rate in the risk-neutral world q as well as in the real world, where the pricing kernel (risk premium) enters the valuation (Vincent-Humphreys & Noss, 2012). S_0 denotes the spot rate at the time of signing the contract, S_t denotes the spot rate at time t , r and q denote the domestic and foreign risk-free interest rate and rp denotes the risk premium.

$$S_0 = E^q S_t e^{-(r-q)t} = E^p S_t e^{-(r-q+rp)t} \quad (1.37)$$

$$rp = \frac{1}{t} \ln \left(\frac{E^q S_t}{E^p S_t} \right) \quad (1.38)$$

Results in Figure 1.12 show that for most of the observed period, the positive value of the risk premium with a predominantly constant size prevails from the point of view of the Czech currency. However, a notable turning point occurred in 2020, coinciding with the onset of the COVID-19 pandemic. During this period, the risk premium spiked dramatically, reaching almost 0.2%, reflecting heightened market uncertainty and a sharp response from investors. Subsequently, the risk premium began to decline, indicating that the initial exaggerated reaction by investors was gradually corrected as the situation stabilized over time. This would correspond to the perception of the Czech Republic as a small open economy and a CEE country, and such countries are considered riskier by most investors. It is also important to realize that we observe a risk premium for the exchange rate. Therefore, it is not surprising that the obtained risk premium may be close to zero or even negative in certain periods.

Figure 1.12: Risk premium



Source: Own calculation, Refinitiv

To evaluate the robustness and predictive power of the obtained subjective densities, the Probability Integral Transformation (PIT) was performed. However, as illustrated in Figure 1.13, the results do not show significant improvement compared to the risk-neutral density. On one hand, this result can be interpreted as an incorrect specification of the model, i.e., rejection of the Ross Recovery theorem. On the other hand, an alternative explanation is that the market is not able to predict future movements, even though the model is correct. To assess the robustness of the achieved results, the period of the COVID-19 pandemic was taken from the data sample, but the results remained essentially the same. (Figure 1.14) Formally, the Kolmogorov-Smirnov statistic was used to test the obtained distribution, where the null hypothesis about the uniformity of the obtained distribution was rejected.

Although this may seem like an overwhelming rejection, it should be noted that we are primarily concerned with a relatively illiquid market. We also focus on a relatively short period of time, especially the COVID-19 pandemic period, which can be seen as a purely exogenous shock with an extremely low probability of occurrence. Even after some time, it can be said that the reaction of monetary and fiscal authorities was unprecedented, which may have partially affected the results due to the difficulty of predicting these events. On the other hand, the dynamics of the obtained risk premia look intuitively appealing. And partially restores the validity of the Recovery theorem.

Figure 1.13: Subjective Density

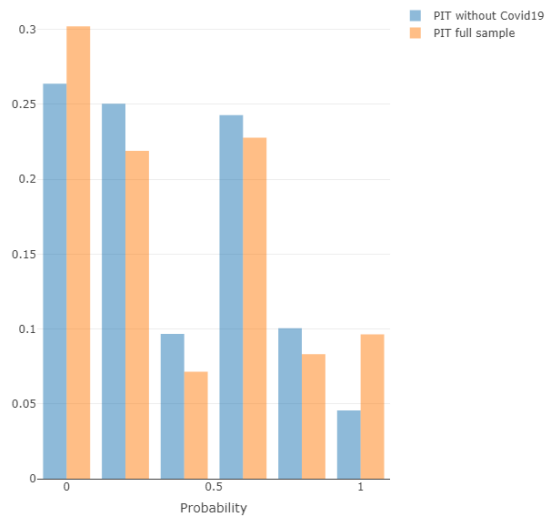
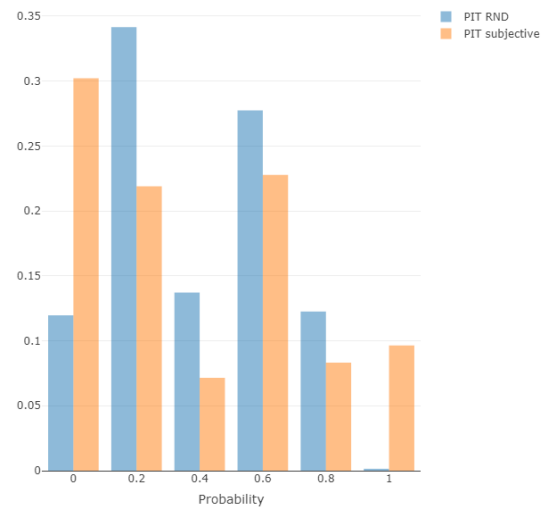


Figure 1.14: RND and Subjective Density



Source: own calculation, Refinitiv

Below, I show the correlation matrix of individual moments from obtained densities. Different correlations indicate additional information is found in different individual moments.

The resulting correlation matrix shows an almost perfect correlation of the first two moments of the subjective and risk-neutral distribution. On the other hand, the third and fourth moments have a correlation of only about 0.6, which suggests that new information about the subjective distribution may be found in these moments. This is also shown by the risk premium correlation, which is the highest with the subjective standard deviation and skewness.

Table 1.1: Correlation matrix

	Risk Premium	RND Mean	RND s.d.	RND Dkewness	RND Kurtosis	Subjective Mean	Subjective s.d.
Risk Premium	1						
RND Mean	-0.11	1					
RND s.d.	0.66	-0.49	1				
RND Skewness	0.24	0.53	-0.39	1			
RND Kurtosis	-0.12	0.59	-0.66	0.81	1		
Subjective Mean	-0.11	1.00	-0.49	0.53	0.59	1	
Subjective s.d.	0.73	-0.45	0.99	-0.34	-0.60	-0.45	1
Subjective Skewness	0.71	0.02	0.35	0.61	0.15	0.02	0.39

1.8 Conclusion

In this study, risk premium, subjective, and risk-neutral densities were estimated using daily options data. Furthermore, the results of this work show that risk-neutral and subjective densities obtained using option pricing methods do not correspond to the historical realization of the real process as is tested by the probability integral transform. This can be interpreted as the market was not able to predict future developments before the COVID-19 pandemic and even lost all predictive power during the first few months of the COVID-19 pandemic. In the case of risk-neutral densities, the existence of bias in expectation is a long-known fact due to the existence of a risk premium, but if we were really able to obtain the subjective probabilities of a representative investor using the Ross recovery theorem, we have shown that the market is unable to predict future developments.

At the same time, we illustrate the time-varying risk premium of investors using data from obtained subjective and risk-neutral densities. The development of risk premium shows an approximately constant value until the period of the COVID-19 pandemic. However, after the beginning of the COVID-19 pandemic, there is a significant growth in risk premium probably due to the growth of global risk aversion. This allows an intuitive explanation of the market's inability to predict future developments as required risk premiums can change dramatically.

However, all the results presented in this study need to be interpreted with great caution because obtaining risk-neutral and subjective probabilities are two ill-posed problems that are connected to each other and are characterized by numerical instability. The fact that the obtained densities do not correspond to the realization of the processes does not necessarily mean that the market is not able to predict future developments, but the chosen method was not able to extract the subjective probabilities of market participants. On the other hand, the time-varying risk premium partially restores the validity of the recovery theorem.

Chapter 2

Deriving Equity Risk Premium Using Dividend Futures

Abstract: In this paper I present a simple stock price decomposition model using the dividend discount model and dividend futures. The main contribution of this paper is the use of dividend futures which represent the risk-adjusted expectations of future dividends. This allows for the calculation of the implied equity risk premium and the decomposition of stock price movements into individual components. Due to the use of daily market data, this method can take into account the structural changes associated with falling interest rates and the Covid-19 pandemic. I empirically show the risk premium development of the SP 500 Index and Euro Stoxx 50 Index in the last decade.

2.1 Introduction

It is generally accepted that equity prices represent the present value of all future dividends.¹ The assumption of discounted dividend valuation is applied in some form or other in all the best known textbook models used for the valuation of individual companies, whether it is the Gordon model or the Free Cash Flow model. The advantage of these types of models is their explicitly forward-looking nature. A disadvantage is the existence of a large number of free variables and an excessive use of expert judgment (Mauboussin *et al.*, 2021).

The studies conducted by academic economists often focus on stock indices as a whole rather than on individual companies. These studies frequently center on the relationship between dividends (earnings) and price, often referred to as the Campbell-Shiller decomposition (Campbell & Shiller, 1988). This decomposition indicates the existence of co-integration relations between these variables. Alternatively, the co-integration assumptions are implicitly built into the model, and their predictive powers are investigated. The different regressions with market ratios such as P/D, P/E, P/S and CAPE are examples of this approach. Although I do not question the existence of co-integration relations, it is possible that the decline in the term structure of interest rates, especially in the last ten years, has led to a structural change in these ratios. Thus, these variables cannot take into account the forward-looking nature of prices, and it is necessary to use different approaches to explain price movements.

This paper contributes to the existing literature by augmenting a simple dividend discount model with dividend futures. This framework allows me to use forward-looking variables with predictive power regarding future developments. This approach also allows for the use of non-

¹At least in the long run when we do not allow for the existence of infinitely growing bubbles.

deflated and daily data and makes it possible to measure the time-varying equity risk premium of investors in the model. I see this as a great advantage as daily frequency data can capture a more detailed view of the market, especially the overall dynamics. Certain events, such as the market panic take place within a few days, and researchers are often interested in how expectations and risk aversion change during this period. Using lower frequency data can mask significant movements, for example, due to the rapid response of central banks and governments.

For this reason, this study does not focus on excess returns or overvaluation, but rather on a structural interpretation of price movements, with an emphasis on the changing perception of risk. Market sentiment is often measured using survey data or statistical methods (e.g. factor models). The advantage of equity risk premium, estimated in this paper, is its simple interpretation, which directly enters the expression for valuation. Overall, this approach can provide useful information about future risks and allows for a narrative-based approach which explains movements in the stock market and even the calibration of possible stress test scenarios. The study relates to Brennan (1998) which argues that a market for dividend derivatives would promote rational pricing in stock markets. It also generally relates to Cochrane (2011) as the research topic is “asset pricing” and not “asset expected returning”.

The rest of this paper is structured as follows: In section 2, I present the existing literature on the equity risk premium. I then present a combination of the dividend discount model and dividend futures from a theoretical perspective. Finally, in sections 4 and 5, I perform an empirical analysis and calculate the implied risk premiums for the Euro Stoxx 50 and S&P 500 indices.

2.2 Literature Review

The topic of this study is similar to the innovative work of Shiller (1981) who focused on excessive price volatility and the fundamental stock price and its derivation through realized dividends. This paper is based on several studies which focus explicitly on explaining the level of stock indices (Barsky & De Long (1993a), Panigirtzoglou & Scammell (2005), Inkinen *et al.* (2010) Andrieu (2019) and Kapp & Kristiansen (2021)). The focus of these studies is to explain the level of the given indices using the Dividend Discount Model and market analysts’ predictions of future earnings and dividends. The studies also (at least implicitly) highlight the extrapolative nature of market analysts’ expectations that influence price at a given moment. Moreover, although a given price may seem overvalued in hindsight, it may correspond to a rational valuation at a given moment. These studies also provide estimates of the equity risk premium implied by their models, which is a proxy for market sentiment and an “irrational” component of valuation. Overall, the literature on estimating equity risk premium is relatively sparse and centers mainly on practical valuation.

A good summary on the estimation of risk premium is presented by Damodaran (2020), Duarte & Rosa (2015) and Fernandez (2007), these studies also mention two other commonly used methods for determining risk premium. First, a historical method based on real excess price returns relative to bond returns, for example: Fama & French (2002); Welch & Goyal (2008); Campbell & Thompson (2008). The inherent problem with this approach is its backward looking nature, i.e. it does not take into account, that changes in the underlying factors could be already priced in. For this reason, the underlying assumption is that the equity risk premium is constant or very persistent over time. The second approach is cross-sectional regressions models which often use the Fama-French decomposition (Fama & French, 1995) to calculate the market risk premium. A more recent contribution is, for example, Adrian *et al.* (2015), which uses affine pricing kernel specification. For completeness, let’s also mention the use of survey data as the

last approach for directly measuring equity risk premium (Graham & Harvey (2018)).

Studies also have been conducted in other research fields using option prices and the Breeden & Litzenberger (1978) methodology to determine the risk premium and market expectations. As in this paper, these studies use the pricing kernel and risk-neutral densities to determine the risk premium. The main problem of this approach is the need for strong implicit assumptions regarding the behavior of the model. For example, constraints are placed on the utility function parameters or the development of state prices. Existing studies usually assume that the subjective density distribution is equal to the realized historical distribution (Ait-Sahalia & Lo (1998), Jackwerth (2000), Liu *et al.* (2007), Rosenberg & Engle (2000)). This, however, is again a backward looking approach, as in the case of excess returns. A different approach is presented in a study by Bliss & Panigirtzoglou (2004), where the authors use risk-neutral densities and a specific form of the pricing kernel to derive the risk aversion coefficient. Another approach is put forward by Ross (2015), where he does not place an explicit constraint on the utility function but on the state price process itself. The approach presented in this paper is considerably more straightforward and although it places a constraint on the pricing kernel and other variables, these constraints are explicit and much simpler to interpret. Most similar to this study (in terms of option approach) are papers by Martin (2017); Martin & Wagner (2019), where the authors provide a lower bound on the expected equity risk premium using options data, going as far as arguing the lower bound can be interpreted as the equity risk premium. The authors' approach is based on the computation of risk-neutral variance of gross returns. Therefore, the main difference is that the authors compute equity risk premium conditional on the return horizon chosen. On the other hand, my approach computes unconditional equity risk premium with a theoretically infinite return horizon. This also explains the lower variability of risk premium obtained in this paper.

Gormsen & Koijen (2020) puts forward a different approach which is loosely related to risk premium, where authors use the development of dividend futures contracts to show investor expectations regarding the development of the stock market and the economy. This approach is based on the idea that dividend futures represent risk-adjusted expectations of future dividends that are quoted on daily basis.

2.3 Model Formulation

In this section, I present a general framework for the decomposition of the stock price into individual components S_t^1 , S_t^2 , S_t^3 . The first stage represents the expected dividends calculated from dividend futures. The other two stages represent the final value of the index, which consists of the steady-state values and the convergence to them. The general framework presented below allows for the decomposition of the price with respect to its individual stages. However, as emphasized above, the main advantage of this framework is that it makes it possible to calculate the implied equity risk premium.

I build on the three-stage dividend discount model by Fuller & Hsia (1984), a simple model which allows me to use as much data as possible. The novelty of this study, compared to others, is its use of dividend futures following the example of Gormsen & Koijen (2020). These futures are quoted daily and represent risk-adjusted expectations regarding future dividends. The model can be written as follows:

$$P_t = \sum_{n=1}^{\infty} E_t^p M_{t+n|t} DIV_{t+n|t} = S_t^1 + S_t^2 + S_t^3 \quad (2.1)$$

Equation 5.1 represents the general dividend discounted model, i.e. the price P_t at time t is expressed as the expected infinite discounted sum of future dividends $DIV_{t+n|t}$ modified

by the pricing kernel $M_{t+n|t}$. This, of course, is abstracting from the bubble solution. I wish to emphasize that the pricing kernel does not correspond to the risk-free rate, as it involves the risk perceptions of investors, i.e., it consists of risk-free rates and an equity risk premium.² E_t^p denotes the expectations with respect to the probability measure p at time t , representing the subjective probability of future developments as perceived by the individual investors. I decompose Equation 5.1 into three stages, which I refer to as S_t^i with $i = (1, 2, 3)$.

The model generally assumes the existence of some type of "balanced growth path" to which all variables (in the model) converge. In other words, a representative investor assumes the existence of the steady-state values of nominal interest rates $i_{long|t}$ and the growth rate of the stock index $g_{long|t}$. This condition is necessary as it allows us to use the perpetuity formula in the last stage. The assumption is, of course, an arbitrary simplification, and is assumed here without further theoretical formulation. To properly justify it, it would be necessary, among other things, to define a data-generating process for dividends and interest rates. For example, I later suppose that dividend growth rate follows the growth rate of the whole economy, and it is a commonly used assumption (both implicit and explicit) in most DSGE literature that (productivity) growth is a trend stationary or a difference stationary. For more details, see, for example, Lafourcade & de Wind (2012). Nevertheless, this is still a more transparent and less restrictive approach than the often assumed stationarity of D/P, D/E ratios.

However, I acknowledge that the restrictiveness of this model is the main shortcoming. On the other hand, there is a clear tradeoff. The main advantage of this approach is its clear structure, where the individual parameter specifications have a clear and unique identification. I also always try to substantiate the calibration of individual parameters with which the reader may or may not agree.

Stage 1 of the model represents the discounted sum of dividends up to time $N-1$, where $N-1$ represents the time of the last quoted dividend future.

$$S_t^1 = \sum_{n=1}^{N-1} \frac{FUTURE_{t+n|t}}{(1 + i_{t+n|t})^n} = \sum_{n=1}^{N-1} E_t^p \frac{DIV_{t+n|t}}{(1 + r_{p,t} + i_{t+n|t})^n} \quad (2.2)$$

The calculation itself then proceeds as follows: Stage 1 is calculated directly from dividend futures $FUTURE_{t+n|t}$, where the subscript indicates the price of the dividend future with maturity at time $t + n$ and at time t . I use the no-arbitrage condition between the present value dividend (specifically dividend strip) and the future price (Van Binsbergen *et al.*, 2012). Also note that the present value of dividends can be expressed using a pricing kernel as in the equation 5.1. The no-arbitrage condition can be written as follows:

$$E_t^p M_{t+n|t} DIV_{t+n|t} = E_t^p \frac{DIV_{t+n|t}}{(1 + r_{p,t} + i_{t+n|t})^n} = \frac{FUTURE_{t+n|t}}{(1 + i_{t+n|t})^n} \quad (2.3)$$

I assume a specific form of the pricing kernel that can be expressed using the spot risk-free yield curve and the equity risk premium. The equity risk premium is assumed to be constant at different maturities, i.e., the equity risk premium is fixed across periods within a single valuation, but it varies across valuations computed at a different t . This also implicitly postulates that the liquidity premium is only included in the yield curve. Although this is a significant simplification, it allows for a numerically tractable risk premium calculation. Alternatively, I argue that this computed equity risk premium represents a weighted average of the different equity risk premiums at different maturities, i.e. unconditional equity risk premium.

Stage 2 represents the interpolation between the steady state value and the last quoted dividend future. A 6–10 year convergence period is most commonly used in the literature, but

²Both components are considered to be time-varying.

the length of this convergence period is often chosen arbitrarily (Damodaran, 2020). I decided on a total of 20 years ($L-1$), including the first stage. I selected this duration because I see the possibility of using a larger part of the yield curve as an advantage.³ Stage 2 is calculated as follows:

$$S_t^2 = \sum_{n=N}^{L-1} E_t^p \frac{(1 + g_{n|t}) DIV_{t+N-1|t}}{(1 + rp_t + i_{t+n|t})^n} \quad (2.4)$$

The time-varying equity risk premium is still maintained in the pricing kernel. $DIV_{t+N-1|t}$ represents the last dividend from the previous stage. $g_{n|t}$ represents the growth rate of dividends, which is computed as a linear interpolation between steady-state growth $g_{long|t}$ and the last future-implied dividend growth rate. Interpolation can be written as follows, where $g_{N-1|t}$ is the ratio of the last two implied dividends, i.e., $\frac{DIV_{t+N-1|t}}{DIV_{t+N-2|t}}$.

$$g_{n|t} = g_{N-1|t} + (g_{long|t} - g_{N-1|t}) \left(\frac{n - N + 1}{L - N} \right) \quad (2.5)$$

Stage 3 represents the steady-state stage. I use a perpetuity formula (i.e., the sum of infinite geometric sequences) with constant dividend growth and a constant risk-free interest rate. The intuition behind this formula is the same as that behind textbook Gordon model, where the terminal price is determined by the difference between the growth rate and the discount factor on condition that the growth rate does not exceed the discount factor.

$$S_t^3 = \sum_{n=L}^{\infty} DIV_{t+L-1|t} \frac{(1 + g_{long|t})^{n-L+1}}{(1 + i_{long|t} + rp_t)^{n+1}} = \frac{DIV_{t+L-1|t}(1 + g_{long|t})}{(1 + i_{long|t} + rp_t)^{L-1}(i_{long|t} + rp_t - g_{long|t})} \quad (2.6)$$

$g_{long|t}$ and $i_{long|t}$ denote the steady-state nominal growth rate of the economy and the steady-state interest rate respectively. Regarding the calibration of the long-term interest rate parameter, I use the long end of the yield curve as a proxy for $i_{long|t}$. Furthermore, I continue to maintain the the assumption of time-varying equity risk premium rp_t .

The main idea behind calibration $g_{long|t}$ is that dividends will eventually grow as fast as the whole economy (Damodaran, 2020), but the expected growth of the economy is also unknown. I primarily do not use the historical growth rate because market interest rates are used. This would lead to a methodological dichotomy between a forward-looking and historical approach.⁴ For this reason, long-term growth is approximated by the yield curve. To mitigate short term fluctuations, I use an annual moving average.⁵ This approach is proposed, for example, in Damodaran (2020). An optimal growth model (Ramsey, 1928) can be used to show that interest rates should always be higher than the growth rate of the economy, assuming a dynamically efficient economy. In reality, interest rates are often lower than the growth rate and the average

³Alternatively, I can present my arguments using the average duration of the financial cycle (Drehmann *et al.*, 2012; Claessens *et al.*, 2012). It is mostly accepted that financial cycles have generally a longer duration than business cycles. Given that the topic of this paper is closely related to a stock valuation, it seems appropriate to choose a time horizon for the model long-run stage, which is not affected by the current position in the business or financial cycle and twenty years generally represents an upper bound on the financial cycle. For this reason, no matter what is the current phase of the cycle, the long-run stage is always beyond the cycle horizon. Regarding the length chosen in other articles, 6-10 years duration can be related to mostly the business cycle fluctuation. Also, note that these studies use mostly survey data and are usually not available for more than a 5 years period. This means that in relative terms, the length of the convergence period is similar.

⁴From a purely practical point of view, we require a non-infinite terminal price, i.e., $g_{long|t} > i_{long|t} + rp_t$.

⁵These rates are remarkably similar to the Holston *et al.* (2017) estimate of the natural interest rate with an added inflation target based on the Taylor rule.

spread can be added to compensate for this (see Blanchard & Weil (1992)). Note that this will not significantly change the model-implied dynamics of the equity risk premium; it will mainly change its overall level.

Alternatively, Panigirtzoglou & Scammell (2005) state that the growth rate in a steady state is equal to Return on Equity (ROE) minus the payout ratio (b), i.e., in a steady state, there is no excess return above the required rate of return.

$$g_{long|t} = ROE(1 - b) = (rp_t + i_{long|t})(1 - b) \quad (2.7)$$

The payout ratio b is assumed to be constant and can be determined as the historical ratio of dividends to earnings, which is approximately 50% based on historical data. The problem with this approach is the counter-intuitive development of long-term growth as it reaches its highest values during market crash periods.

Finally, following the example of Barsky & De Long (1993a), I also assume extrapolative expectations regarding long-term future dividend growth. This is achieved assuming that steady-state dividend growth can be approximately expressed as an Exponentially Weighted Moving Average (EWMA) of the difference in log-dividends:

$$g_{long|t} = (1 - \lambda) \left(\sum_{n=0}^{N-1} \lambda^n \Delta \ln E_t^p DIV_{t+N-1-n|t} \right) + \lambda^{N-1} g_0 \quad (2.8)$$

Intuitively, long term growth is the geometric average of past and expected dividend growth when investors use the whole information set. λ denotes the smoothing parameter and g_0 denotes the initial growth trend which can represent, for example, actual historical dividend growth. The equation illustrates that dividends behave like an integrated moving average process of order (1, 1), i.e., the series can be decomposed into permanent shocks and transitory shocks. This also implies that the extrapolative expectations are the same as the rational ones.

This calculation of S_t^i and rp_t consists of a set of simultaneous equations. These are n equations for each $FUTURE_{t+n|t}$, which also include long run stage dividends, i.e. equation 3, 4, 6, and one equation for the price P_t , i.e. equation 1. For example, in the case of 11 future contracts, there are 21 equations and 21 unknowns. The unknown variables are the equity risk premium, implied dividends for the first 10 years directly derived from dividend futures, implied dividends for the next 10 years derived using equations 4 and 5, and long run stage dividends derived using equations 6. However, due to the nonlinearity of the equations, we are forced to resort to numerical optimization. Specifically, I implemented Broyden's method (Broyden, 1965). Alternatively, the model can be used to conduct a sensitivity analysis, for example, to find a long-term value of rp_t . The actual value can then be compared with the model-derived price which can serve as a proxy for a through-the-cycle valuation.

2.4 Data

I performed the calculation using daily data from the S&P 500 Index and the Euro Stoxx 50 Index. In the case of S&P 500, the calculation is performed from mid-December 2015 to February 2021, and in the case of the Euro Stoxx 50 Index the period of the calculation is from January 2013 to February 2021.⁶ The data used are the dividend futures of the given index, the US and EUR IRS yield curves, and the price of the given index. In the case of the broadest US

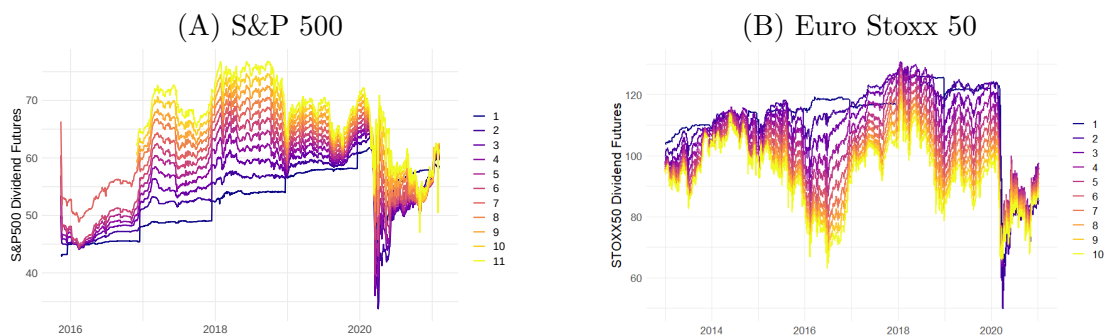
⁶Dividend futures are also listed for individual companies. In the case of large indices, dividend futures for the FTSE and Nikkei are also available.

index, the S&P 500, dividend futures are quoted for up to 11 years.⁷ In the case of the Euro Stoxx 50 Index, they are quoted for up to 10 years. The main disadvantage of the approach presented in this paper is the relative novelty of dividend futures. As a result, the analysis is performed only for a period of slightly over four years, or six years in the case of the latter. Furthermore, due to the novelty of dividend futures, some mispricing may occur, especially at the beginning of trading (Wilkins & Wimschulte, 2010). However, since the analysis always uses current data for a given day, this is not such a big concern, and maintaining the longest period possible seems to be the most transparent solution. Nevertheless, it seemed necessary to draw the reader's attention to this fact.

I use cubic spline interpolation on dividend futures. This is because futures are quoted on a specific date in December, and interpolation is needed to maintain the same maturity, although the future prices themselves are quoted daily. In other words, I am creating a dividend yield curve for easier data manipulation.⁸ I also interpolate the yield curve using the standard NSS model (Svensson, 1994). In both cases, I use IRS data for a period of one month to 20 years.⁹ All data were obtained from the Bloomberg terminal.

As regards dividend futures, Figure 2.1 show their development in the period under review. Sharp transitions are due to the elimination of a matured dividend future and the inclusion of a new one. Note in particular the different dynamics between the European and US indices. The S&P 500 Index is characterized by rising dividend futures, while in the case of the Euro Stoxx 50 Index, dividend futures are mostly in backwardation. The question is to what extent this is due to the influence of institutional investors who may be hedging their position by selling dividend futures (Mixon & Onur, 2016). It is crucial to remember that these are dividend futures, and the expected dividends, according to Equation 5.2, can increase with maturity due to the discount factor. Nevertheless, it can be said that very similar trends have been observed for dividend futures of all maturities in the period up to 2020. I would also like to emphasize the variability in the perception of the growth rate implied by dividend futures.

Figure 2.1: Dividend Futures



Note: Numbers represent different dividend futures contracts by maturity.

Source: Author's calculation, Bloomberg

I use the end of the yield curve for the long-run interest rate for the steady-state parameter values. More specifically, I use a thirty-year year swap rate.¹⁰ I do not use use historical values

⁷This has been the case since December 2017.

⁸Other interpolation methods such as NSS or linear interpolation did not have a significant effect on results.

⁹I also use treasury data in my estimations, but there is no universally agreed benchmark for Europe. Nevertheless, this did not have a significant effect on the results in the US case.

¹⁰I also experimented with a twenty-year swap rate. The results don't differ much in terms of overall dynamics. I opted for a longer tenor, given some anecdotal evidence that liquidity is similar, at least in the case of the US treasury rate (Fleming & Ruela, 2020).

due to the forward-looking nature of prices. The problem is even worse given the possible unit root behavior of interest rates. (Bauer & Rudebusch, 2020). In the case of $g_{long|t}$, which expresses potential growth, I use the same value as that used for the long-term interest rate smoothed by the annual moving average, with an additional 50 basis points representing the average historical spread. Alternatively, I use the endogenous long-term growth rate defined in Equation 5.7 and Equation 5.8, where λ is equal to 0.97 as in Barsky & De Long (1993a), and g_0 is equal to 4%, which reflects historical dividend growth. Regarding the length of the individual stages, I decided to use all the available dividend futures, although a higher equity risk premium may be observed due to lower liquidity, especially for later futures. Regarding the price decomposition into individual stages, S_t^1 represents a component of the price for an 11-year period in the case of the S&P 500 Index and 10 years in the case of Euro Stoxx 50 Index. S_t^2 represents a price component for 12–20 years or, in the case of the latter, 13–20 years, and S_t^3 represents a price component from the 21st year to infinity.

2.5 Empirical Results

Figure 2.2 shows the relationship between the calculated equity risk premium and the development of the S&P 500 Index. The model clearly identified the first half of 2020 as a period of increased equity risk premium owing to the Covid-19 pandemic. Increased uncertainty can also be observed in 2016, which is again in line with the price development of the index and may be related to Brexit and the US election. The decline in the years that followed may have been related to optimism generated by the Trump administration’s tax cuts. The average level of the equity risk premium is approximately 3%, which is similar to Andrlé (2019) and approximately two percentage points lower than in Damodaran (2020), as is shown in Figure 2.7, although it can be observed that the development itself is almost identical. The differences in the level of equity risk premium are due mainly to different steady-state growth assumptions and data sources used for expected dividends, as dividend futures imply lower dividend growth than indicated in surveys.

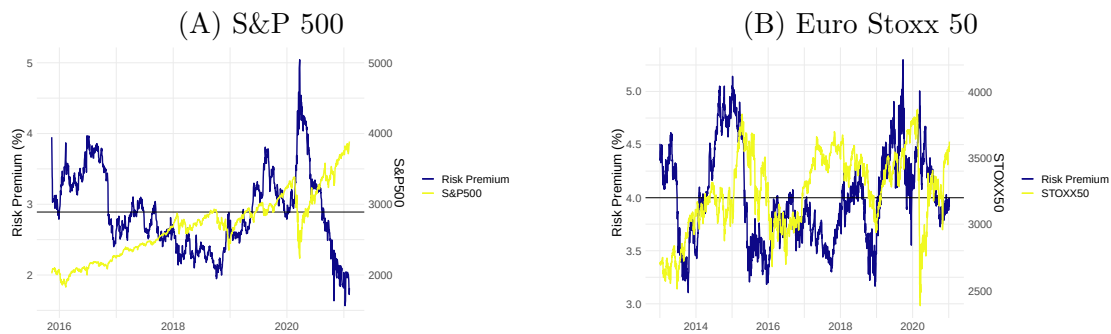
In the case of the Euro Stoxx 50 Index, the equity risk premium is relatively higher and a little less volatile than in the previous case (see Figure 2.2, panel B). This is caused by lower interest rates and the assumption regarding the estimation of long-term growth. Generally speaking, the lower interest rates would indicate less discounting in the future. However, this is offset by a higher equity risk premium. The average level of the equity risk premium is approximately 4%. One can also notice that in both cases the equity risk premium is highly persistent, with the first autocorrelation lag close to unity.

Also, note that during the second half of 2020, the implied S&P500 equity risk premium declines substantially, while the implied Euro Stoxx 50 equity premium slightly increases. This can be partly explained by the course of the Covid 19 pandemic in Europe. The reason may also be the previously indicated endogeneity of the risk premium, which would indicate the targeting of required (excess) nominal returns by investors. For a more detailed discussion, see Kapp & Kristiansen (2021).

As a robustness analysis, Figure 2.8 (panel B) shows a sensitivity analysis regarding an alternative terminal dividend growth rate in the case of the S&P 500 Index (see Appendix A). It can be seen that both alternative specification models produce very similar results to the primary model in terms of the equity risk premium, although the level may differ. However, it should be noted that the two alternative models, both of which use endogenous and extrapolative growth rates, imply excessive and counter-intuitive volatility of the growth rate, especially during the pandemic, as shown in Figure 2.8 (panel A). On the other hand, robustness checks of the Euro

Stoxx 50 Index show that the risk premium is gradually increasing over time, contrary to the baseline specification (Figure 2.9). This is probably due to a decline in interest rates associated with an only slight increase in the index. A possible explanation for this is that an increase in the risk premium offsets the decrease in interest rates. This would suggest that investors are using a nominal return benchmark rather than an excess return benchmark. Nevertheless, the equity risk premium still shows a very similar development to that observed in previous cases.

Figure 2.2: Implied Equity Risk Premium

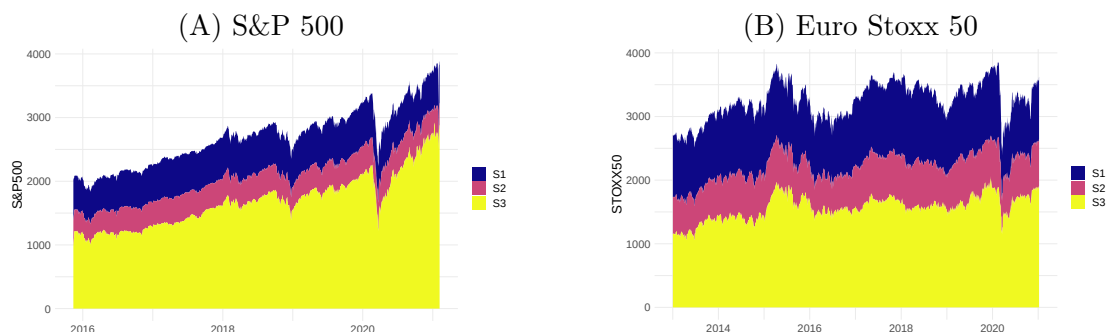


Note: The black line represents the average equity risk premium over the whole sample.

Source: Author's calculation, Bloomberg

Furthermore, I show the decomposition of the indices into individual stages. S_t^1 is given directly by dividend futures. S_t^1 and S_t^2 together form the sum of the present value of expected dividends in the next 20 years and, as we can see in both cases, this represents less than one half of the total price of the index. In the case of the S&P 500 Index, most of its value is formed by the S_t^3 as can be seen in Figure 2.3. I should point out that this stage represents the discounted sum of future dividends twenty and more years into the future and, as such, is associated with the most significant level of uncertainty. It should also be noted that this stage was the main driver in the growth of this index in 2020. This was mainly due to the flattening of the yield curve and a decrease in the risk premium. In the case of the Euro Stoxx 50 Index, a much higher value is implied in the first 20 years, as can be seen in Figure 2.3. This would suggest a much more pessimistic view of the future and a higher risk aversion on the part of investors. Generally speaking, dividend payouts are usually higher in Europe, while many companies in the US do not pay dividends or keep the payout ratio low. Also one can notice that changes during the peak of the Covid-19 pandemic were driven mainly by S_t^1 .

Figure 2.3: Index Decomposition by Stage



Source: Author's calculation, Bloomberg

2.5.1 Counterfactual Analysis

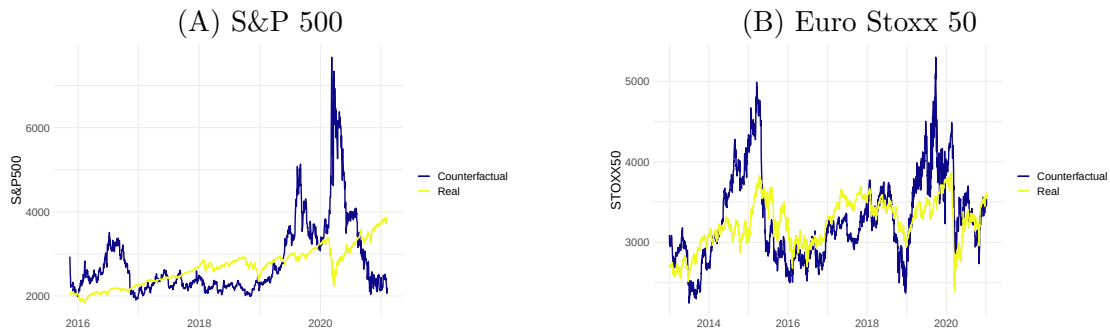
Given the relative simplicity of the model, it is easy to perform counterfactual simulations. These are what-if simulations, which expand our understanding of the impact of underlying factors. In other words, it allows me to quantify the importance of time-varying equity risk premium, expected dividends, and interest rates to the stock market dynamics. Also, note the similarity with previous research dealing with the influence of underlying factors on the stock prices. For example, further shown (principal component) decomposition of index changes is similar to paper by Campbell & Mei (1993), as I essentially decompose the market beta.

The decomposition is first done by changing only a specific component of the model. All else being equal, this shows a hypothetical development. It should be noted that I do not claim these to be predictions, given the undoubted correlation of all the components. For example, a decline in the term structure of interest rates is an endogenous response of the system to the decline in the real economy and the growth of risk aversion.

Figure B1 (panel A) shows the theoretical development of the S&P 500 Index, assuming that the risk premium is constant (in this case the average value over the entire sample). The development of all the other factors is consistent with reality. Notice the extreme price hikes during the Covid-19 pandemic, caused by the sharp decline in the entire yield curve. The following development during 2020 was mainly driven by the decline in dividend futures in combination with a slight increase in the yield curve. This is a clear indicator of the prevailing optimistic market sentiment because the risk premium was basically the only source of index growth in the given period. Considered in more detail, the risk premium is the only part of the model which is not fundamentally linked to the real economy but expresses the required investor mark-up for risk and can be irrational. This notwithstanding, the assurance of policymakers that they will do their utmost to maintain the functioning of the market cannot be considered purely irrational.

Figure B1 (panel B) shows a simulation with a constant risk premium for the Euro Stoxx 50 Index. Figure B1 also illustrates that there was a significant increase in the index in 2016 due to the decline in the overall interest rate structure. In the context of the Covid-19 pandemic, the theoretical development is similar to the reality of the situation. This is probably due to the monetary authority's limited ability to react, as interest rates were already unprecedentedly low even before the Covid-19 pandemic. Another factor is a more stable development of the risk premium which prevented a more significant drop in prices at the beginning of the pandemic but also prevented higher growth.

Figure 2.4: Counterfactual Simulation I

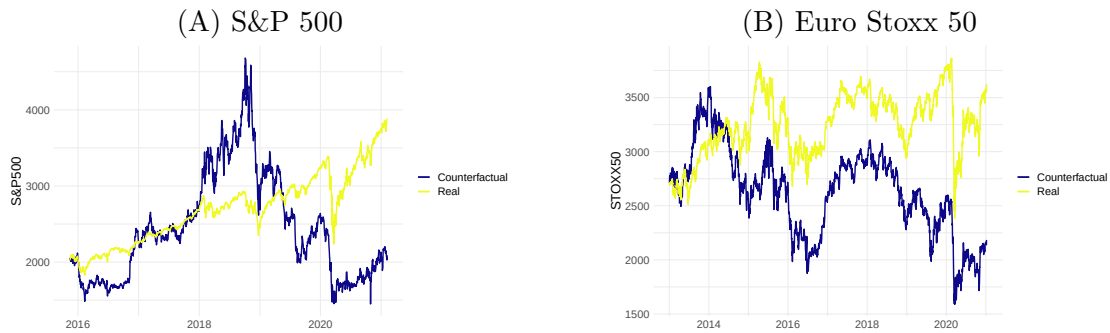


Note: The equity risk premium is fixed at the average value over the sample period.

Source: Author's calculation, Bloomberg

Next, I show the theoretical development of the S&P 500 and Euro Stoxx 50 indices while maintaining a fixed yield curve and keeping the other components time-varying, as in reality. Figure 2.5 support the general view that falling interest rates are one of the main contributors to the growth of the above indices. In the case of the S&P 500 Index, lowering interest rates has significantly boosted index growth since the year 2019, and especially at the beginning of the Covid-19 pandemic. In the case of the Euro Stoxx 50 Index, this is true for the whole period under review; otherwise, the index would be declining without falling interest rates.

Figure 2.5: Counterfactual Simulation II



Note: The yield curve is fixed at the beginning of the sample period.

Source: Author's calculation, Bloomberg

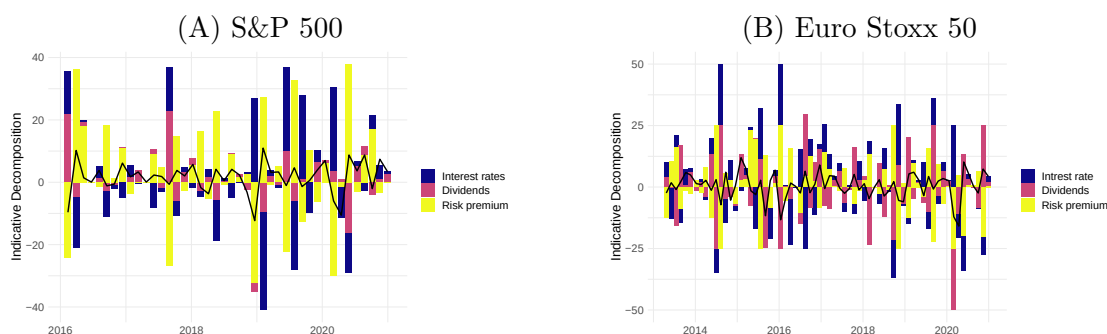
Finally, I also focus on changes in the level of the given indices. Due to non-linearity, I cannot calculate individual percentage contributions. For this reason, I performed indicative decomposition using the results of the individual counterfactual analyses. This decomposition is calculated as follows: two variables are kept fixed at their values from the previous period (month) and one is maintained at real value in the given period. I recognize three primary drivers (variables) behind price movements. These variables are: change in the yield curve, change in expected dividends (dividend futures), and change in the risk premium. These three changes do not correspond to a real change in the index due to non-linearity, and for that reason, I decided to re-scale them to match real change. Alternatively, I calculated decomposition using the regression of monthly index changes on the principal components of dividend futures, the yield curve, and the risk premium (Appendix A). I use Principal Component Analysis decomposition to reduce the number of factors and allow for a more straightforward interpretation of the results; generally, the results are similar to those obtained using indicative decomposition (Figure 2.10).

Figures 2.6 show the decomposition of the monthly changes into individual components.

The black line represents the real monthly change in the given index and the bars represent the counterfactual analysis results. Figure 2.6 (panel A) indicates that the most positive returns were as a result of low-risk premiums and that change in expected dividends had only a limited effect on positive returns. Figure 2.6 also confirms my previous intuition regarding the Covid-19 pandemic when the initial shock in 2020 Q1 was caused mainly by an increase in the equity risk premium and mitigated by a decrease in interest rates (similar to what occurred in the 2016 stock market selloff). However, there was a revision in dividend futures in the period that followed, i.e., expected dividends have been falling since March 2020, but the decline in the risk premium outweighs this decrease.

Euro Stoxx 50 Figure 2.6 (panel B) illustrates the limited effect of the equity risk premium in the pandemic; this is probably due to the increased equity risk premium level even before the pandemic. However, dividend expectations have changed significantly during this period. Also, the yield curve's counter-cyclical movement during this period, although smaller than in the previous case, is also illustrated. Moreover, we can also observe higher dividend volatility than in the previous case, which seems to be an important driver of price changes.

Figure 2.6: Indicative Decomposition



Note: The black line represents the nominal monthly percentage change of the given index.

Source: Author's calculation, Bloomberg

2.6 Conclusion

In this paper, I present a simple way of extracting the implied equity risk premium using the dividend discount model and the equity index dividend futures. The main contribution of this paper is the use of dividend futures which represent the risk-adjusted expectations of future dividends. This approach allows for the computation of the equity risk premium on a daily basis and, as such, can serve as another indicator for measuring market sentiment which provides an easy-to-understand interpretation. This is due to the fact that risk premium is the only part of the model which is not fundamentally linked to the real economy but expresses the required investor mark-up for risk. However, it is still important to realize that the equity risk premium calculated in this way is residual and, much like “Solow residual,” is a measure of our ignorance regarding the stock market.

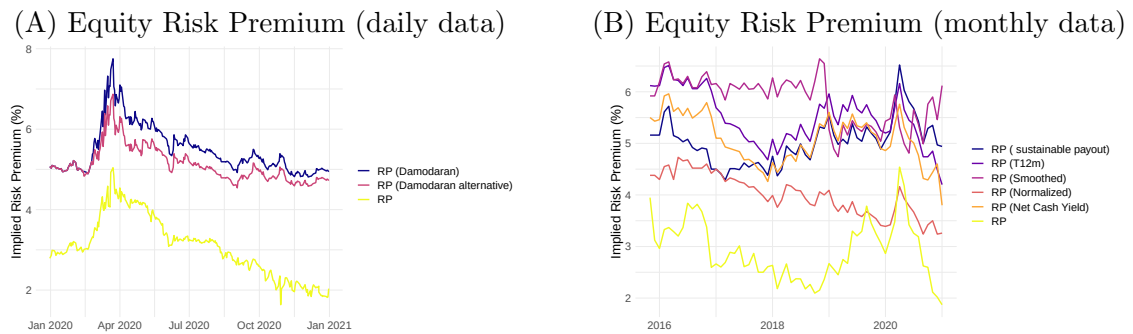
The study empirically shows the development of the equity risk premium in the S&P 500 and Euro Stoxx 50 markets in the last decade using daily data. I can use this model to show the period of the Covid-19 pandemic and Brexit, and the immediate reaction of the investor risk premium to this period, as implied by the model. I illustrate the high variability of the equity risk premium – one of the main drivers of price development – during the period under review. I also show the importance of a time-varying structure of interest rates on price

development. Furthermore, I present different sensitivity scenarios of alternative stock market developments using counterfactual analyses. These scenarios illustrate the impact of individual model components on the final price, including the importance of monetary policy decisions.

Overall, the study allows for a better structural interpretation of price movements which can be used, for example, to calibrate stress tests or other sensitivity analyses in relation to stock prices. In the context of financial stability, central banks and other policymakers should not be concerned with forecasting market developments per se but should be aware of the sources of price movements and the potential risks arising from them. Specifically, the presented model can be used for modeling stock market indices under various adverse scenarios, as it offers a possibility to base future projections on easy-to-understand economic narratives.

2.7 Appendix - Robustness check

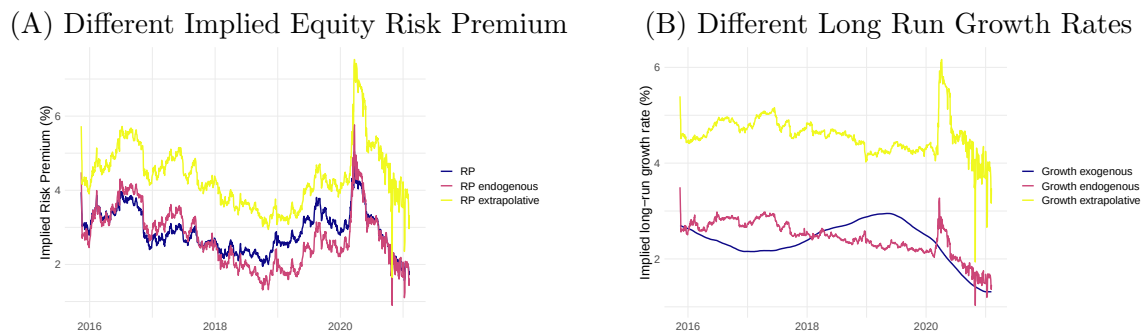
Figure 2.7: S&P500 Implied Equity Risk Premium Comparison



Note: The yellow line represents Implied Equity Risk Premium calculated in this paper.

Source: Author's calculation, Bloomberg, Aswath Damodaran web page (<http://pages.stern.nyu.edu/~adamodar/>)

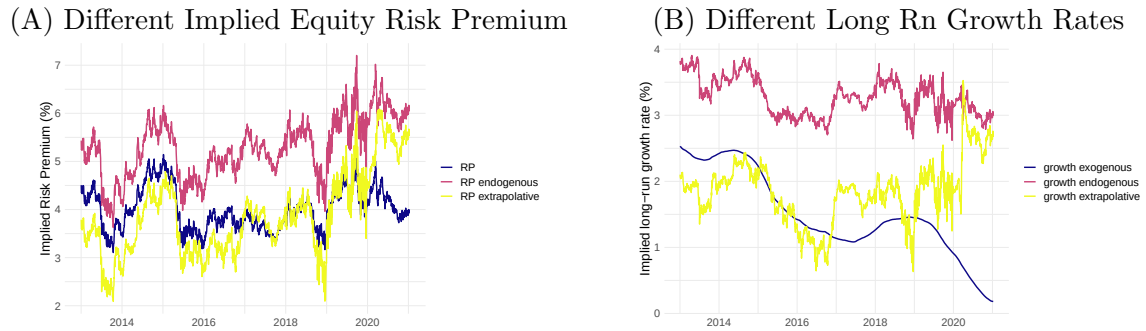
Figure 2.8: S&P500 Robustness checks



Note: Endogenous and extrapolative risk premium, long run growth rate corresponds to equations 7 and 8.

Source: Author's calculation, Bloomberg

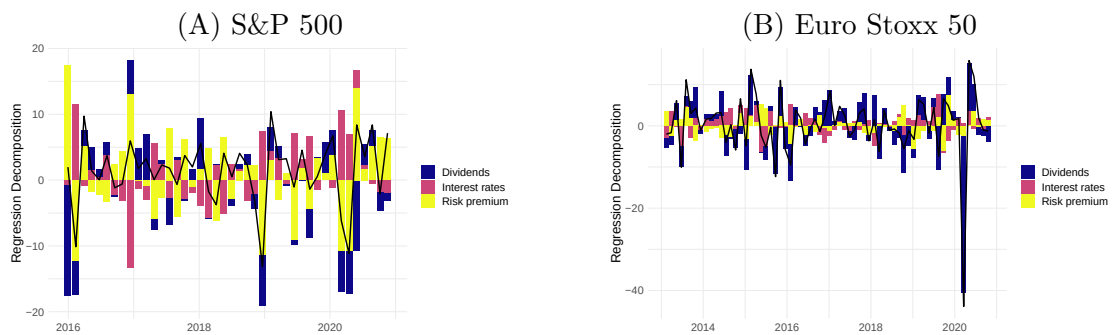
Figure 2.9: Euro Stoxx 50 Robustness checks



Note: Endogenous and extrapolative risk premium, long run growth rate corresponds to equations 7 and 8.

Source: Author's calculation, Bloomberg

Figure 2.10: Regression Decomposition



Note: Black line represent nominal monthly percentage change of given index. Decomposition was calculated using regression of monthly index percentage changes on principal components of dividend futures, yield curve and the implied equity risk premium.

Source: Author's calculation, Bloomberg

Chapter 3

Inflation, Interest Rates and the Predictability of Stock Returns

Abstract: This paper focuses on the predictive role of inflation and interest rates in forecasting stock returns. We report economically significant in-sample evidence that stock returns are predictable using cyclical components of inflation and interest rates. The out-of-sample analysis results show that these cyclical components have reasonable out-of-sample predictability for the equity risk premium when the comparison is made against a historical average benchmark. We believe this is due to the fact that both inflation and interest rates contain stochastic trends. Accounting for changes in trend inflation (and equilibrium real interest rate) is essential for forecasting stock returns.

3.1 Introduction

Forecasting stock returns has long been recognized as a highly challenging task, and achieving consistent out-of-sample out-performance against average historical return benchmarks remains highly difficult. This well-known fact has not changed much since the seminal works of Goyal and Welch (Goyal & Welch, 2003; Welch & Goyal, 2008; Goyal *et al.*, 2021). Conversely, numerous studies assert the existence of predictive power in certain variables. These studies report economically significant in-sample evidence that stock returns are predictable. However, the out-of-sample performance is often mixed. But even in this case, the results differ significantly depending on the methodology and variables used, as is shown for example in Rapach *et al.* (2010), Dangl & Halling (2012), Pettenuzzo *et al.* (2014), Yin (2019), Pan *et al.* (2020), Tsiakas *et al.* (2020) and Wang *et al.* (2021). Overall, the predictability of stock returns has been the subject of much research and debate.

A wide range of economic variables has been proposed in the literature as potential predictors¹ of stock returns. In this article, our primary focus is on nominal interest rate variables and the inflation rate. Previous research by Campbell (1987), Fama & French (1989), Keim & Stambaugh (1986), Ang & Bekaert (2007), among others, have shown that stock returns can be predicted using nominal interest rates. Similarly, Nelson (1976), Fama & Schwert (1977), Campbell & Vuolteenaho (2004), and others have identified predictability based on the inflation rate. However, the forecasting power of these variables has deteriorated significantly since the 1980s (Goyal *et al.*, 2021).

¹Note that this list is not comprehensive, and many other predictors were suggested. For more information, see Rapach & Zhou (2013), where authors provide an extensive survey of the literature.

This article aims to explain the diminishing forecasting power of interest rates and inflation. In line with Denga *et al.* (2022), we emphasize the significance of expectations in asset pricing, as investors price assets based on their beliefs. In more detail, the agents distinguish between long-run trends and cyclical (transitory) fluctuations. According to asset pricing theory (Rapach & Zhou, 2013), the predictability of stock returns can arise from the influence of time-varying aggregate risk. Therefore, if forecasting models consistently incorporate this time-varying aggregate risk premium, their effectiveness will likely endure over extended periods. This reinforces the idea that incorporating the influence of such risk factors is crucial for maintaining forecasting model effectiveness in predicting stock returns.

Our study builds upon this premise, asserting that cyclical deviations from the long-term trends of standard macro-predictors significantly influence investors' risk perception. This is because the trends regarding inflation and interest rates consist of (slowly changing) steady-state expectations of investors, which are reflected in the valuation of all instruments on the financial market. On the other hand, short-term deviations from these trends primarily affect the risk aversion of individual agents, for example, because the cyclical fluctuations represent expectation forecast errors², or due to the existence adjustment cost, etc. For a theoretical explanation of cyclical variation in risk aversion and prices of risky assets, see, for example, Campbell & Cochrane (1999) or Campbell *et al.* (2017). Alternatively, we can simply state that cyclical components of interest rates and inflation serve as proxies for changing risk aversion.³

More specifically, inflation has, in theory, no effect if it affects nominal dividend growth rates and the nominal discount factor in the same way⁴. However, as our article demonstrates, the cyclical component of inflation negatively affects excess returns, indicating that these cyclical fluctuations influence risk aversion and enable the prediction of excess returns. On the other hand, it is widely acknowledged the stock can be considered an inflation hedge in the long run (Boudoukh & Richardson, 1993). This is possible only if the nominal dividend growth rate slowly adjusts to the inflation trend, which is reflected in changing nominal stock returns and stable real returns. In other words, we state that inflation can be described as an I(1) series and its long-term trend also affects other variables, i.e., interest rates load upon the inflation trend. Accounting for changes in trend inflation is, therefore, essential for understanding stock returns.

A potential criticism of our approach is that the FED following inflation target suggests that inflation is stationary. However, this was not always the case; and the inflation targeting itself depends on the FED's credibility. Our reasoning also aligns with the changing stock-bond correlation discussion (Andersson *et al.*, 2008; Pericoli, 2020). The negative correlation between stock and bond returns seen in the last two decades could be caused by the stabilization of long-term inflation expectations. Therefore, results concerning last two decades are mainly driven by cyclical fluctuations. Conversely, the positive correlation witnessed in the latter half of the twentieth century combined cyclical movements with simultaneous shifts in long-term

²For instance, because agents update their information set sluggishly or due to how the expectations are formed, i.e., extrapolative/adaptive expectations, habit formation, behavioral inattention, etc.

³From an empirical point of view, the models may suffer from endogeneity issues as changing risk aversion may affect both dependent and explanatory variables. Also, given the fact that these cyclical components may serve as proxies for some other unobserved factor, the result may not indicate true causal relation. For example, Baker *et al.* (2019) and Chiang (2023) argue that inflation tends to co-move with stock market volatility, which has a negative effect on stock returns.

⁴In contrast, inflation has a negative impact on bonds, whose payoffs are set in nominal terms. An ambiguous impact on stocks is caused by the fact that their payoffs vary in relation to changes in inflation. However, inflation also influences the nominal pricing kernel, and the impact of inflation, therefore, depends on the relative sizes of these effects. Our preferred explanation of the results obtained below is that by focusing on cyclical components, we obtain variables that mainly influence the pricing kernel and cause changes in time-varying risk aversion.

expectation, making its interpretation highly complex. For example, Andersson *et al.* (2008) argues that stock and bond prices move in the same direction during periods of high inflation expectations, while epochs of negative stock-bond return correlation seem to coincide with subdued inflation expectations.

Differing from Denga *et al.* (2022), our study does not rely on survey data for extracting expectations but instead employs an exponentially weighted moving average (EWMA) detrending method to decompose investor expectations (regarding inflation and interest rates) into long-term⁵ and cyclical components. This can be justified by assuming, for example, extrapolative expectations of inflation and interest rates.⁶

In this regard, our work draws inspiration from the shifting endpoint literature, exemplified by studies conducted by Kozicki & Tinsley (2001), Cieslak & Povala (2015), Bauer & Rudebusch (2020). These studies illustrate that empirical properties, interpretability, and predictions significantly improve when shifting endpoints are included, i.e., detrending. For instance, Kozicki & Tinsley (2001) demonstrate that a substantial portion of inflation rate variation can be explained by the inflation trend. Similar findings are observed for bond yields in the study by Cieslak & Povala (2015). Bauer & Rudebusch (2020) argue for including the change in real interest rate trend to fully capture the trend component in interest rates, as accounting solely for the inflation trend leaves a highly persistent component of interest rates unexplained. Our study also shares similarities with the works of Cooper & Priestley (2009), Yu *et al.* (2023) and Atanasov *et al.* (2020), where authors also concentrate on the cyclical components of commonly used explanatory variables and highlight their enhanced predictive power compared to non-detrended variables.

The remainder of this article focuses on an in-sample and out-of-sample forecasting exercise for excess returns. We employ a standard bivariate regression approach, following the methodology outlined by Welch & Goyal (2008).

3.2 Data

We use the well-known dataset of Welch & Goyal (2008). More specifically, we work with updated (monthly) data for the variables used in their study, which are available from Amit Goyal’s webpage. This is to establish a connection between our discoveries and the extensive literature studying the predictability of market returns. In other words, this allows us to conduct a comparative analysis between the forecasting capability of our predictors and other predictor variables outlined by Welch & Goyal (2008). More specifically, we incorporate the following predictors: the interest rate on a three-month Treasury bill (secondary market) (TBY) and long-term government bond yield (LTY). To measure the inflation rate, we use the year-on-year inflation rate calculated from the Consumer Price Index (π) for all urban consumers obtained from the U.S. Bureau of Labor Statistics. This is because year-over-year change is the preferred inflation measure closely watched by markets and central bankers. For comparison, we also use the inflation rate (π^{GW}) from Goyal’s dataset. Consistent with the existing literature, we focus on predicting the log excess return on an S&P 500 Index. We compute excess returns by subtracting the return on the Treasury bill (a proxy for risk-free rate) from the market return.

As explained above, we assume that interest rates and inflation variables can be described as $I(1)$ series.⁷ In more detail, we primarily assume that both inflation and interest rates load

⁵This approach allows for steady states to be time-varying.

⁶Barsky & De Long (1993b) and Evans & Honkapohja (2009) show that, in many cases, this approach coincides with rational expectations and represents the optimal prediction rule.

⁷This is generally confirmed by ADF and KPSS test results.

upon the inflation trend. We follow Cieslak & Povala (2015) approach to obtain the inflation trend. Specifically, we construct the inflation trend as exponentially-weighted moving average of past values of year-over-year inflation:

$$\pi_t^{trend} = (1 - \lambda) \sum_{i=0}^{t-1} \lambda^i \pi_{t-i} \quad (3.1)$$

where π_t denotes year-over-year inflation, π_t^{trend} is inflation trend. We set λ to 0.987 as in Cieslak & Povala (2015). The detrended variables are obtained by subtracting the inflation trend. This can be written as follows: $TBY_t^{detr} = TBY_t - \pi_{t-1}^{trend}$, $LT Y_t^{detr} = LT Y_t - \pi_{t-1}^{trend}$, $\pi_t^{detr} = \pi_t - \pi_t^{trend}$.⁸

Furthermore, following Bauer & Rudebusch (2020), we also account for the shift in real natural/equilibrium interest rate. The change in real interest rate trend is again computed using EWMA:

$$r_t^{trend} = (1 - \lambda) \sum_{i=0}^{t-1} \lambda^i r_{t-i} \quad (3.2)$$

where $r_t = TBY_t - \pi_t$ and we set λ to 0.999. This alternative detrending is obtained as follows: $TBY_t^{detr2} = TBY_t - r_{t-1}^{trend} - \pi_{t-1}^{trend}$, $LT Y_t^{detr2} = LT Y_t - r_{t-1}^{trend} - \pi_{t-1}^{trend}$.

The analysis is performed on the monthly⁹ data ranging from 1951 (M1) to 2022 (M12). This is due to the fact that interest rate data are hard to interpret before the 1951 Treasury Accord, as the FED effectively pegged interest rates during the 1930s and the 1940s. We, therefore, focus on the post-Accord period.

3.3 Methodology

We consider following standard univariate predictive regression model to analyze aggregate excess stock return predictability:

$$r_{t+h}^e = \alpha + \beta x_t + \epsilon_{t+h} \quad (3.3)$$

where $r_{t+h}^e = r_{t+h} - r_{f,t+h}$ denote the equity risk premium at time $t + h$, r_{t+h} is the total log return (including dividends) at time $t + h$, $r_{f,t+h}$ is the log return of risk-free rate, i.e., we measure r_{t+h}^e as the h-period continuously compounded log return less the corresponding h-period continuously compounded log risk-free rate. x_t is a predictor at time t , and ϵ_{t+h} denote error term. The coefficients α and β are estimated with OLS.

Due to well-known biases in standard OLS regressions, given the use of persistent explanatory variables, we use Newey-West standard errors. Furthermore, we calculate wild bootstrap p-values with the null hypothesis of unpredictable returns. This is because Newey-West standard errors deteriorate with longer horizon forecasts, especially when overlapping observations are used. The procedure account for persistence in explanatory variables and correlations between excess return and explanatory variables. The bootstrap procedure we apply follows that of Maio (2013).

As is standard in stock return literature, we run an out-of-sample forecasting exercise using a recursive window estimation, i.e. for every period, we add the actual realization of the excess return and other explanatory variables (interest rate, inflation) as additional data points at

⁸We account for the delay in CPI data releases when testing the predictive ability of inflation and lagged this variable by an additional month.

⁹For forecasting excess returns above a one-month period, we use overlapping observations.

the end of the sample, and re-estimate the model. An h period-ahead forecast is generated at each step. We use the sample of the first 72 periods (6 years) as a training period for the estimation.¹⁰ Following Welch & Goyal (2008), the historical average of the excess returns is used as a benchmark forecast model. The average is also estimated using a recursive window (since 1927).

Following Campbell & Thompson (2008); we also introduce an economically-motivated restriction on individual predictive regression models, which restricts the sign of the predictor coefficients and forecasted excess returns. According to this approach, the predictor coefficients must have the theoretically expected sign, and the equity risk premium must be positive. Placing the above-mentioned restrictions may somewhat improve out-of-sample performance.

To compare the out-of-sample forecasts, we use out-of-sample R^2 statistic (R_{OS}^2). The R_{OS}^2 statistic has a similar properties to the standard in-sample R^2 statistic and can be written in following way:

$$R_{OS}^2 = 1 - \frac{MSFE_{model}}{MSFE_{benchmark}} \quad (3.4)$$

where $MSFE_{model}$ and $MSFE_{benchmark}$ denotes the mean-square-forecast error of the predictive regression and the benchmark historical average model. When $R_{OS}^2 > 0$, the predictive regression model have a lower $MSFE$ than the benchmark.

The test statistics¹¹ (CW) by Clark & West (2007) was also employed for a more formal approach. The null hypothesis is that the root-mean-square-forecast error (RMSFE) of the predictive regression model and the historical benchmark are equal. The alternative hypothesis is that the predictive regression outperforms the historical benchmark.

3.4 Results

Table 5.1 display the in-sample predictive regression results. The table is divided into two panels for one-month (Panel A) and three-month (Panel B) excess returns¹². The results regarding different predictors are shown in separate columns. In every column, we show the coefficient of the point estimate, the corresponding Newey-West standard error (in parentheses), bootstrapped p-value of the regression coefficient (square brackets), and the adjusted and un-adjusted R^2 .

We find that the estimated coefficients regarding the transitory components of interest rates and inflation are negative and have an economically sizable impact on future excess returns. The coefficients' estimates are strongly statistically significant in both one-month and three-month horizons, and the associated R^2 is significantly higher than that of the non-detrended predictors.¹³ The results are also confirmed by the bootstrapped p-values of the regression coefficients. Table 5.1 (Panel B) also shows that predictability is increasing with the horizon both in terms of the size of the estimated coefficients and R^2 statistics. Thus, expected returns are high when cyclical components of inflation and interest rates are low. Therefore, the periods of cyclical low inflation and cyclical low-interest rates (relative to the perceived long-term trends) are also periods of high-risk aversion and slower/negative (more uncertainty about) economic growth, i.e. high excess returns are generally associated with a negative output gap, lower

¹⁰The length of the training period balances having enough observations for a reasonably precise estimation of the initial parameters with a relatively long out-of-sample period for forecast evaluation. Most of the studies set an initial period in the range of 5 to 15 years.

¹¹We used the HAC estimator to calculate the variance.

¹²Due to the well-known problems with long-horizon forecasts, we do not include them. See Boudoukh *et al.* (2008)

¹³Accounting also for the change in real interest rate trend does not bring significant benefits relative to including only inflation trend.

inflation and low nominal interest rates. In a similar fashion, Andrieu *et al.* (2017) show that regarding business cycle fluctuations, there is mostly positive co-movement of output, interest rate and inflation. This can be explained by the fact that the main driver of business cycle fluctuations are demand shocks. Pericoli (2020) similarly argues that the dynamics of inflation has gone from counter-cyclical to pro-cyclical and assesses that inflation is likely to be pro-cyclical when it is propelled by demand rather than supply shocks.

Table 3.1: In-Sample Results

Panel A: One-Month									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
π^{detr}	-0.198** (0.077) [0.0087]								
π		-0.137** (0.061) [0.0283]							
π^{GW}			-0.920** (0.462) [0.0322]						
TBY^{detr}				-0.232*** (0.070) [0.0155]					
TBY^{detr2}					-0.243*** (0.069) [0.0123]				
TBY						-0.116** (0.049) [0.0702]			
LTY^{detr}							-0.260*** (0.094) [0.0286]		
LTY^{detr2}								-0.245*** (0.086) [0.0529]	
LTY									-0.090* (0.054) [0.1636]
<i>Constant</i>	0.005*** (0.001)	0.010*** (0.002)	0.008*** (0.002)	0.006*** (0.001)	-0.001 (0.002)	0.010*** (0.002)	0.011*** (0.002)	0.003** (0.002)	0.011*** (0.003)
Observations	863	863	863	863	863	863	863	863	863
R ²	0.012	0.008	0.006	0.013	0.014	0.007	0.010	0.009	0.004
Adjusted R ²	0.010	0.007	0.005	0.012	0.013	0.006	0.009	0.008	0.002
Panel B: Three-Month									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
π^{detr}	-0.551** (0.214) [0.0002]								
π		-0.378** (0.166) [0.0023]							
π^{GW}			-2.391* (1.316) [0.0101]						
TBY^{detr}				-0.611*** (0.179) [0.0006]					
TBY^{detr2}					-0.640*** (0.175) [0.001]				
TBY						-0.304** (0.126) [0.0094]			
LTY^{detr}							-0.694*** (0.228) [0.0017]		
LTY^{detr2}								-0.651*** (0.212) [0.0049]	
LTY									-0.239* (0.136) [0.0434]
<i>Constant</i>	0.016*** (0.004)	0.030*** (0.006)	0.023*** (0.006)	0.019*** (0.004)	0.001 (0.006)	0.029*** (0.006)	0.031*** (0.006)	0.011** (0.005)	0.030*** (0.008)
Observations	861	861	861	861	861	861	861	861	861
R ²	0.029	0.021	0.014	0.031	0.032	0.016	0.023	0.021	0.008
Adjusted R ²	0.028	0.020	0.012	0.030	0.031	0.015	0.022	0.020	0.007

Note: The table presents one-month ahead (Panel A) and three-month ahead (Panel B) predictive regressions for the equity risk premia. The Newey-West standard errors (in parentheses) with h lags were used. π is inflation rate computed from Consumer Price Index, TBY denotes the three-month treasury bill, and LTY is the long-term government bond yield. The analysis is performed on the data ranging from 1951 (M1) to 2022 (M12). *p<0.1; **p<0.05; ***p<0.01 (Newey-West standard errors). Bootstrapped p-values of the regression coefficients are displayed in square brackets.

Table 3.2 shows the results of the out-of-sample exercise. The table is again divided into two panels for one-month and three-month forecast horizons. In every panel, the first two rows display out-of-sample R_{OS}^2 and the p-value of Clark West (CW) test statistics. The third and fourth rows show the same statistics but take into account Campbell & Thompson (2008) restrictions. (We set the regression coefficient to zero whenever it has the “wrong” sign and rule out a negative risk premium.) This out-of-sample forecasting exercise shows that the detrended interest rates and inflation have reasonable out-of-sample predictability for the risk premium compared to the historical average benchmark and the non-detrended predictors. This is true even for the one-month forecast horizon. Campbell & Thompson (2008) argue that even small positive R_{OS}^2 , such as 0.5% regarding monthly data, signal an economically meaningful degree of return predictability. In our case, the detrended treasury bill yield satisfies this condition even without taking into account Campbell & Thompson (2008) restrictions.

Table 3.2: Out-of-Sample Results

Panel A: One-Month									
	π^{detr}	π	π^{GW}	TBY^{detr}	TBY^{detr2}	TBY	LTY^{detr}	LTY^{detr2}	LTY
R_{OS}^2	0.004	0.000	-0.002	0.005	0.006	-0.010	0.002	0.002	-0.012
CW	0.103	0.199	0.106	0.007	0.012	0.017	0.074	0.120	0.167
$R_{OS,Campbell-Thompson}^2$	0.007	0.006	-0.001	0.004	0.005	0.004	0.003	0.003	0.001
CW _{Campbell-Thompson}	0.012	0.020	0.110	0.014	0.018	0.020	0.075	0.109	0.107
Panel B: Three-Month									
	π^{detr}	π	π^{GW}	TBY^{detr}	TBY^{detr2}	TBY	LTY^{detr}	LTY^{detr2}	LTY
R_{OS}^2	0.020	0.011	0.002	0.018	0.020	-0.011	0.014	0.013	-0.010
CW	0.023	0.047	0.031	0.001	0.001	0.001	0.007	0.007	0.011
$R_{OS,Campbell-Thompson}^2$	0.025	0.020	0.003	0.012	0.015	0.015	0.016	0.016	0.012
CW _{Campbell-Thompson}	0.020	0.035	0.067	0.011	0.009	0.013	0.019	0.015	0.020

Note: The table shows the results of the out-of-sample forecasting exercise regarding one-month ahead (Panel A) and three-month ahead (Panel B) equity risk premia. π is inflation rate computed from Consumer Price Index, TBY denotes the three-month treasury bill, and LTY is the long-term government bond yield. CW denote the p-value of Clark West test statistics. Subscript Campbell-Thompson denote that restrictions as in Campbell & Thompson (2008) were used. The analysis is performed on the data ranging from 1957 (M1) to 2022 (M12).

3.5 Conclusion

Welch & Goyal (2008) showed that standard macro-finance predictors mostly have poor out-of-sample properties. This article attempts to show that this does not always have to be the case. More specifically, this paper's main contribution to the existing literature is an in-sample and out-of-sample forecasting exercise using a standard bivariate regression approach following Welch & Goyal (2008), where we illustrate that there exists substantial in-sample and out-of-sample predictability for the equity risk premium using cyclical components of inflation and interest rates when the comparison is made against a historical average benchmark. We also observe a significant forecast improvement in contrast to the non-detrended variables. The results can be rationalized by the existence of time-varying risk premia and the fact that inflation and interest rates contain stochastic trends.

Chapter 4

Supply Shocks, Demand Shocks and Yield Curve Dynamics

Abstract: This paper presents a framework for extracting the supply and demand shocks using the revisions of expectations obtained from a Survey of Professional Forecasters. The main contribution of this paper is the identification of these shocks and the subsequent assessment of their impact on yield curve dynamics. This allows for the asymmetry in the bond yield movements, where we show that yields react differently to supply and demand shocks. The analysis is performed on the US data ranging from 1969 (q4) to 2022 (q1).

4.1 Introduction

The practical thinking of many analysts is still dominated by the IS-LM and AD-AS framework and the associated thinking about the business cycle in terms of supply and demand shocks. This reasoning is somewhat diminished in the current academic literature. Nevertheless, also most of the New Keynesian model, which is a workhorse of modern macroeconomics, can be interpreted similarly as these models can be expressed in the most straightforward concept using only three equations, e.g., canonical three-equation New Keynesian model. The main difference is the treatment of expectation, which substantially impacts the model solution, but the supply-demand interpretation remains standard.

This paper returns to basics and examines supply and demand shocks and their impact on financial variables, specifically bond yields and Fed funds rate. In terms of empirical work, our study follows many other papers examining the impact of the macroeconomic variables on yield curve dynamics (Joslin *et al.* (2014), Bauer & Rudebusch (2017)). The fundamental difference compared to many other studies is that we try to identify supply and demand shocks directly and, therefore, distinguish their impact on these financial variables. In this regard, most similar to this paper is a study by Bekaert *et al.* (2021), which identifies demand and supply shocks based on higher-order moments in the data. For a more theoretical treatment of this topic, see Michailat & Saez (2022). Hence, our research basically offers a different view of a time-varying term premium and natural rate of interest, where we seek a more structural perspective on the given subject, as the observed variability in the bond yields documented by many other studies may be due to different shocks affecting the economy rather than a structural change of the term premium or equilibrium interest rate. For this reason, our paper have close connection to the studies that identified the monetary policy rule (Taylor (1993), Clarida *et al.* (2000), Orphanides & Williams (2004), Coibion & Gorodnichenko (2011), Carvalho *et al.* (2021), Bauer *et al.* (2022)) or evolution of the term premium (Adrian *et al.* (2013), Bauer & Rudebusch

(2020)). Accordingly, our paper also connects to studies that deal with the importance of asymmetries in the economy (Conrad *et al.* (2013), Baumeister & Hamilton (2018), Bauer & Chernov (2021)).

In terms of methodology, we freely follow studies dealing with estimating monetary policy surprise. These studies try to isolate this shock based on high-frequency data and assess its impact on other variables. For example: Kuttner (2001), Gürkaynak *et al.* (2005), Bernanke & Kuttner (2005), Hanson & Stein (2015), Bauer & Swanson (2022) estimate the effects of monetary policy on financial variables. Our study is (in terms of methodology) most similar to the studies of Jarociński & Karadi (2020) and Jarocinski (2020), which distinguish monetary shocks according to the high-frequency response of the stock market and Fed fund futures. As a result, the authors can distinguish between two different types of shocks, which have the same impact on Fed funds futures, but a different impact on the stock market. Our approach is very similar, although we do not use high-frequency data. Instead, we use the revisions of expectations regarding the output and the price level of individual analysts, assuming that different types of shocks affecting a real economy cause different revisions in expectations. This is a similar approach to the one used in studies by Andrade *et al.* (2016), Carvalho *et al.* (2021).

Specifically, we use a Survey of Professional Forecasters to identify supply and demand shocks, which we identify using the revision of expectation in the price level and GDP/GNP. Our identification is generally straightforward: If there is an upward revision of expectations about GDP/GNP and inflation, we assume that a (positive) demand shock hits the economy in the given period. On the other hand, if there is a revision of the GDP/GNP and inflation, which is in the opposite directions, then we assume that supply shocks prevail in the economy. Given that we rely only on the differences of the expected values, these are rather to be considered perceived shocks as they do not necessarily have to coincide with the real economy. Also, as some recent research shows (Bauer & Swanson (2021), Steffensen *et al.* (2021)), central banks and financial markets do not possess significantly different information. Therefore, supply and demand shocks can be expected to be perceived similarly by the monetary authority, and we can also examine how monetary policy responds to these shocks by examining the movements in the short-end of the yield curve.

Concerning our identification strategy, a natural concern is that some of the movement in bond yields could be unrelated to supply and demand shocks and could instead reflect some type of yield shock. This is a valid argument, but we argue that it is a more likely explanation that markets generally respond to the current macroeconomic environment. The same applies to central banks, which are more likely to respond to current news rather than intentionally deviating from their monetary policy rules. Therefore, this paper does not consider these types of shocks. This assumption is in line with Bauer & Swanson (2021), which shows that publicly available news causes the private sector to revise its forecasts and the Fed to change monetary policy. To conclude, the bias in our estimation due to the existence of yield/monetary shocks depends on the size of these shocks. If these shocks explain only a small fraction of variation macroeconomic series, then the bias will be small. This is supported by the research from Faust *et al.* (2004), or Carvalho *et al.* (2021).

4.2 Data

We use US data from the Survey of Professional Forecasters (SPF) and additional data regarding Fed funds rate (from FRED) and bond yields. We use zero-coupon Treasury yields from Gürkaynak *et al.* (2007). We include maturities in the range of 1-15 years. Regarding SPF data, we use aggregated mean value (growth rate) data for the following series: GDP/GNP and

GDP/GNP price deflator. Note that research by Mboup & Wurtzel (2018) shows that there are basically no differences between the mean and median forecasts for real GDP and inflation. We use all available data, i.e., the period from 1969 (q4) to 2022 (q1). Regarding frequency, the data are available only on a quarterly frequency. To determine revisions of expectations, we use the difference between the expected value for the quarter in which the Survey is conducted, and we compare this value with the expectations for this quarter in the previous round of the Survey, i.e., $E_t x_t - E_{t-1} x_t$, where x_t is a (forecast of) either GDP/GNP (y_t) or inflation (π_t). The Survey sets the deadlines for responses in the second to the third week of the middle month of each quarter when the Survey is conducted. Therefore, we use the end-of-month data for other series. Due to the importance of expectations, we use data for these nearest forecast periods, which should be the least sensitive to changes in expectations.

Regarding yield curve analysis, we use multiple representations of the yields. As a starting point, we include the 10-year yield together with the Fed funds rate. To generalize the results for the whole yield curve, we replace the single yield¹ by the level, slope, and curvature of the yield curve using Nelson & Siegel (1987) methodology:

$$y_t(\tau) = L_t + S_t \left(\frac{1 - e^{\lambda\tau}}{\lambda\tau} \right) + C_t \left(\frac{1 - e^{\lambda\tau}}{\lambda\tau} - e^{\lambda\tau} \right) \quad (4.1)$$

$y_t(\tau)$ denotes the yield with time to maturity τ . L_t , S_t and C_t denote the level, slope and curvature of the yield curve, λ is a parameter fixed to the value of 0.0609 (Diebold & Li, 2006). Table A1 presents a summary of the data used:

¹We omit the Fed funds rate. The slope factor reflects the fluctuations on the short end of the yield curve and should provide sufficient information regarding the short rate movements.

Table 4.1: Data Description

Name (abbreviated)	Ticker	Source
Federal Funds Effective Rate	FEDFUNDS	FRED
GDP/GNP price index	PGDP	SPF
Real GDP/GNP	RGDP	SPF
The U.S. Treasury Yield Curve: 1961 to the Present	SVENYXX	federalreserve.gov
Name (abbreviated)	Mean	Standard Dev.
GDP/GNP price index revision	0,01	0,66
Real GDP/GNP revision	-0,30	1,70
Federal Funds Rate	4,98	3,90
10y gov bond yield	6,10	3,15
Level	6,70	2,76
Slope	-1,62	1,90
Curvature	-2,51	3,09

Note: The upper part of the table shows the data sources used. The lower part of the table shows the summary statistics of the actual data used in the estimation.

4.3 Methodology

In terms of methodology, we generally follow Jarociński & Karadi (2020).² We use structural VAR with signs restriction to identify the impact of supply and demand shocks. More specifically, we use sign restrictions in conjunction with revisions of expectations to identify these shocks. Note that this decomposition is generally not unique but allows supply and demand shocks to hit the economy simultaneously. We use Minnesota-style priors as in Jarociński & Karadi (2020) to prevent the overfitting of the model. However, the estimation does not differ much from the frequentist version. Specifically, we take the prior for all coefficients of each variable to be zero, and we use standard Minnesota prior for their variance. We set the tightness parameter equal to 5 and decay parameter equal to 1, i.e., $\lambda_1 = 5$, $\lambda_2 = 1$. Furthermore, for the VAR covariance matrix, we use Inverse Wishart prior $IW(S, v)$, where we set $v = N + 2$, where N denotes the number of series used, and S is a diagonal matrix with the standard errors. We also impose block exogeneity on the revision of expectations, which means that the forecast revisions are unpredictable, i.e., $A_t, B_t = 0$.³ Equation 5.2 represents the VAR model:

$$\begin{pmatrix} REV_t \\ X_t \end{pmatrix} = \sum_{n=1}^p \begin{pmatrix} A_t & B_t \\ C_t & D_t \end{pmatrix} \begin{pmatrix} REV_{t-n} \\ X_{t-n} \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha \end{pmatrix} + \begin{pmatrix} e_t^{rev} \\ e_t^X \end{pmatrix} \sim N(0, \Sigma) \quad (4.2)$$

REV_t is a vector of surprises which consist of revision of GDP and inflation, i.e. $E_t x_t - E_{t-1} x_t$, where $x_t = (\pi_t, y_t)$. X_t is a vector of variables of interest which consist either of Fed funds rate, and 10-year yield or yield curve factors (slope, level, curvature). We use two lags ($p = 2$), but the results are generally robust to any number of lags (1-4). We further impose the zero and sign restrictions but only to the revisions of expectations to identify supply and demand shocks, i.e. $shock_{supply} = (E_t \pi_t - E_{t-1} \pi_t) > 0 \wedge E_t y_t - E_{t-1} y_t < 0 \wedge X_t = 0$ and $shock_{demand} = (E_t \pi_t - E_{t-1} \pi_t) > 0 \wedge E_t y_t - E_{t-1} y_t > 0 \wedge X_t = 0$. Other series (X_t) remain unrestricted.⁴

²We only slightly modified their (publicly available) original code.

³The results don't differ much if we relax this assumption.

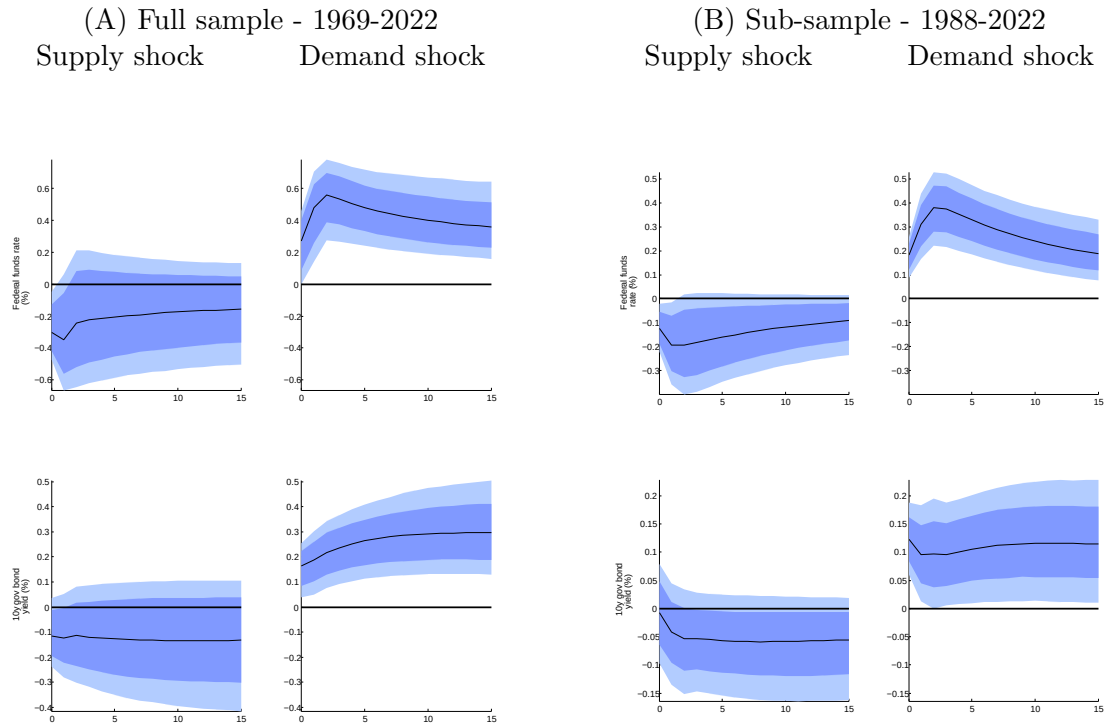
⁴We are aware of the criticism of VAR with signs restriction by Baumeister & Hamilton (2020) about the Haar prior (un)informativeness for the structural impulse responses (as well as the response by Inoue & Kilian (2021)). Alternatively, we could directly divide supply and demand shock based on the sign restrictions: $shock_{supply(demand),t} = (E_t \pi_t - E_{t-1} \pi_t)$ if $(E_t y_t - E_{t-1} y_t) * (E_t \pi_t - E_{t-1} \pi_t) < (>) 0$, else 0. And then use these shocks as instruments.

4.4 Empirical Results

Figure 4.1 left panel shows the model's results using full-sample data. A (negative) supply shock is associated with a 0,415 % increase in the revision of the inflation forecast and a 1,1578 % drop in the revision of the output forecast. This is due to the model construction, where these responses have to satisfy the sign restrictions. In the case of the supply shock, the responses are not clearly identified, but there generally seems to be some downward pressure on the Fed funds rate and the 10-year yield. Specifically, in the case of the Fed funds rate, the downward pressure is equal to 0,2 % for all quarters. In the latter case, the downward movement is 0,1 %. The impact of demand shocks on monetary policy and yields is generally more clearly identified. A demand shock is associated with a 0,485 % increase in the revision of the inflation forecast and a 1,28 % increase in the revision of the output forecast. The model confirms the usual institution that the rise of inflation (due to the demand shock) causes the rise of the Fed funds rate, which peaks in the first year and then slowly fades away. The 10-year yield also rises, but the increase is approximately half of the previous case. However, this increase seems to be longer-lasting.

Figure 4.1 right panel shows the results of the robustness check, which uses only a sub-sample of data. The sub-sample covers the period from 1988 (q1) to 2022 (q1). This is motivated by the desire to cover a more homogeneous period characterized by predominantly stable inflation. The general finding is that the impact of supply/demand shocks is similar to full-sample estimation. Our finding shows that a (positive) demand shock, associated with a 0,197 % increase in the revision of the inflation forecast and a 0,904 % increase in the revision of the output forecast, causes a rise in the Fed fund rates. The 10-year yield rises but significantly less than the Fed funds. In the case of the (negative) supply shock, associated with a 0,329 % increase in the revision of the inflation forecast and a 1,151 % drop in the revision of the output forecast, there seems to be some downward pressure. Contrary to the full-sample, there seems to be less significant downward movement in bond yields. Generally, the shock impacts (and the shocks) seem less significant in size than in the full-sample estimation.

Figure 4.1: Impulse Responses - Fed funds rate, 10-year yield

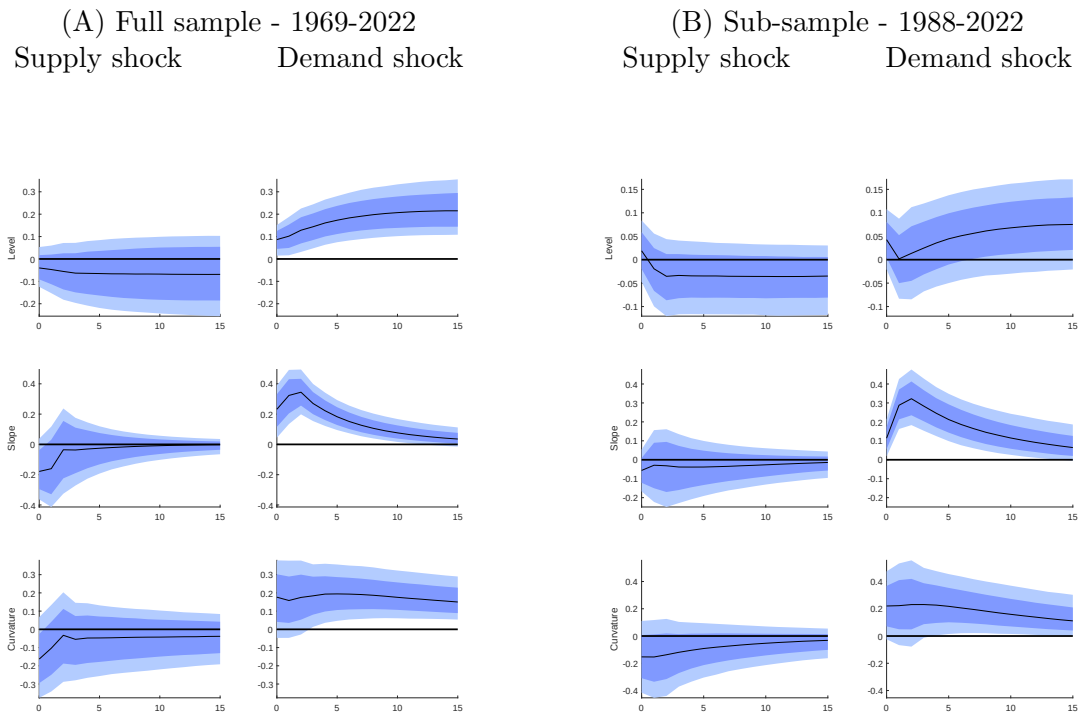


Note: Median (line), percentiles 16–84 (darker band), percentiles 5–95 (lighter band).

Figure 4.2 left panel shows the impulse-responses using Nelson-Siegel decomposition. In the case of the supply shock, associated with a 0,44 % increase in the revision of the inflation forecast and a 1,171 % drop in the revision of the output forecast, the yield curve response is not clearly identified. However, there seems to be some downward pressure on the level. Specifically, the downward pressure is equal to approximately 0,05 % for all quarters. On the other hand, supply shock has only a short-lived effect on the slope and curvature. This means that supply shock causes a parallel shift in the yield curve. A demand shock is associated with a 0,485 % increase in the revision of the inflation forecast and a 1,273 % increase in the revision of the output forecast. The results show that demand shock is associated with the rise of the level, which seems to be long-lasting. Also, slope and curvature rise, this increases short and medium maturities more than long yields, but this effect slowly fades away. Specifically, slope response peaks in the first year and then returns to zero.

Figure 4.2 right panel shows the results using only a sub-sample of data. This generally confirms the results of the full-sample estimation. A negative supply shock is associated with a 0,198 % increase in the revision of the inflation forecast and a 0,89 % drop in the revision of the output forecast. Contrary to the full-sample, there seems to be even less significant downward movement in level and slope. Interestingly, the main difference is regarding curvature, with a more negative and longer-lasting response. Positive demand shock is associated with a 0,33 % increase in the revision of the inflation forecast and a 1,51 % increase in the revision of the output forecast; it causes a rise in all the factors, and results generally do not differ much from the full-sample estimation. The main difference is a less-significant response of the level.

Figure 4.2: Impulse Responses - Yield Curve



Note: Median (line), percentiles 16–84 (darker band), percentiles 5–95 (lighter band).

4.5 Conclusion

In this paper, we present a simple way of extracting the supply and demand shock using the revisions of expectations of analysts using the Survey of professional forecasters. The main contribution is the identification of these shocks and the subsequent assessment of their impact on Fed funds rate and bond yields. Specifically, we use structural VAR with signs restrictions to assess the impact of these shocks. The study empirically shows the different impacts of supply and demand shocks on bond yields. Overall, the study allows for a better structural interpretation of yield curve dynamics.

Chapter 5

Forecasting Nominal Exchange Rates Using a Dynamic Model Averaging Framework

Abstract: This paper focuses on the predictive role of inflation and interest rates in forecasting stock returns. We report economically significant in-sample evidence that stock returns are predictable using cyclical components of inflation and interest rates. The out-of-sample analysis results show that these cyclical components have reasonable out-of-sample predictability for the equity risk premium when the comparison is made against a historical average benchmark. We believe this is due to the fact that both inflation and interest rates contain stochastic trends. Accounting for changes in trend inflation (and equilibrium real interest rate) is essential for forecasting stock returns.

5.1 Introduction

Exchange rate forecasting is generally considered highly challenging, and most studies illustrate that consistently beating the random walk benchmark is extremely difficult. This was demonstrated in the early studies of Meese and Rogoff (Meese & Rogoff, 1983a,b) and, until recently, has remained essentially unchanged. On the contrary, this paper presents evidence that there is significant predictability of nominal exchange rates in the medium and long run and even some predictability in the short-term horizon, at least regarding the last two decades considered and the currencies considered. The primary contribution of this study lies in the out-of-sample prediction exercise, wherein we provide forecasts of nominal exchange rates utilizing a dynamic model averaging (DMA) approach. This method outperforms the random walk benchmark, particularly in the medium and long run.

The field of exchange rate forecasting has traditionally centered around achieving optimal point forecasts, driven by the widely acknowledged difficulty of predicting exchange rates. Consistently outperforming the random walk benchmark in forecasting has posed a significant challenge, a fact that until the recent decade has remained largely unchanged since the seminal papers by Meese and Rogoff (Meese & Rogoff, 1983a,b). Consequently, much of the literature has concentrated on the objective of obtaining the best point forecasts. However, considering the uncertainty surrounding exchange rate forecasts is crucial for economic agents when making decisions. While point forecasts provide valuable information about the most likely outcome, they do not convey the complete picture of the range of possible outcomes and associated risks. By incorporating measures of uncertainty, such as the full probability distribution, economic

agents can better assess the potential variability and downside risks associated with different exchange rate scenarios.

We offer several explanations for this result. We present substantive evidence that the nominal exchange rates may be mean-reverting, and the inclusion of other explanatory (fundamental) variables implied by different theories does not significantly change the forecast results, at least in the short run and in the medium-term horizon; their contribution is relatively modest. In the long run, we, however, present some evidence that the inclusion of additional variables significantly improves forecasting results. We illustrate these findings by using a dynamic model averaging framework, which implicitly accounts for most of the possible specifications used in exchange rate forecasting, e.g., the forecast of exchange rate based on a monetary model, the Taylor rule model, uncovered interest rate parity model, and purchasing power parity model of exchange rates.¹ This framework also allows us to study parameters and model uncertainty in exchange rate forecasting, which is closely related to the so-called 'scapegoat theory of exchange rates'.

From the practical point of view, most of the studies regarding exchange rate predictions are based on the idea that there exists a cointegrating relation between the exchange rate and other variables, and exchange rates converge back to their fundamental value. On the other hand, we argue that an exchange rate forecast capable of defeating a random walk benchmark is sufficient to use lagged exchange rate values and also the use of a direct forecasting approach, which allows filtering out the non-fundamental noise in the short-run periods. We stress that this finding can be considered novel; although we are not the first to make this claim, the use of the dynamic model averaging approach and the possibility to examine time-variation in parameters and the posterior inclusion probabilities further expands the results recently obtained by Engel & Wu (2023) regarding the fact that level of the exchange rate appears to have significant forecasting power. The results also indicate that the inclusion of additional fundamental variables can further improve exchange rate predictions.

The rest of this paper is structured as follows: In section 2, we present the existing literature regarding exchange rate forecasting. In sections 3 and 4, we describe the data and the methodology used. Finally, in sections 5 and 6, we present empirical results obtained using a dynamic model averaging approach. In this part, we perform out-of-sample forecasting exercise for the AUD/USD, CAD/USD, EUR/USD, CHF/USD, GBP/USD, NOK/USD, NZD/USD, SEK/USD, and JPY/USD nominal exchange rates and compare them to several other commonly used benchmark models (random walk without drift and benchmark models based on simple OLS regression). The rationale behind the inclusion of these currency pairs follows the study by Engel & Wu (2023), which also examined a similar time period. These currencies are considered the non-dollar G10 currencies, broadly representing the most heavily and most liquidly traded currencies in the world.

5.2 Literature Review

Empirical studies regarding exchange rate development are, in many cases, loosely based on economic theory. Most of the empirical forecasting literature is therefore based on the monetary theory of exchange rates, some version of the Taylor (monetary policy) rule, or the articles are based on some version of the validity of purchasing power parity or uncovered interest rate parity. The body of literature on exchange rate forecasting is extensive. Consequently, this article does not provide an exhaustive summary but instead focuses on a select subset of key

¹We offer a brief description of these models in the next section.

contributions. For a more comprehensive review of the literature, refer to, for example, Rossi (2013).

Exchange rate models grounded in monetary theory (Dornbusch, 1976; Frankel, 1979; Mussa, 1977, 1982) have been among the most frequently tested empirically. However, these models typically underperform when compared to other forecasting methods, a result first documented in the seminal works of Meese & Rogoff (1983a,b). More recent research, however, indicates that the predictive power of these models can be somewhat enhanced through alternative approaches, such as panel regression techniques (Mark & Sul, 2001; Groen, 2005; Engel *et al.*, 2007, 2015), or the application of machine learning and Bayesian methods (Carriero *et al.*, 2009; Beckmann & Schüssler, 2016; Amat *et al.*, 2018). It is widely recognized that forecasting performance varies with the time horizon selected, and the literature generally suggests that some degree of predictability emerges over the long term (Macdonald & Taylor, 1993; Mark, 1995; Chinn & Meese, 1995; Macdonald, 1995; MacDonald, 1998; Chinn & Quayyum, 2012; Huber, 2016; Ren *et al.*, 2021).

From a practical point of view, defining money empirically poses significant challenges, and many authors use monetary aggregates without considering the composition of these aggregates. Consequently, this article draws on the rapidly growing body of literature examining the role of credit—the primary counterpart to money—and the mechanics of credit creation. Numerous articles within international economics and finance have integrated the concept of credit into their frameworks. For instance, Drehmann *et al.* (2012) present a two-country model emphasizing the distinction between net payment flows and financial (gross) flows. Bruno & Shin (2015) adopt a portfolio theory-based approach, while Cesa-Bianchi *et al.* (2019) and Kumhof *et al.* (2020) extend the new Keynesian structural model with a banking sector and endogenous money creation into a two-country model. These studies highlight how exchange rates can be shaped by the credit and activities of the banking sector. Casta *et al.* (2022) then shows that credit to corporations has forecasting power in predicting exchange rates, as it is a key driver in the creation of new deposits (money) connected to real economic activity.

The Taylor (monetary policy) rule inspires other models often used in forecasting literature. This approach is based on the fact that central banks adjust nominal interest rates according to perceived or expected inflation and the output or unemployment gap.² Consequently, this policy rule can be incorporated into the uncovered interest rate parity relation, providing an alternative expression for expected exchange rate movements.³ The key (albeit simplified) distinction between this Taylor rule-based approach and the monetary model is that the endogenous monetary policy rate replaces money demand. Several studies have examined this approach and found evidence of forecasting power, including Engel & West (2006), Molodtsova *et al.* (2008), Molodtsova & Papell (2009), Molodtsova *et al.* (2011), Molodtsova & Papell (2013), Byrne *et al.* (2016), and others. However, the forecasting power is found mostly in the medium and long-run forecasting horizon.

The models based on the validity of purchasing power parity (PPP) are, in theory, the oldest, but from the forecasting point of view, they only recently came back into prominence. Ca’Zorzi *et al.* (2016) show that there exists mean reversion in the real exchange rates due to the long-run convergence towards purchasing power parity, which occurs through nominal exchange rate adjustments. Ca’Zorzi & Rubaszek (2020) build upon previous results claiming that if real exchange rates are predictable, then the nominal ones are also predictable if one can reasonably extrapolate what drives the relative price indices (differences in the price levels of

²Molodtsova *et al.* (2011) and Molodtsova & Papell (2013) generally achieved better results with the unemployment rates/gaps as they found out that the use of unemployment rates/gaps increase out-of-sample exchange rate predictability for almost all of the specifications.

³Alternatively, interest rates can also be used directly as explanatory variables.

the given countries). In short, as the authors point out, the building blocks of this approach are that the purchasing power parity holds in the long run and that the nominal exchange rate drives most of the real exchange rate adjustment. In other words, the real exchange rate is mean-reverting.

However, the results of exchange rate predictability are frequently subject to criticism, primarily due to the highly persistent nature of the independent variables (Kilian, 1999; Berkowitz & Giorgianni, 2001; Cheung *et al.*, 2005, 2019; Engel *et al.*, 2019). From a theoretical point of view, the situation is even more difficult because if we assume rational expectations, these models generally imply a forward-looking solution. This is a standard asset pricing approach where the exchange rate is a function of discounted values of some (expected) fundamental variables. If the fundamentals (explanatory variables) are highly persistent and the discount factor approaches unity, then the nominal exchange rate behaviour will follow random walk behaviour closely. (Engel & West, 2005; Evans & Lyons, 2005; Engel *et al.*, 2007, 2009). Although, this does not negate the fact that these factors may indeed influence exchange rate determination. If the model is correctly specified, the error correction term could serve, in some cases, as an indirect observation of the (mean-reverting) FX risk premium.⁴ For example, Engel (2014, 2016) claim that the FX risk premium may be a significant determinant and often an omitted variable in forecasting literature.

In terms of the methodology used in this article, we build upon literature that studies parameter and model uncertainty in exchange rate forecasting. Specifically, the studies based on the Bayesian model averaging approach are the most similar ones. These studies often use a more eclectic view regarding theoretical justification and focus more on out-of-sample properties of forecasts. The articles using the BMA/DMA approach are, for example, Wright (2008), De Bruyn *et al.* (2015), Beckmann *et al.* (2020), and Dybka (2022). These studies generally show that these approaches outperform more common methods used for exchange rate predictions, primarily as a result of the shrinkage properties of estimators used. Specifically, Beckmann & Schüssler (2016) employ the DMA framework similar to ours for exchange rate forecasting together with a broad array of competing models to examine parameters and model uncertainty. However, unlike us, the authors focus only on short-term predictions. We, on the other hand, explore the predictability in different forecasting horizons and document significant improvements relative to the random walk benchmark only in the medium and long run. More generally, articles by Sarno & Valente (2009), Berge (2014), and Kouwenberg *et al.* (2017) also examine the role of model uncertainty in exchange rate forecasting.

The parameter and model uncertainty is often theoretically based on the so-called 'scapegoat theory of exchange rates'⁵, see, for example, Bacchetta & Van Wincoop (2004), Bacchetta & Van Wincoop (2013), or Fratzscher *et al.* (2015). This line of reasoning requires a forecasting technique that can handle rapid shifts in parameters and allows the relevant subset of variables to change over time. The DMA framework used in this paper encompasses this theory, as it is able to accommodate both parameter instability and model uncertainty. As emphasized by Sarno & Valente (2009), since the true data-generating process linking fundamentals to the future exchange rate is unknown, a reasonable approach to producing exchange rate forecasts

⁴Both models can be derived using uncovered interest parity relation, which includes the risk premium. For example, the foreign exchange (FX) risk premium can be interpreted as a convexity adjustment of the lognormal pricing kernels, specifically reflecting the difference between the convexity adjustments of the home and foreign countries (Backus *et al.*, 2001a; Engel, 2016).

⁵The parameter instability is based on a scapegoat story: some (fundamental) variable is given an 'excessive' weight during a certain period due to the belief that it influence the exchange rate. The exchange rate may change for reasons that have nothing to do with the observed (fundamental) variable. Facing incomplete and heterogeneous information, investors attach excessive weight to an observed variable. Over time, different variables can be taken as scapegoats so that the weights attributed to the variables change.

involves searching for an adequate model specification among all possible models believed to be capable of predicting the exchange rate.

In terms of results obtained and the reasoning behind them, the most similar findings are presented in a recent study by Engel & Wu (2023). More specifically, the authors present a similar conclusion to our research that the level of nominal exchange rate is the best predictor of future exchange rate changes. However, the authors also present a more skeptical perspective on whether the results obtained may be due to the small sample. In our opinion, this further emphasizes the importance of researching this topic. The use of the dynamic model averaging approach further strengthens the results regarding the forecasting power of the lagged level of exchange rate obtained by Engel & Wu (2023), at least regarding the short and medium-term horizon. However, our results also show that the DMA approach provides a significantly better forecast in the long run in comparison to the alternative benchmark model, which uses a simple direct multi-step OLS forecast in which the change in the exchange rate is a function of a constant and the lagged level of the real or nominal exchange rate.

5.3 Data

The data used for our forecasting exercise are the spot nominal exchange rates (AUD/USD, CAD/USD, CHF/USD, EUR/USD, GBP/USD, JPY/USD, NOK/USD, NZD/USD, SEK/USD) expressed as USDs per unit of domestic currency and other explanatory (fundamental) variables.⁶ The selection of explanatory variables was primarily guided by the theoretical models explaining exchange rate movements described above and also by empirical studies that were at least partially successful in exchange rate forecasting (Engel *et al.* (2007), Molodtsova & Papell (2013), Ince (2014), and Cheung *et al.* (2019)). More specifically, the data used as additional explanatory variables are the countries' price levels, credit, unemployment rates, and interest rates. All variables are transformed to logarithmic form with the exception of interest rates. We use only data in (log) levels; however, adding any data transformation is straightforward in the DMA approach, as is shown later.

Regarding the other explanatory (fundamental) variables used in this study, the rationale for their inclusion is as follows: The motivation behind including interest rates is based on uncovered interest rate parity. The inclusion of unemployment rates is based on monetary policy rule models and, thus, implicitly, on uncovered interest rate parity. The unemployment rate in this regard acts as a proxy for the output/unemployment gap⁷ as in Molodtsova *et al.* (2011) and Molodtsova & Papell (2013), although we recognize that the time-series properties of these two series are somewhat different. The use of the unemployment rate allows us to avoid various methodological problems associated with measuring the natural rate of unemployment (Crump *et al.*, 2019). Additionally, we prefer to use the unemployment rate rather than output (gap) due to the monthly periodicity of the forecasting exercise, as output data is only available quarterly. The inclusion of price levels is motivated by studies by Ca'Zorzi *et al.* (2016);

⁶For clarification, we use the data regarding other explanatory (fundamental) variables for the countries of the respective exchange rates, which are listed above.

⁷The motivation behind the inclusion of this variable is based on the monetary policy rule. Thus, we indirectly forecast interest rate changes, which, based on uncovered interest rate parity, cause movements in the exchange rate. Also, the implicit assumption is that we assume a constant natural rate of unemployment or that the use of time-varying coefficients also implicitly accounts for changes in the natural rate of unemployment. Molodtsova & Papell (2013) use the non-accelerating inflation rates of unemployment (NAIRU) from the OECD Economic Outlook and unemployment rates to construct the unemployment gaps. On the other hand, Molodtsova *et al.* (2011) simply use (forecasted) unemployment rates. The authors argue that given the use of rolling regressions where the constant, as well as the coefficients, change each period, using the real-time unemployment rate is similar to using the real-time unemployment gap.

Ca'Zorzi & Rubaszek (2020) and including these variables, therefore, implicitly accounts for the real exchange rate term in the fundamentals. The inclusion of the proxy variable for money is based on the monetary model of exchange rates. In this case, use credit to non-financial corporations of the given countries denominated in national currencies as a proxy for money, as in Casta *et al.* (2022). Loans to corporations can serve as a proxy for corporate deposits and are expected to have the greatest impact on the exchange rate, as they are most likely to be involved in international transactions. More generally, corporate loans are a key determinant of real economic dynamics, as highlighted by Biggs & Mayer (2013), and the creation of new deposits has the most direct connection to real economic activity.⁸

Data were obtained from Datastream and a range of national and supranational statistical databases, including those of the Bank of Canada (BoC), Bank of Japan (BoJ), ECB Statistical Data Warehouse, Eurostat, FRED, Reserve Bank of Australia (RBA), Reserve Bank of New Zealand (RBNZ), Statistics Sweden (SCB), and the Swiss National Bank (SNB). We primarily use seasonally unadjusted end-of-period monthly data.⁹ Detailed information about the data can be found in Table A1 in the Appendix. Table A2 in the Appendix presents the summary statistics for the final dataset used in the estimation. We should point out that the standard unit root tests mainly indicate the time series' nonstationarity.

We performed our calculations using monthly data from January 1997 to December 2022 in the case of AUD/USD and CAD/USD, in the case of CHF/USD¹⁰ from January 2002 to December 2022, in case of EUR/USD from January 2003 to December 2022, in the case of GBP/USD from September 1997 to December 2022, in the case of JPY/USD from October 2000 to December 2022, in the case of NZD/USD from June 1998 to December 2022, and in case of SEK/USD from December 2001 to December 2022.

Due to the unavailability of non-revised data for all variables, we relied on the revised dataset available as of December 2022. This choice is unlikely to pose significant issues, as most of the data has not undergone substantial revisions. However, we acknowledge that the unemployment data may have been somewhat affected by revisions. For consistency, we opted not to use earlier versions of the unemployment data, although older vignettes were employed for sensitivity analysis, which yielded results that did not differ significantly. We also abstract from the fact that macroeconomic variables are frequently published with a lag.

The selection of sample periods used in the forecasting exercise was generally made for two reasons. First of all, we were limited by data availability regarding some fundamentals and currencies.¹¹ The second reason for the choice of the sample period is to focus on periods

⁸From an accounting perspective, these loans serve as asset counterparts to a portion of monetary aggregates. If we were to use total loans or broad monetary aggregates such as M2 or M3, which closely align with total loans, we would also incorporate items like mortgages. Mortgages, which have an average maturity exceeding 20 years in developed countries and account for approximately half of total credit, would introduce substantial persistence into the credit time series due to their long-term nature, effectively creating autocorrelation over two decades. An alternative approach would be to use only new (corporate) loans; however, such time series are not available for all countries.

⁹When monthly data were unavailable, we employed piecewise constant interpolation to convert the series to monthly values. This approach ensures that unpublished data are not used in the forecasting exercise. For New Zealand, the monthly unemployment rate was constructed by using the registered unemployment figures, which are published monthly, alongside quarterly data on the labor force population to derive the rate. In the case of Japan, a time series of loans to all corporations was used as a proxy for money. This is because data on loans to non-financial corporations are not available.

¹⁰Note that the Swiss National Bank implemented a one-sided exchange rate target zone with respect to the euro from September 2011 to January 2015. This policy may influence the results obtained and should be taken into consideration when interpreting the findings.

¹¹We were mainly limited by the inclusion of the EUR/USD currency pair as one of the most active currencies and also, in some cases, the availability of credit data.

dominated by a stable inflation environment and inflation-targeting regimes. However, not including longer time series may somewhat enhance the small sample properties problem.

5.4 Methodology

We estimate the forecasting model using a dynamic model averaging approach. We primarily use a direct (multi-step) forecast, and due to the dynamic model averaging framework, we use a recursive (expanding) window approach, i.e., every period, we add the actual realization of the exchange rate and other explanatory variables. This is due to the fact that the model averaging is done recursively, which means that the estimation procedure allows for different models to be selected in every period, and this probability is based on the posterior model probability criterion. At each step, an h -period-ahead forecast is produced, utilizing a distinct set of models corresponding to each forecasting horizon. In the out-of-sample forecasting exercise, the initial 72 periods, equivalent to six years, are employed as the training dataset. The forecast horizon ranges from one month to 36 months.¹² We stress that in the out-of-sample forecasting exercise, the selection of explanatory variables and updating of the coefficients are also done in a (pseudo) real-time setting. More specifically, we estimate the following set of models (expressed in the state-space form) for every currency pair (see Equations 5.1 and 5.2):

$$\Delta_{t+h}e_{i,t} = A_{i,t} + B_{i,t}e_{i,t} + C_{i,t}f_{i,t} + \varepsilon_{i,t+h} \quad (5.1)$$

$$\gamma_{i,t+1} = \gamma_{i,t} + \xi_{i,t} \quad (5.2)$$

where $e_{i,t}$ denotes chosen currency pair¹³ i (AUD/USD, CAD/USD, CHF/USD, EUR/USD, GBP/USD, JPY/USD, NOK/USD, NZD/USD, SEK/USD), Δ_{t+h} denotes the h step forward difference operator, and h is the number of leads, i.e., $\Delta_{t+h}e_{i,t} = e_{i,t+h} - e_{i,t}$.¹⁴ Furthermore, $e_{i,t}$ denote nominal exchange rate, $A_{i,t}$ is the constant term, and $B_{i,t}, C_{i,t}$ are regression parameters, and $\varepsilon_{i,t+h}$ denotes stochastic term in observation equation following normal distribution and $var(\varepsilon_{i,t+h}) = \sigma_{i,t+h,\varepsilon}^2$. We emphasize that $A_{i,t}, B_{i,t}, C_{i,t}$ are parameters that are time-varying. Furthermore, $f_{i,t}$ denotes the vector of other potential explanatory variables (fundamentals), i.e., $f_{i,t}$ is constructed from the following variables: $(m_{US}, m_{domestic}, p_{US}, p_{domestic}, un_{US}, un_{domestic}, ir_{US}, ir_{domestic})$, where index US denote United States fundamentals and index $domestic$ denote the fundamentals of the domestic country related to the chosen currency pair i . In more detail, m_{US} is the loans to non-financial corporations (money) in the US, and $m_{domestic}$ is the loans to (non-financial) corporations in the domestic country in the chosen currency pair. p_{US} and $p_{domestic}$ denote the price level of countries, un_{US} and $un_{domestic}$ are the unemployment rates regarding the chosen currency pair, ir_{US} and $ir_{domestic}$ denote interest rates regarding the chosen currency pair. To simplify notation, we also introduced $\gamma_{i,t} = (A_{i,t}, B_{i,t}, C_{i,t})$, where $\gamma_{i,t}$ follows a random walk. Therefore, regarding state equation $\xi_{i,t}$ denotes the error term, and also set $var(\xi_{i,t}) = \sigma_{i,t,\xi}^2$. The errors, $\varepsilon_{i,t+h}$ and $\xi_{i,t}$, are assumed to be mutually independent at all leads and lags.

Based on the economic theory described in the literature section, we can also a priori (and with a large amount of simplification) determine which signs we expect for the coefficients (or

¹²To simplify the interpretation of the text, we often describe the results in terms of short, medium, and long-run periods. We generally consider a period of up to and including 6 months as a short-run period. A period from 7 to 12 months is considered a medium-run period, and more than one year is considered a long-run period.

¹³Therefore, equation 5.8 and 5.9 estimated separately for each currency. Also, in this part, for better readability, we omit the subscript k , which indicates the different models regarding every currency pair i . This is because the DMA approach uses many different model specifications.

¹⁴Since a distinct model is estimated for each forecast horizon and each currency, the lead length h varies according to the selected forecast horizon.

combination of coefficients) regarding fundamentals.¹⁵ The description below also focuses on coefficients regarding the medium and long-term forecasting horizon, where we believe there should be at least some convergence to a fundamental value. Furthermore, it should be noted that due to the importance of the US economy, variables regarding the US economy may act as proxies for general risk aversion. Based on purchasing power parity (PPP), we expect the coefficient regarding p_{US} to be positive and the coefficient regarding $p_{domestic}$ to be negative. In other words, the currency i depreciates if the (domestic) country $domestic$ has a higher price level than the US price level, i.e., $e_{i,t} = p_{US,t} - p_{domestic,t}$. Based on the monetary theory of exchange rates, we again expect the coefficient regarding m_{US} to be positive and the coefficient regarding $m_{domestic}$ to be negative, as the model is an extension of the PPP theory. Based on the uncovered interest rate parity (UIRP) and standard formulation of monetary policy rules, we expect the coefficients regarding un_{US} and ir_{US} to be positive and the coefficients regarding $un_{domestic}$ and $ir_{domestic}$ to be negative. This is based on the following UIRP relation: $\Delta_{t+h}e_{i,t} = ir_{US,t} - ir_{domestic,t}$. However, an increase in the interest rate, whether current or expected in the future, should cause an immediate appreciation of the dollar, followed by forecasted (and actual) depreciation. For a comprehensive discussion on this topic, see, for example, Molodtsova & Papell (2009), where authors argue for opposite signs regarding $(un_{US}, un_{domestic})$ due to the (interest rate) smoothing. Also, previous studies, for example by (Kojien *et al.*, 2018), have also shown that carry trades are generally profitable, which suggests the invalidity of uncovered interest rate parity in the medium and short run.¹⁶ Furthermore, since the fundamental variables enter the model separately, we do not restrict their coefficients, and given the effect of, for example, risk premium, we do not expect cointegration vectors of exactly $(1, -1)$ for any of the above relation.

Note that specification 5.1 represents the most parsimonious form, where we do not impose a cointegration relation a priori; rather, it is implicitly accounted for in the estimation process. To illustrate this, the expression can be rewritten as follows, where $c_{i,t}, d_{i,t}$ are the new constant terms, and $\gamma_{i,t}, \delta_{i,t}$ are the new regression parameters: $\Delta_{t+h}e_{i,t} = c_{i,t} + \delta_{i,t}(e_{i,t} - \gamma_{i,t}f_{i,t} + d_{i,t}) + \varepsilon_{i,t+h}$. This specification (error correction model) is often used to calculate forecasts based on the monetary and Taylor rule models.¹⁷ As Mark (1995) points out, this specification is a projection of the h-step change in the nominal exchange rate on the current deviation from the fundamental value. In summary, each time series regarding the explanatory variables enters the model separately as we do not impose cointegration vectors a priori.

As mentioned above, the approach used for the estimation is dynamic model averaging (DMA), which is a relatively novel approach in exchange rate forecasting literature. This approach uses a set of approximations to evaluate a large number of models in every period, which significantly reduces the computational burden. In short, the DMA approach combines the merits of time-varying parameter models with the Bayesian model averaging framework. This is especially suitable for this forecasting exercise given a large uncertainty about which set of variables best characterizes the development of exchange rates. Moreover, we also suspect that the relevance of individual variables or combinations may vary over time, for example, due to different economic conditions over the business and financial cycle. Similar to the argument by Carriero *et al.* (2009), we consider this a natural way to reduce instability problems. More

¹⁵However, we abstract from the fact that if we assume rational expectations, the theoretical models described above generally imply forward-looking solutions. The results also depend on the rate of adjustment, i.e., if we assume that all adjustments are immediate or if there exists some smoothing. For example, interest rate smoothing, sticky prices, etc.

¹⁶Therefore, the signs regarding these coefficients are generally unclear.

¹⁷Most studies (Mark, 1995; Cheung *et al.*, 2005, 2019) employ a single-equation error correction model (ECM) striped of short-term adjustment terms, as this approach facilitates exchange rate forecasting using only the past values of other variables.

specifically, dynamic model averaging (DMA) uses sets of approximations regarding Kalman filter equations and Markov-switching models to reduce computational complexity. DMA framework primarily builds upon the Kalman filter equations for the estimation of individual models, but the prediction formula for the variance in the Kalman filter is simplified.¹⁸ In more detail, the traditional prediction formula for the variance of the prediction error can be written as follows (see Equation 5.3):

$$P_{i,t|t-1} = P_{i,t-1|t-1} + \sigma_{i,t,\xi}^2 \quad (5.3)$$

The formula can be simplified (replaced) in the following way using forgetting factor λ (see Equation 5.4):

$$P_{i,t|t-1} = \frac{1}{\lambda_{i,t}} P_{i,t-1|t-1} \quad (5.4)$$

where the $\lambda_{i,t}$ regulates the uncertainty regarding the state (i.e. time-varying coefficient) evolution, and it is typically set slightly below 1. (Values close to 1 lead to a more stable model.) Therefore, we can write the following (see Equation 5.5):

$$\sigma_{i,t,\xi}^2 = \left(1 - \frac{1}{\lambda_{i,t}}\right) P_{i,t-1|t-1} \quad (5.5)$$

We also follow Koop & Korobilis (2012); Baxa *et al.* (2017) and estimate the forgetting factor $\lambda_{i,t}$ as a time-varying parameter. This forgetting factor used in approximation regarding the prediction error variance $\lambda_{i,t}$ is set to be a time-varying parameter between 0.9 and 1. More specifically, we estimate $\lambda_{i,t}$ in following way (see Equation 5.6):

$$\lambda_{i,t} = \lambda_{min} + (1 - \lambda_{min}) L^{f_{i,t}} \quad (5.6)$$

λ_{min} denotes the minimum value of the forgetting factor, and L defines the sensitivity of the coefficients' variation to prediction errors. We set $\lambda_{min} = 0.9$ and $L = 1, 1$ and we further define f_t as one-step-ahead prediction error squared. $\sigma_{i,t+h,\varepsilon}^2$ is then obtained using an exponentially weighted moving average estimate, which can be written as follows (see Equation 5.7):

$$\sigma_{i,t+h,\varepsilon}^2 = \kappa \sigma_{i,t+h-1,\varepsilon}^2 + (1 - \kappa) \varepsilon_{i,t+h}^2 \quad (5.7)$$

where κ denotes another decay factor used in the DMA framework, which is set to a value of 0.96. However, the results were generally robust to different specifications between 0.9-0.98. We also use diffuse priors to initialize the Kalman filter recursion. Another set of approximations is to avoid specifying the transition matrix between different model specifications. The simplification is based on replacing the traditional transition matrix by introducing another forgetting factor λ , which controls the 'speed of model-switching'. The following approximation replaces the model prediction equation (see Equation 5.8):

$$\pi_{i,t|t-1,k} = \frac{\pi_{i,t-1|t-1,k}^\alpha}{\sum_{l=1}^M \pi_{i,t-1|t-1,l}^\alpha} \quad (5.8)$$

where $\pi_{i,t|t-1,k}$ denotes the probability of model k being the correct one at time t . We also set the forgetting factor λ in the approximation to the value 0.95¹⁹, and M is the number of competing models. In other words, we estimate a set of M models, which are characterized by having different subsets of predictors. There are $2^K - 1$ possible models to construct, where K denotes all possible explanatory variables (including constant term). Note, however, that

¹⁸For ease of exposition, we restrict our attention only to the relevant Kalman filter formulas.

¹⁹The results were generally robust to different specifications.

in some of the models considered in the DMA framework, the economic theories of exchange rate determination may occur in some combinations, but other models will not correspond to such concepts. We could easily restrict the models considered and allow a priori only a subset of these models based on economic theories. We, however, prefer to rely on a more agnostic approach regarding the predictor variables and include all possible combinations. Alternatively, we can consider these (non-stationary) variables a proxy for drift/trend.²⁰

In our forecasting exercise, we thus evaluate all possible combinations of explanatory variables. Updated model probabilities can then be written as follows (see Equation 5.9):

$$\pi_{i,t|k}^{t-1} = \frac{p_k(\Delta_{t+h}e_{i,t}|\Delta_{t+h}e_i^{t-1})\pi_{i,t|t-1,k}^\alpha}{\sum_{l=1}^{l=M} p_l(\Delta_{t+h}e_{i,t}|\Delta_{t+h}e_i^{t-1})\pi_{i,t|t-1,l}^\alpha} \quad (5.9)$$

where $p_l(\Delta_{t+h}e_{i,t}|\Delta_{t+h}e_i^{t-1})$ denote the predictive density for model l regarding currency i obtained by the Kalman filter. We set $\pi_{i,0|0,k}$ to $1/M$, which can be considered un-informative prior. A significant advantage of this approach is that, due to the use of a set of approximations, it is quite easy to calculate the posterior inclusion probabilities on individual models, i.e., all possible combinations of explanatory variables. The out-of-sample forecasting is done by employing the prediction equation from the Kalman filter and the posterior model probabilities, which are lagged by one period to avoid 'data snooping'. For more technical details regarding dynamic model averaging, see Koop & Korobilis (2012), Raftery *et al.* (2010), or Baxa *et al.* (2017).

The predictive accuracy was assessed through several methods. Firstly, we compared the root-mean-square forecast error (RMSFE) of the DMA model to that of a random walk benchmark, where the random walk without drift generates forecasts according to: $\hat{e}_{t+h} = e_t$. The model is considered superior to the random walk benchmark if the RMSFE ratio is less than one. Additionally, we utilized the mean absolute forecast error (MAFE) as a comparison metric. MAFE is included because it assigns less weight to large forecast errors compared to RMSFE. The interpretation of the MAFE ratio is analogous to that of the RMSFE ratio. In the robustness check section, we replace the random walk benchmark with an alternative benchmark based on OLS regression.

We also utilized the test statistics (DMW) proposed by Diebold & Mariano (2002) for a more formal evaluation. This test examines the null hypothesis that the DMA model's root-mean-square forecast error (RMSFE) is equal to that of the random walk (or alternative OLS) benchmark. The alternative hypothesis posits that the DMA model outperforms the benchmark model. Additionally, we employed the test statistics (CW) developed by Clark & West (2007).²¹ Clark & West (2007) demonstrate that the asymptotic distributions of the sample difference between two RMSFE ratios are not identical and are biased downward from zero. This adjustment accounts for the fact that when comparing two models, where one model is nested within the other, the nested model has fewer parameters to estimate, leading to reduced estimation error. However, it is important to note that our recursive approach does not fully satisfy all the assumptions required by these tests.

²⁰A related point is that some of the variables considered are non-stationary in levels. This is not a problem when considering the usual error-correction model specification but may be problematic because the cointegrating relationships are not present in all combinations in the DMA framework. For this reason, in the empirical part, we also focus on the stationarity of residuals.

²¹We employ HAC estimator to calculate the variance

5.5 Empirical Results

The results of our forecasting exercise are displayed in Table 5.1. The table is divided into sections for each nominal currency pair.²² The first row shows the ratio of RMSFE of the obtained forecast and the random walk benchmark in different time horizons. The next row of the table displays the same ratio for MAFE. The table then displays the Diebold Mariano and Clark West statistics in the next two rows. We report the corresponding p-value of the DMW and CW test statistics of the one-sided test with the alternative hypothesis that a given model performs better than the random walk. In the last row, we display the number of forecast periods. The individual columns (Mx) denote the forecast results for the x month ahead.

Our main findings are as follows: The DMA approach clearly defeats the random walk benchmark based on the RMSFE and MAFE ratio in the medium and long-term horizon. This result is generally robust as it applies to all exchange rate pairs. Regarding the period of the first six months, the random walk benchmark and DMA forecasts are approximately equal in terms of forecasting. There are, of course, some differences in forecasting power regarding different currencies. This can be summarized that the various estimated models are more or less successful depending on the forecast horizon and exchange rate chosen. Generally, the overall results support our claim of significant nominal exchange rate predictability in the medium and long run.

In more detail, we observe that no model can beat the random walk at the one-month horizon. The same applies for the three-month horizon with the exception of the AUD/USD and NZD/USD currency pairs. The results regarding the short-term forecasts are in accordance with the existing literature, where our results support the claim that it is very difficult to find forecasts from a model that can consistently beat the random walk benchmark using standard RMSFE and MAFE criteria. However, as mentioned before, the forecasting results according to the RMSFE and MAFE ratios do not differ much from the random walk benchmark. In the period over six months, according to the RMSFE and MAFE ratios and CW statistics, the DMA approach beat the random walk benchmark for all currencies. Also, according to DMW statistics, the DMA approach is statically distinguishable from the random walk benchmark in most cases. For example, in 12 months period, except for the CHF/USD currency pair, DMA provides better predictions according to DMW statistics at the five percent significance level.

Furthermore, we find strong predictability in the long-term horizon, in accordance with Rossi (2013), specifically at two-year and three-year forecast horizons. This can be illustrated by inspecting the RMSFE and MAFE ratios, which appear to be declining, i.e., there is growing predictability with an increasing time horizon. Therefore, the DMA predictions in this forecasting horizon produce significantly better forecasts than the random walk benchmark according to RMSFE and MAFE ratios. The results are further confirmed by CW test statistics, where the CW test in the majority of cases distinguishes (at a five percent significance level) between the DMA forecast and random walk benchmark. The results are, however, a bit more mixed according to DMW statistics, but in a majority of cases, still within a five percent significance level. For comparison purposes, the long-term forecast results are comparable, and in some cases better (three-year forecast horizon), than the nominal exchange rate forecast conducted in most studies published in the literature (Ca'Zorzi & Rubaszek, 2020; Rubaszek *et al.*, 2022). This suggests that the DMA delivers outstanding forecasting performance in the very long run.

Figure B1 in the Appendix provides further intuition on the results obtained.²³ It is generally difficult to assess the quality of the forecast through the chart. For this reason, we consider

²²The table has the same structure in every section.

²³The obtained out-of-sample forecasts were transformed back to levels to provide further intuition on results.

Figure B1 more a supplement to Table 5.1 as this chart allows us to access the forecast quality in different time periods. However, we can observe that the forecast obtained consistently beat the random walk in the long run. And more importantly, we can observe that the prediction accuracy is approximately constant over the period considered.

How to interpret the results obtained, given that the nominal exchange rate seems predictable contrary to much of the forecasting literature? We offer several possible explanations. One possible explanation is that by using a large number of variables, we implicitly identify the cointegration vector and the nominal exchange rate converges to the equilibrium value defined by this vector. However, this result contrasts with the theoretical finding by Engel & West (2005) because the explanatory variables are highly persistent, and the discount factor is probably close to one. An alternative explanation can be based on the 'scapegoat theory of exchange rates' or on the claim that the forecasts obtained are, in reality, based on the stationarity and predictability of risk premia, which is indirectly estimated by the fundamental factors (explanatory variables). Another explanation is that the (nominal) exchange rates are mean-reverting. This reasoning aligns with Ca'Zorzi & Rubaszek (2020), where authors present a similar argument for real exchange rates. It seems reasonable that even the nominal exchange rates can be considered stationary in the period examined, which is characterized by mostly low inflation rates and central banks targeting an explicitly set inflation target.²⁴ It should also be noted that these possible explanations are not mutually exclusive. The nominal exchange rate may be stationary, but other variables may help determine its (fundamental) value.

²⁴The implicit argument here is that both countries have the same inflation target. At the same time, due to the interconnectedness, inflationary pressures spill over between respective countries, and the central banks react similarly.

Table 5.1: Out of Sample Forecast Evaluation - Random Walk Benchmark

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	0.98	0.89	0.82	0.79	0.62	0.54
MAFE_M/MAFE_RW	1.00	0.98	0.92	0.90	0.86	0.79	0.72
DMW	0.70	0.37	0.01	0.00	0.01	0.00	0.00
CW	0.09	0.01	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	0.99	0.95	0.90	0.88	0.70	0.63
MAFE_M/MAFE_RW	1.00	1.00	0.97	0.95	0.92	0.84	0.80
DMW	0.89	0.39	0.12	0.01	0.03	0.04	0.02
CW	0.60	0.03	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.03	1.10	1.06	0.92	0.78	0.57	0.51
MAFE_M/MAFE_RW	1.01	1.04	1.00	0.96	0.90	0.77	0.70
DMW	0.97	0.99	0.76	0.20	0.06	0.08	0.13
CW	0.91	0.94	0.19	0.01	0.01	0.01	0.03
N	179	177	174	171	168	156	144
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.04	1.04	0.87	0.81	0.69	0.60	0.70
MAFE_M/MAFE_RW	1.01	1.01	0.92	0.88	0.83	0.76	0.81
DMW	0.98	0.80	0.04	0.04	0.01	0.03	0.12
CW	0.95	0.21	0.00	0.00	0.00	0.01	0.05
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.02	1.01	0.90	0.79	0.71	0.58	0.56
MAFE_M/MAFE_RW	1.01	1.00	0.94	0.90	0.86	0.78	0.76
DMW	0.96	0.57	0.08	0.05	0.03	0.04	0.01
CW	0.81	0.06	0.00	0.00	0.00	0.01	0.01
N	231	229	226	223	220	208	196
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.04	1.03	0.90	0.78	0.71	0.50	0.37
MAFE_M/MAFE_RW	1.02	1.03	0.96	0.88	0.83	0.69	0.61
DMW	0.96	0.73	0.09	0.03	0.01	0.02	0.02
CW	0.67	0.14	0.00	0.00	0.00	0.00	0.00
N	194	192	189	186	183	171	159
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.02	1.03	0.96	0.88	0.84	0.71	0.66
MAFE_M/MAFE_RW	1.01	1.00	0.96	0.91	0.91	0.86	0.81
DMW	0.97	0.78	0.27	0.04	0.07	0.03	0.09
CW	0.74	0.13	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	0.97	0.92	0.83	0.75	0.76	
MAFE_M/MAFE_RW	1.01	1.00	1.00	0.95	0.91	0.82	0.87
DMW	0.86	0.58	0.21	0.02	0.04	0.04	0.03
CW	0.51	0.03	0.00	0.00	0.00	0.00	0.00
N	222	220	217	214	211	199	187
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.04	1.04	0.91	0.86	0.77	0.61	0.62
MAFE_M/MAFE_RW	1.01	1.01	0.95	0.91	0.88	0.75	0.78
DMW	0.92	0.78	0.01	0.00	0.01	0.03	0.05
CW	0.71	0.25	0.00	0.00	0.00	0.00	0.01
N	180	178	175	172	169	150	140

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) denote the results for the x month ahead forecast. RMSFE_M/RMSFE_RW denote the ratio of root mean squared forecast error of the DMA framework and random walk benchmark. MAFE_M/MAFE_RW is the ratio of root mean absolute forecast error of the DMA framework and random walk benchmark. DMW and CW denote p-values of Diebold Mariano and Clark West statistics. N denotes the number of estimated forecasts. DMA framework was used for the estimation.

Furthermore, we believe the second important modelling decision regarding the results obtained, which show significant nominal exchange rate predictability in the medium and long-term horizon, is the use of direct forecast. This is due to the fact that the exchange rate dynamics is obscured by high-frequency 'non-fundamental' noise that is hard to predict. If a model successfully filters out this noise, it may prove to be a good predictor in the medium to long run precisely because it does not target unpredictable high-frequency fluctuations. This is why we prefer to use direct forecast, where the prediction errors do not accumulate. This is because, in the case of the recursive forecast, a small perturbation in one period forecast can cause a significant change regarding prediction for several periods.

Which of the proposed explanations for exchange rate predictability can be considered the most likely? The dynamic model averaging (DMA) approach provides us with posterior model probability, allowing us to access the importance of individual variables, which can help us identify the reason for exchange rate predictability. That being said, it can be reasonably expected that the variables selected the most (with the highest posterior inclusion probability²⁵) are the reason behind the exchange rate predictability. Table 5.2 displays the posterior probability of inclusion of the given explanatory variable. Given that probability is time-varying, we show the median value over the whole horizon (not including the training period). The results also indicate fast model switching (not shown). Table 5.2 indicates that the variable with the highest posterior probability of being included is the nominal exchange rate value. The probability of including any other explanatory variable is significantly lower. The probability of including a nominal exchange rate in estimation is highest in the medium and long run, i.e., in periods where the model performs the best²⁶. The result support the claim that nominal exchange rates are mean reverting in the period under consideration. On the other hand, this in itself does not mean that the lagged exchange rate is the only and best predictor for predicting changes in exchange rates. It follows from the definition of the co-integration vector, i.e., if we were able to identify the deviation from fundamental value, that even in this case, the exchange rate should be included. Therefore, most economically reasonable specifications include a lagged exchange rate as an explanatory variable.

Interestingly, explanatory variables that act as a proxy for money also have a fairly high probability of being included in the estimates, especially in the medium and long-term horizon. This suggests that money (loans to corporations), which act as a proxy for corporate deposits, could also play some role in determining the nominal exchange rate. This indicates some validity of the monetary model, i.e., either through the determination of the fundamental value or through the indirect estimation of the risk premia or as a 'scapegoat' variable.²⁷ Price levels, interest rates, and unemployment rates have somewhat lower inclusion probability than explanatory variables that act as a proxy for money. Regarding price levels, a possible explanation is the stable rate of inflation in the period and countries examined. Regarding unemployment rates and interest rates, these results suggest the invalidity of forecasting exchange rates based strictly on monetary policy rules or uncovered interest rate parity. On the other hand, we remind the readers that we work only with the data in (log) levels; a possible extension is to include growth rates or gap estimates.

²⁵Posterior inclusion probabilities for individual variables are based on posterior model probabilities that contain the given variable.

²⁶However, the inclusion probability of fundamental variables is generally slightly increasing with the time horizon.

²⁷From a practical point of view, we offer an explanation that corporate deposits are the main component of money that affects the nominal exchange rate because they have the highest chance of entering international transactions. For this reason, we are able to filter out 'noise' commonly included in monetary aggregates.

Table 5.2: Posterior Inclusion Probability

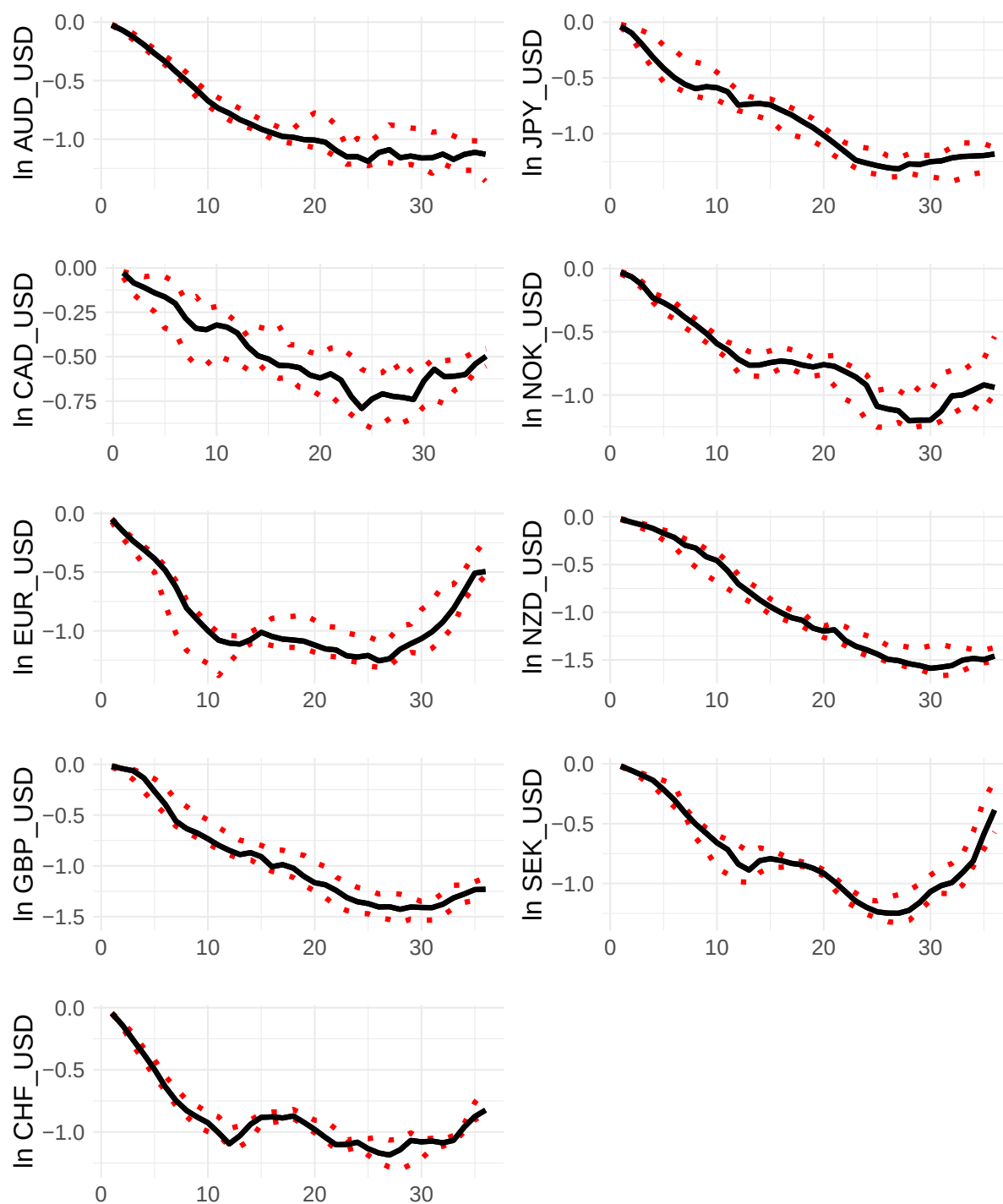
AUD/USD	M1	M3	M6	M9	M12	M24	M36
AUD/USD	0.53	0.73	0.97	1.00	1.00	1.00	1.00
US money	0.44	0.52	0.76	0.85	0.87	0.89	0.55
AUD money	0.48	0.57	0.68	0.79	0.74	0.73	0.78
US unemployment	0.48	0.47	0.49	0.55	0.49	0.48	0.35
AUD unemployment	0.47	0.50	0.47	0.43	0.41	0.48	0.46
US price level	0.44	0.48	0.54	0.51	0.49	0.35	0.44
AUD price level	0.43	0.50	0.53	0.46	0.46	0.41	0.35
US interest rate	0.46	0.44	0.45	0.53	0.46	0.22	0.48
CA interest rate	0.45	0.45	0.52	0.56	0.40	0.50	0.71
constant	0.45	0.48	0.54	0.52	0.48	0.44	0.42
CAD/USD	M1	M3	M6	M9	M12	M24	M36
CAD/USD	0.53	0.72	0.86	0.92	0.94	0.97	0.98
US money	0.46	0.49	0.52	0.51	0.48	0.41	0.31
CA money	0.48	0.56	0.52	0.54	0.47	0.30	0.31
US unemployment	0.44	0.50	0.49	0.50	0.50	0.43	0.38
CA unemployment	0.44	0.47	0.47	0.47	0.37	0.31	0.38
US price level	0.48	0.52	0.54	0.54	0.46	0.47	0.42
CA price level	0.47	0.48	0.49	0.46	0.44	0.46	0.42
US interest rate	0.46	0.49	0.40	0.33	0.28	0.13	0.15
CA interest rate	0.47	0.46	0.48	0.45	0.51	0.45	0.46
constant	0.42	0.51	0.53	0.54	0.54	0.57	0.66
CHF/USD	M1	M3	M6	M9	M12	M24	M36
CHF/USD	0.58	0.86	0.97	1.00	1.00	1.00	0.99
US money	0.44	0.43	0.53	0.54	0.56	0.70	0.40
CH money	0.44	0.48	0.50	0.46	0.35	0.59	0.46
US unemployment	0.33	0.44	0.50	0.50	0.60	0.64	0.40
CH unemployment	0.45	0.49	0.40	0.41	0.40	0.44	0.38
US price level	0.45	0.52	0.71	0.82	0.91	0.53	0.37
CH price level	0.45	0.44	0.50	0.37	0.39	0.49	0.61
US interest rate	0.42	0.37	0.22	0.22	0.21	0.66	0.28
CH interest rate	0.42	0.43	0.44	0.43	0.39	0.57	0.33
constant	0.46	0.55	0.65	0.76	0.88	0.60	0.84
EUR/USD	M1	M3	M6	M9	M12	M24	M36
EUR/USD	0.55	0.84	0.96	1.00	1.00	1.00	0.69
US money	0.37	0.40	0.44	0.47	0.36	0.61	0.49
EU money	0.36	0.45	0.68	0.67	0.66	0.70	0.40
US unemployment	0.27	0.45	0.52	0.51	0.50	0.44	0.70
EU unemployment	0.39	0.38	0.39	0.32	0.44	0.27	0.38
US price level	0.40	0.46	0.52	0.59	0.53	0.43	0.41
EU price level	0.40	0.45	0.49	0.50	0.50	0.44	0.40
US interest rate	0.37	0.38	0.48	0.55	0.65	0.91	0.22
EU interest rate	0.36	0.38	0.43	0.30	0.39	0.20	0.28
constant	0.43	0.45	0.42	0.40	0.38	0.38	0.49
GBP/USD	M1	M3	M6	M9	M12	M24	M36
GBP/USD	0.50	0.70	0.89	0.98	0.99	1.00	1.00
US money	0.44	0.60	0.85	0.93	0.94	0.91	0.50
GB money	0.39	0.49	0.38	0.36	0.39	0.39	0.41
US unemployment	0.44	0.43	0.32	0.38	0.38	0.44	0.53
GB unemployment	0.40	0.37	0.26	0.21	0.32	0.43	0.57
US price level	0.42	0.47	0.44	0.38	0.67	0.60	0.59
GB price level	0.42	0.51	0.44	0.41	0.46	0.51	0.59
US interest rate	0.43	0.43	0.41	0.40	0.37	0.41	0.44
GP interest rate	0.43	0.43	0.40	0.37	0.39	0.40	0.38
constant	0.42	0.47	0.44	0.44	0.46	0.44	0.50
JPY/USD	M1	M3	M6	M9	M12	M24	M36
JPY/USD	0.48	0.83	0.97	0.95	0.98	1.00	1.00
US money	0.41	0.38	0.40	0.45	0.44	0.30	0.25
JPN money	0.41	0.46	0.48	0.52	0.61	0.54	0.72
US unemployment	0.39	0.38	0.35	0.34	0.32	0.40	0.37
JPN unemployment	0.39	0.35	0.25	0.28	0.29	0.45	0.45
US price level	0.42	0.41	0.46	0.41	0.43	0.38	0.38
JPN price level	0.43	0.44	0.46	0.44	0.41	0.47	0.52
US interest rate	0.42	0.35	0.25	0.31	0.30	0.37	0.46
JPN interest rate	0.42	0.41	0.49	0.74	0.45	0.75	0.52
constant	0.43	0.48	0.50	0.46	0.49	0.55	0.55
NOK/USD	M1	M3	M6	M9	M12	M24	M36
NOK/USD	0.51	0.72	0.90	0.97	0.98	0.98	0.82
US money	0.40	0.51	0.70	0.79	0.89	0.41	0.35
NO money	0.44	0.49	0.59	0.56	0.60	0.55	0.34
US unemployment	0.39	0.35	0.33	0.34	0.40	0.42	0.64
NO unemployment	0.45	0.41	0.33	0.41	0.38	0.42	0.45
US price level	0.44	0.46	0.46	0.39	0.46	0.53	0.36
NO price level	0.43	0.46	0.44	0.44	0.47	0.38	0.40
US interest rate	0.43	0.43	0.31	0.39	0.39	0.43	0.21
NO interest rate	0.43	0.43	0.31	0.39	0.39	0.43	0.21
constant	0.44	0.46	0.51	0.45	0.40	0.48	0.67
NZD/USD	M1	M3	M6	M9	M12	M24	M36
NZD/USD	0.53	0.70	0.85	0.96	0.99	1.00	1.00
US money	0.43	0.53	0.44	0.61	0.65	0.81	0.63
NZD money	0.45	0.45	0.45	0.42	0.44	0.46	0.54
US unemployment	0.42	0.46	0.42	0.42	0.40	0.55	0.58
NZD unemployment	0.44	0.44	0.40	0.39	0.35	0.62	0.48
US price level	0.45	0.51	0.54	0.50	0.61	0.53	0.46
NZD price level	0.45	0.51	0.45	0.45	0.57	0.43	0.47
US interest rate	0.42	0.46	0.44	0.52	0.81	0.73	0.53
NZD interest rate	0.46	0.46	0.46	0.38	0.42	0.61	0.52
constant	0.45	0.51	0.46	0.43	0.49	0.85	0.84
SEK/USD	M1	M3	M6	M9	M12	M24	M36
SEK/USD	0.48	0.65	0.83	0.98	0.99	1.00	0.66
US money	0.39	0.42	0.65	0.60	0.51	0.49	0.41
SWE money	0.42	0.48	0.48	0.43	0.52	0.45	0.45
US unemployment	0.39	0.41	0.46	0.57	0.52	0.42	0.53
SWE unemployment	0.44	0.43	0.43	0.47	0.47	0.43	0.35
US price level	0.43	0.50	0.53	0.48	0.58	0.45	0.50
SWE price level	0.41	0.48	0.47	0.48	0.52	0.53	0.51
US interest rate	0.38	0.38	0.46	0.48	0.58	0.50	0.32
SWE interest rate	0.37	0.42	0.44	0.41	0.37	0.38	0.42
constant	0.41	0.41	0.46	0.50	0.77	0.58	0.53

Note: The table displays variable inclusion probability for different explanatory variables. The individual columns (Mx) denote the results for the x month ahead forecast. DMA framework was used for the estimation.

We also expect that the coefficients regarding explanatory variables that are most stable over time are most likely to show a true relation. Gradual change within an (economic or financial) cycle is understandable, but coefficients characterized by high variability probably capture mostly noise than anything meaningful. Alternatively, the possible explanation of parameter instability is offered by the 'scapegoat theory of exchange rates', i.e., the existence of significant variation in the parameters is due to the fact that these variables act as 'scapegoats' and the true relationship is non-existent. For this reason, we also look at the stability of the regression coefficients. Given the fact that we use a large number of explanatory variables, the time-varying nature of these parameters, and the general number of models estimated, we present the results as charts that display the median value and 25%–75% quantiles of the estimated regression coefficients. In more detail, the solid line represents the median value, and the dashed lines represent the 25%–75% quantiles.

Firstly, we focus on the parameter related to the (lagged) value of the nominal exchange rate. This is because, as mentioned before, it is the most likely for this explanatory variable to be included in our forecasts. Figure 5.1 displays the regression coefficients regarding the spot nominal exchange rate level for different forecasting horizons. The coefficients are clearly decreasing with the increasing time horizon of the forecast. These results suggest that there exists a positive auto-correlation in nominal exchange rates, which, however, can turn slightly negative for certain currencies (and only in the long run). It can be observed that the values of this coefficient in most cases do not differ much from the value implied by nominal exchange rates behaving according to AR(1) process. To see this fact, equation 1 can be rewritten as follows: $e_{i,t+h} = A_i + (1 + B_i)e_{i,t} + C_i f_{i,t} + \varepsilon_{i,t+h}$. Furthermore, given that the value of these coefficients for the forecast for one period ahead is close to 0 in all cases, a recursive forecast implies very different long-run forecast values given small perturbations in the parameter estimate. Also, although there exists some variability/instability in these parameter estimates, we consider obtained results encouraging, as parameters generally do not change dramatically. Also, if we inspect the time-varying nature of this parameter (not shown), the movement generally seems to be slow-moving. This could be reflecting some slow movements in the underlying economic process.

Figure 5.1: Regression Coefficient - Nominal Exchange Rate



Note: The graph shows the median of the regression coefficient regarding the nominal exchange rate level for different h periods ahead forecast. Dashed lines represent the interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

For other variables, see Figures B2, B3, B4, B5, B6, B7, B8, B9, B10 in the Appendix which display the regression coefficients regarding these variables. In general, these coefficients exhibit a much higher degree of instability compared to the coefficient for the (lagged) value of the nominal exchange rate. In many cases, the range of 25%–75% quantiles of these coefficients includes the zero value. This is especially true for short-term forecasts, highlighting the instability of these coefficients. Even when the median values of these coefficients for fundamentals are different from zero, the quantiles indicate high instability in the estimates. Furthermore, the coefficients for the same variable often differ across countries, i.e., the strength of the link between exchange rates and fundamentals varies across currencies. In other words, the relationship between fundamentals and exchange rates is not stable over time and often varies across currencies. As the forecast horizon lengthens, the coefficients (both median and 25%–75% quantiles) are more often different from zero, suggesting that these variables can have additional explanatory power in certain forecasting horizons.

In more detail, the coefficients regarding the price levels seem to be very close to zero in the short and medium run, and we again observe the issue that the coefficient signs of these variables indicate different relationships for different currencies. In the long run, they are distinguishable from zero and mostly have the correct sign. This suggests a very slow pass-through of price levels to the nominal exchange rate and, therefore, some validity of the PPP theory. In the case of the coefficients regarding the credit to corporations (a proxy for money), we can observe that these variables have mostly non-zero values, especially in the medium and long run. The issue is that the coefficient signs of these variables indicate different relationships for different currencies, and in many cases, the sign of these coefficients is opposite to what the theory implies. As explained in the methodology section, we expect the coefficients m_{US} and p_{US} to be positive and the coefficients $m_{domestic}$ and $p_{domestic}$ to be negative. This generally follows from the purchasing power parity (PPP) relation.

Furthermore, the coefficients on the unemployment rate of the non-US country appear to fluctuate between positive and negative signs²⁸, but US unemployment appears to be a significant predictor as it mostly differs from zero, indicating that this variable can function as a proxy for the risk premium.²⁹ The coefficients regarding interest rates have the same issue regarding their signs as they again often fluctuate between negative and positive values. Also, the size of these coefficients is, in practical terms, quite negligible, which indicates that they do not significantly affect the forecast. Finally, the changing constant term in some cases implies that there may be some underlying (medium or long-term) process in the economy in which the exchange rate depreciates or appreciates systematically, such as financial deregulation. This seems to be the case for CHF/USD, NZD/USD, and SEK/USD.

In summary, the parameter instability suggests that there is no stable cointegration relation in the data. However, these variables can still have forecasting power in certain periods. This may be because they act as 'scapegoats' and/or serve as proxies for changing risk premiums. The difficulty of selecting the best predictive model can be due to frequent shifts in the set of fundamentals driving exchange rates, reflecting swings in market expectations over time (Sarno & Valente, 2009).

It is important to note that, given the use of an overlapping data and the potential nonsta-

²⁸We again generally expect the coefficients un_{US} and ir_{US} to be positive and the coefficients $un_{domestic}$ and $ir_{domestic}$ to be negative, following uncovered interest rate parity and monetary policy rules. However, the issue regarding the correct signs is more complicated, as an increase in the interest rate should cause an immediate appreciation of the dollar, followed by forecasted (and actual) depreciation.

²⁹US unemployment can serve as a proxy for global economic conditions. As such, it may be a significant predictor of the exchange rate not only indirectly through the uncovered interest rate parity but also directly, as it serves as a proxy for global risk aversion and phenomena such as flight to quality.

tionarity of many explanatory variables, the results must be interpreted with caution. Their reliability hinges on either the stationarity of the explanatory variables or the presence of cointegration. Failure to meet these conditions, combined with the large overlapping period, can induce positive autocorrelations in both the dependent and explanatory variables, potentially leading to spurious regression issues (Berkowitz & Giorgianni, 2001; Kilian, 1999). To assess the robustness of our findings, we examine the stationarity of residuals. Table B1 presents the results of the Augmented Dickey-Fuller (ADF) test applied to residuals, reporting the median p-values from all estimated DMA models in the out-of-sample exercise. The results indicate that most residuals can be considered stationary at the five percent significance level, which we find promising given the use of overlapping differences that introduce autocorrelation up to order 36. Additionally, we employed the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test as a further robustness check. The median p-values from all the estimated models are reported in Table B2. In contrast to the ADF results, the KPSS test yields more mixed results, with residuals generally failing to be considered stationary. On the other hand, it should be noted that DMA in our approach estimates all possible models³⁰, even the ones that make no sense from a theoretical perspective.

Finally, to further support our result regarding significant mean reversion in exchange rates. We estimate the DMA model using only a subset of explanatory variables. More specifically, the only included explanatory variables are the lagged nominal exchange rate, the constant term, and price levels. The price levels are included to account for possible mean reversion in real exchange rates (Ca’Zorzi & Rubaszek, 2020).

The results of this DMA model with the subset of explanatory variables are shown in Table 5.3. The table has the same structure as before and is again divided into sections for each nominal currency pair. The main result is that the model has similar forecasting power in the short run, but in the medium and especially in the long run, it provides significantly worse results for most of the currencies than the full DMA model using all of the variables. In other words, the smaller model has somewhat worse forecasting power. This contrasts with the fact that the posterior inclusion probabilities suggest that the nominal exchange rate forecast could be done with only the lagged value of the nominal exchange rate as an explanatory variable (along with the constant term). Given these results and also taking into account the instability of parameters regarding fundamental variables, all of this suggests some validity of the ‘scapegoat’ theory of exchange rates, which offers a reconciliation of these findings. The parameters are generally unstable, and the preferred model often changes. However, these variables still have forecasting power in certain periods. The results, therefore, indicate that the inclusion of additional fundamental variables can further exchange rate predictions, which may act as a ‘scapegoat’. The DMA approach is then valuable for its ability to adjust to changes in the fundamentals driving exchange rates, accommodating the dynamic nature of the market.

³⁰We used all possible combinations of explanatory variables. There are $2^K - 1$ possible models to construct, where K denotes all possible explanatory variables (including constant term).

Table 5.3: Out of Sample Forecast Evaluation - Different DMA Specification

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.00	0.99	0.98	0.89	0.80	0.77
MAFE_M/MAFE_RW	1.00	0.99	0.98	0.98	0.93	0.90	0.90
DMW	0.70	0.49	0.38	0.11	0.09	0.09	0.00
CW	0.26	0.05	0.01	0.00	0.00	0.01	0.00
N	239	237	234	231	228	216	204
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.00	0.98	0.93	0.92	0.77	0.66
MAFE_M/MAFE_RW	1.00	1.01	0.98	0.97	0.94	0.88	0.82
DMW	0.83	0.54	0.35	0.02	0.01	0.07	0.02
CW	0.61	0.09	0.02	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.04	0.99	0.97	0.95	0.76	0.54
MAFE_M/MAFE_RW	1.00	1.02	0.99	0.99	0.98	0.86	0.75
DMW	0.93	0.83	0.37	0.25	0.16	0.11	0.14
CW	0.89	0.73	0.11	0.04	0.03	0.01	0.04
N	179	177	174	171	168	156	144
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.00	0.96	0.95	0.97	0.89	0.78
MAFE_M/MAFE_RW	1.00	0.99	0.96	0.98	1.00	0.94	0.90
DMW	0.72	0.50	0.10	0.16	0.30	0.16	0.20
CW	0.48	0.18	0.01	0.00	0.01	0.06	0.08
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.00	0.98	0.96	0.92	0.62	0.72
MAFE_M/MAFE_RW	1.00	1.00	0.99	0.98	0.96	0.78	0.87
DMW	0.98	0.74	0.10	0.05	0.02	0.04	0.04
CW	0.97	0.48	0.02	0.01	0.00	0.01	0.01
CW	0.81	0.06	0.00	0.00	0.00	0.01	0.01
N	231	229	226	223	220	208	196
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.02	1.03	0.98	0.92	0.83	0.62	0.51
MAFE_M/MAFE_RW	1.01	1.02	0.99	0.95	0.90	0.75	0.70
DMW	0.97	0.91	0.25	0.03	0.00	0.01	0.01
CW	0.87	0.57	0.01	0.00	0.00	0.00	0.00
N	194	192	189	186	183	171	159
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.03	1.02	0.97	0.94	0.79	0.69
MAFE_M/MAFE_RW	1.01	1.02	0.99	0.97	0.94	0.91	0.83
DMW	0.99	0.87	0.64	0.23	0.13	0.11	0.12
CW	0.97	0.57	0.25	0.02	0.00	0.00	0.00
N	239	237	234	231	228	216	204
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.00	1.01	1.01	1.00	1.01	1.10	1.06
MAFE_M/MAFE_RW	1.00	1.00	1.01	1.01	0.99	1.02	1.01
DMW	0.81	0.68	0.69	0.48	0.55	0.97	0.66
CW	0.64	0.20	0.06	0.03	0.02	0.00	0.27
N	222	220	217	214	211	199	187
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_RW	1.01	1.01	0.98	0.96	0.91	0.80	0.67
MAFE_M/MAFE_RW	1.00	1.00	0.99	0.97	0.95	0.89	0.84
DMW	0.85	0.60	0.29	0.19	0.08	0.13	0.06
CW	0.70	0.22	0.01	0.00	0.00	0.00	0.00
N	180	178	175	172	169	150	140

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) represent the results for the x month ahead forecast. RMSFE_M/RMSFE_RW is the ratio of root mean squared forecast error of the DMA framework and random walk benchmark. MAFE_M/MAFE_RW represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation and we used only subset of variables.

5.6 Robustness Check

In this section, we further build upon our results presented in the previous section that exchange rates can be considered mean-reverting and lagged exchange rates have significant forecasting power. This line of reasoning generally aligns with research by Ca'Zorzi & Rubaszek (2020); Engel & Wu (2023), where authors present similar conclusions. We thus further test this hypothesis by comparing the forecasts from the DMA model to benchmarks other than the random walk without drift.

The alternative benchmark model uses a simple direct multi-step forecast (local projection equation) in which the change in the nominal exchange rate from t to $t + h$ is a function of a constant and the nominal exchange rate at time t . Therefore, an h period-ahead forecast is again generated from a different model for every forecasting horizon. The alternative benchmark model is primarily estimated using a standard OLS method. Therefore, the results indicate to what extent additional fundamentals other than the exchange rate help forecast exchange rates. In other words, this allows us to assess if the DMA approach delivers value-added compared to the above simple model. For comparability, we once more use the first 72 periods as the training period. The forecast horizon is again the same as above (one month up to 36 months). However, we use a rolling window estimation approach, where, for each period, the actual realizations of the time series are added as an additional data point at the end of the sample, while the earliest observations are removed. This ensures that the model's training period remains fixed in length throughout the estimation process.³¹ More specifically, we estimate the following model for every currency (see Equation 5.10):

$$\Delta_{t+h}e_{i,t} = \alpha_{i,t} + \beta_i e_{i,t} + \varepsilon_{i,t+h} \quad (5.10)$$

where α_i is the constant term, and β_i denotes the slope coefficient. Other notation is the same as in the DMA case. Also, note that for this alternative benchmark model, we do not assume any other explanatory variable except the nominal exchange rate. Furthermore, unlike in the DMA case, the parameters are not considered to be time-varying.³²

The results of the baseline robustness check are shown in Table 5.4. The table has the same structure as in the previous section and is again divided into sections for each nominal currency pair. The main results are as follows: The DMA approach generally delivers forecasts beating the OLS benchmark according to RMSFE and MAFE ratios in the medium and long run.³³ This is especially true in the long run, where the results are even, in most cases, confirmed by DMW statistics. However, in the short-term horizon (up to 6 months), the results of the DMA method relative to the alternative benchmark model are generally comparable. The analysis supports the conclusion that additional fundamentals other than the exchange rate do not help predict exchange rates in the short term from an out-of-sample forecasting perspective. In the medium-term horizon, the DMA contribution to forecasting power is generally modest. However, the DMA method has significantly more forecasting power in the long run than this alternative OLS benchmark, suggesting that additional variables have some forecasting power in these periods. In other words, this suggests that in the long run, fundamental factors have an effect on the exchange rate. From another point of view, the results also suggest that the alternative benchmark results are similar to the random walk benchmark regarding the short-term forecasting horizon. Furthermore, in the medium and long-term horizon, the simple OLS

³¹We also experimented with the recursive (expanding) window approach, but the results with the recursive were significantly worse than with the rolling window. In short, the rolling window approach provides better predictions relative to the recursive window regarding RMSFE and MAFE ratios.

³²However, the use of rolling window estimation in practical terms enables time-varying coefficients.

³³Also, CW test statistics confirm these results.

benchmark defeats the random walk benchmark in all cases.³⁴ Furthermore, we believe that this supplementary analysis further fosters our argument that direct forecasting greatly helps ensure predictability. In other words, we argue that for an exchange rate forecast capable of defeating a random walk benchmark in the short and medium run is sufficient to use lagged exchange rate values and also the use of a direct forecasting approach, which allows filtering out the non-fundamental noise in the short-term horizon.

Table 5.4: Out of Sample Forecast Evaluation - Different Benchmark Model 1

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.97	0.89	0.83	0.81	0.64	0.59
MAFE_M/MAFE_B	0.98	0.96	0.90	0.88	0.86	0.79	0.73
DMW	0.35	0.33	0.17	0.14	0.16	0.02	0.02
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.01
N	239	237	234	231	228	216	204
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.98	0.94	0.90	0.88	0.75	0.76
MAFE_M/MAFE_B	0.99	0.99	0.96	0.94	0.92	0.90	0.89
DMW	0.27	0.28	0.24	0.17	0.20	0.14	0.17
CW	0.00	0.00	0.00	0.00	0.00	0.01	0.04
N	239	237	234	231	228	216	204
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.09	1.07	0.95	0.82	0.77	0.79
MAFE_M/MAFE_B	1.00	1.03	1.00	0.95	0.90	0.89	0.84
DMW	0.74	0.97	0.71	0.35	0.10	0.12	0.12
CW	0.09	0.23	0.05	0.00	0.00	0.01	0.00
N	179	177	174	171	168	156	144
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.02	0.86	0.85	0.78	0.75	0.87
MAFE_M/MAFE_B	1.00	1.02	0.93	0.91	0.89	0.87	0.89
DMW	0.65	0.58	0.15	0.22	0.17	0.12	0.05
CW	0.15	0.02	0.01	0.01	0.01	0.03	0.04
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	1.00	0.93	0.83	0.76	0.65	0.62
MAFE_M/MAFE_B	1.00	1.01	0.97	0.92	0.88	0.79	0.78
DMW	0.35	0.49	0.17	0.07	0.05	0.01	0.05
CW	0.02	0.02	0.00	0.00	0.00	0.00	0.00
N	231	229	226	223	220	208	196
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.98	0.95	0.84	0.74	0.69	0.54	0.49
MAFE_M/MAFE_B	0.99	0.99	0.93	0.85	0.82	0.73	0.67
DMW	0.22	0.21	0.04	0.03	0.02	0.03	0.01
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	194	192	189	186	183	171	159
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.04	1.00	0.94	0.91	0.77	0.76
MAFE_M/MAFE_B	1.01	1.01	0.98	0.95	0.96	0.91	0.88
DMW	0.64	0.72	0.50	0.35	0.33	0.10	0.12
CW	0.08	0.07	0.01	0.02	0.01	0.01	0.01
N	239	237	234	231	228	216.00	204
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.03	1.05	1.03	0.96	0.92	0.89
MAFE_M/MAFE_B	1.01	1.01	1.00	0.98	0.95	0.88	0.91
DMW	0.64	0.65	0.68	0.57	0.38	0.22	0.07
CW	0.06	0.01	0.00	0.00	0.00	0.00	0.00
N	222	220	217	214	211	199	187
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.02	1.06	1.01	1.02	0.96	0.79	0.78
MAFE_M/MAFE_B	1.01	1.02	0.99	0.98	0.95	0.87	0.86
DMW	0.73	0.75	0.53	0.56	0.39	0.20	0.14
CW	0.12	0.09	0.02	0.02	0.01	0.03	0.04
N	180	178	175	172	169	150	140

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) represent the results for the x month ahead forecast. RMSFE_M/RMSFE_B is the ratio of root mean squared forecast error of the DMA framework and OLS benchmark. MAFE_M/MAFE_B represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation. A rolling window approach was used for the estimation of the benchmark model.

We also experimented with imposing the constant term to a zero value, i.e., $\alpha_i = 0$. However, the results of this alternative benchmark model are significantly worse than the DMA approach, as is displayed in Table B3 in the Appendix, indicating the importance of keeping a constant term that captures the changes in mean caused by other factors or which captures some long-

³⁴Due to the article's length, the results are not shown in a separate table. Nevertheless, they (RMSFE and MAFE ratios) can be obtained using results from Table 5.1 and Table 5.4.

term trend. More importantly, the results are even worse than the random walk benchmark in the long run, further reinforcing the importance of including a constant.

Furthermore, we also used the panel data approach (fixed effects regression) as another benchmark model, i.e., $\Delta_{t+h}e_{i,t} = \alpha_{i,t} + \beta e_{i,t} + \varepsilon_{i,t+h}$. Note that in this case, β is without currency index i . The rationale behind this approach is that the panel data method may help control estimation errors. The results are shown in Table B4 in the Appendix. The main findings are as follows: The results are comparable to the DMA methodology only in the case of AUD/USD, GBP/USD, and JPY/USD. On the other hand, regarding the rest of the currencies, the results are significantly worse. In these cases, they are even only significantly worse than the simple OLS benchmark or the random walk benchmark. This suggests different rates of mean reversion regarding different currencies, which the inclusion of different regression terms β_i for every currency allows us to capture, i.e., separate regression for every currency.

Moreover, for an additional robustness check, we replace the explanatory variable nominal exchange rate in the benchmark model with the real exchange rate. This can be written as follows (see Equation 5.11):

$$\Delta_{t+h}e_{i,t} = \alpha_{i,t} + \beta_i q_{i,t} + \varepsilon_{i,t+h} \quad (5.11)$$

where $q_{i,t}$ denote the real spot exchange rate. The results of this model are displayed in Table B5 in the Appendix. It can be said that the results are generally very similar to the benchmark with a nominal exchange rate. In some cases, the results are slightly better, and in some cases, and in others, somewhat worse, but the differences can be considered negligible. This suggests that differences in price levels do not have significant additional forecasting power in the period and countries examined in the simple regression framework.³⁵ Again, the argument is that the period under consideration is characterized by mostly low inflation rates and inflation targeting. Therefore, the results suggest that with explicitly set inflation targets, nominal exchange rates and relative price indices can be considered mean reverting.

Finally, we also again experimented with the panel data approach (fixed effects regression) to control for the estimation error. We show the results in Table B6 in the Appendix. The results are roughly similar to the DMA methodology only in the case of AUD/USD, EUR/USD, and JPY/USD. In other cases, they are once more significantly worse than the DMA approach. Also, the simple OLS benchmark using either nominal or real exchange rate as the explanatory variable generally provides significantly better results. This suggests a different rate of mean reversion even after controlling for the difference in the price levels.

In the robustness check, we, therefore, used five distinct specifications regarding alternative benchmark models other than the random walk benchmark. Overall, only the simple OLS regression benchmark with the real or nominal exchange rate as the explanatory variable, unrestricted constant term, and estimated for each currency separately turns out to be consistently better than the random walk benchmark. Furthermore, short-term forecasts are broadly comparable to the predictions from the DMA model. However, the DMA framework generally produces somewhat better results in the medium run and significantly better results in the long run.

³⁵We, however, remind the readers that in this specification, the coefficients regarding price levels are fixed and are the same for every forecasting horizon.

5.7 Conclusion

This paper presents a dynamic model averaging (DMA) approach for forecasting exchange rates. The main contribution of this article is the use of this method, which is relatively novel in exchange rate forecasting literature and allows us to study parameters and model uncertainty in exchange rate forecasting. Specifically, the method allows us to assess many different model specifications and posterior inclusion probabilities of all the variables considered.

The study empirically shows the medium and long-term predictability of nine liquid currencies (AUD/USD, CAD/USD, CHF/USD, EUR/USD, GBP/USD, JPY/USD, NOK/USD, NZD/USD, and SEK/USD) in approximately the last two decades using monthly data. In the out-of-sample forecasting exercise, we use a direct multi-step forecast combined with a dynamic model averaging framework. This approach delivers a forecast that defeats the random walk without drift benchmark based on the RMSFE and MAFE ratios since the nine-month ahead forecast period for all currencies. Overall, the study presents substantive findings regarding statistically and economically significant exchange rate predictability in the medium and long run for all nine currency pairs. We also present some findings on predictability and forecasting performance, even for the short run, although in this case, the results are much less impressive.

Specifically, the long-term forecasts—namely, those for two and three years ahead—appear to be particularly impressive. The model outperforms the random walk without drift, even when the specification is altered to include only the lagged nominal exchange rate, the constant, and price levels. Our results show that this simplified model, however, performs slightly worse in the medium and long term for most of the currency pairs examined. Our results suggest that, at a minimum, studies making medium and long-term exchange rate forecasts should include lagged nominal exchange rates, a constant term, and price levels as explanatory variables. The results further indicate that the inclusion of additional variables can improve exchange rate predictions. However, the parameter values are generally unstable and country-dependent. Despite this variability, these variables can still possess forecasting power during certain periods, which is accommodated by the DMA framework.

On the contrary, the short-term predictive power is not particularly strong; however, the results are comparable to the random walk benchmark when evaluated using the RMSFE and MAFE ratios. However, our approach still defeats random walk in eight out of the nine cases in six months ahead forecast. In short, in forecasting for very short-term periods, typically one to three months, standard exchange rate models, even in the DMA framework, are not particularly successful. Therefore, the random walk model likely remains the best benchmark in these scenarios. For these periods, future research may need to use a completely different methodology or focus on variables other than 'fundamental' factors, as these variables may not sufficiently capture the short-term drivers of exchange rates. For longer periods, the DMA framework can yield better results, although outcomes can vary depending on the currency.

In general, the model's performance improves as the forecasting horizon extends, although the results may still be affected by the typical limitations associated with long-horizon forecasts. In more detail, Engel & Wu (2023) highlight that the results of many prediction models may be exaggerated due to a lack of awareness of small sample bias, which is a significant drawback in long-horizon forecasts. When time series are highly persistent, a sample size of 20 years may be considered relatively small, leading to test statistics that are subject to considerable biases not easily corrected with standard statistical methods. Additionally, as the literature has shown, it is also challenging to assess this small sample bias using simulation methods. This issue is even more pronounced within the DMA framework, as we are not aware of any study that has attempted to address this bias using simulation methods within this framework.

In addition, we also present several possible explanations for our results regarding exchange

rate predictability. One potential explanation is that by using a large number of variables, we implicitly identified the cointegration vector, which means that there exists some fundamental value to which the nominal exchange rate converges. The long-run forecasts suggest some validity of this theory. On the other hand, the instability of many parameters considered in the DMA framework challenges this explanation. The possible reconciliation of these findings can be offered by the 'scapegoat theory of exchange rates'. However, based on the short and medium-term results, we also claim that there exists significant mean reversion in nominal exchange rates, at least in the period examined, i.e., approximately the last two decades. This is in line with Ca'Zorzi & Rubaszek (2020) view of stationary real exchange rates. In a mostly low inflation environment and with central banks following explicitly set inflation targets, it makes sense that the nominal rate is also stationary. Also, note that the 'scapegoat theory of exchange rates' and the mean reversion in exchange rates are not contradictory. The exchange rate may be stationary, but other variables may help to predict its value.

Furthermore, the use of a direct multi-step forecast is another important modeling decision regarding significant exchange rate predictability in the medium and long run. This is because exchange rate dynamics are often obscured by high-frequency 'non-fundamental' noise, which is inherently difficult to predict. By using this forecasting approach, we were able to filter out this noise. In other words, the prediction errors do not accumulate over time due to the use of direct forecasts. This interpretation is further strengthened in the robustness check section, where we used several different benchmark models. Most successful alternative benchmark models use a simple direct multi-step OLS forecast in which the change in the exchange rate is a function of a constant and the lagged level of the nominal or real exchange rate. These models achieve similar results in the short and medium term, but the DMA approach in the long-term horizon defeats them. These findings again suggest that, in the long run, fundamental factors have some effect on the exchange rate, but their impact is relatively modest in the short and medium term.

.1 Data Description

Table A1: Data Description

Name (abbreviated)	Ticker	Source	Country
U.S. Dollars to Australian Dollar Spot Exchange Rate	DEXUSAL	FRED	AUD/USD
Canadian Dollars to U.S. Dollar Spot Exchange Rate	DEXCAUS	FRED	CAD/USD
Swiss Francs to US Dollar Spot Exchange Rate	DEXSZUS	FRED	CHF/USD
U.S. Dollars to Euro Spot Exchange Rate	DEXUSEU	FRED	EUR/USD
U.S. Dollars to U.K. Pound Sterling Spot Exchange Rate	DEXUSUK	FRED	GBP/USD
Japanese Yen to U.S. Dollar Spot Exchange Rate	DEXJPUS	FRED	JPY/USD
Norwegian Kroner to U.S. Dollar Spot Exchange Rate	DEXNOS	FRED	NOK/USD
U.S. Dollars to New Zealand Dollar Spot Exchange Rate	DEXUSNZ	FRED	NZD/USD
Swedish Kronor to U.S. Dollar Spot Exchange Rate	DEXSDUS	FRED	SWD/USD
Credit; Business Banks	D2 LENDING AND CREDIT AGGREGATES	RBA	AU
Credit liabilities of private non-financial corporations		BoC	CA
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5A,T1,F	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5A,T1,B	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5B,T1,F	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5B,T1,B	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5C,T1,F	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5C,T1,B	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5D,T1,F	SNB	CH
Domestic loans (utilisation and credit lines)	EPB@SNB.bakredbetgrbmAV1,KC5D,T1,B	SNB	CH
Loans vis-a-vis euro area NFC reported by MFI	BSI.M.U2.N.A.A20.A.1.U2.2240.Z01.E	ECB	EU
Outstanding Loans and Bills Discounted/Corporations	MD11'DLCLAEDBLTCO	BoJ	JAP
Total Credit to Non-Financial Corporations	CRDQNOANUBIS	FRED	NO
Sector lending	CRDS.MALP3	RBNZ	NZL
Lending, Swe, Non-financial corporations		SCB	SWE
UK M4 LENDING: PRIVATE NON-FINANCIAL	UKBANKLPB	Datastream	UK
Commercial and Industrial Loans, All Commercial Banks	BUSLOANSNSA	FRED	US
Consumer Price Index of All Items in Australia	AUSCPIALLQINMEI	FRED	AU
Consumer Price Index for All Urban Consumers	CPIAUCSL	FRED	CA
Consumer Price Index: All Items for Switzerland	CHECPIALLMINMEI	FRED	CH
HICP - monthly data (index)	PRC.HICP.MIDX	Eurostat	EU
Consumer Price Index of All Items in Japan	JPNCPPIALLMINMEI	FRED	JAP
Consumer Price Index: All Items for Norway	NORCPPIALLMINMEI	FRED	NO
Consumer Price Index: All Items for New Zealand	NZLCPPIALLQINMEI	FRED	NZL
Consumer Price Index: All Items for Sweden	SWECPIALLMINMEI	FRED	SWE
Consumer Price Index of All Items in the United Kingdom	GBRCPPIALLMINMEI	FRED	UK
Consumer Price Index for All Urban Consumers	CPIAUCSL	FRED	US
Harmonized Unemployment Rate: Australia	LRHUTTTTAUM156N	FRED	AU
Unemployment Rate: Aged 15 and Over: Canada	LRUNTTTTTCAM156N	FRED	CA
Unemployment by sex and age	UNE.ERT.M	Eurostat	CH
Unemployment by sex and age	UNE.ERT.M	Eurostat	EU
Unemployment Rate: Aged 15-64: All Persons for Japan	LRUN64TTJPM156N	FRED	JAP
Registered Unemployment Level for Norway	LMUNRRTTNOM156N	FRED	NO
Registered Unemployment Level for New Zealand	LMUNRLTTNZM647N	FRED	NZL
Unemployment Rate: New Zealand	LRUNTTTTTNZQ156S	FRED	NZL
Harmonized Unemployment Rate: Total: Sweden	UNRATENSA	FRED	SWE
Harmonized Unemployment Rate: United Kingdom	LRHUTTTTGGBM156N	FRED	UK
Unemployment Rate, Percent, Monthly	UNRATENSA	FRED	US
Interest Rates: 3-Month or 90-Day Rates and Yields: Australia	IR3TIB01AUM156N	FRED	AU
Interest Rates: 3-Month or 90-Day Rates and Yields: Canada	IR3TIB01CAM156N	FRED	CA
Interest Rates: 3-Month or 90-Day Rates and Yields: Switzerland	IR3TIB01CHM156N	FRED	CH
Interest Rates: 3-Month or 90-Day Rates and Yields: Euro Area	IR3TIB01EZM156N	FRED	EU
Japanese Yen TIBOR	3MONTH	jbatibor	JAP
3-Month or 90-day Rates and Yields: Norway	IR3TIB01NOM156N	FRED	NO
3-Month or 90-day Rates and Yields: New Zealand	IR3TIB01NZM156N	FRED	NZL
3-Month or 90-day Rates and Yields: Sweden	IR3TIB01SEM156N	FRED	SWE
3-Month or 90-day Rates and Yields: United Kingdom	IR3TIB01GBM156N	FRED	UK
3-Month or 90-day Rates and Yields: United States	IR3TIB01USM156N	FRED	US

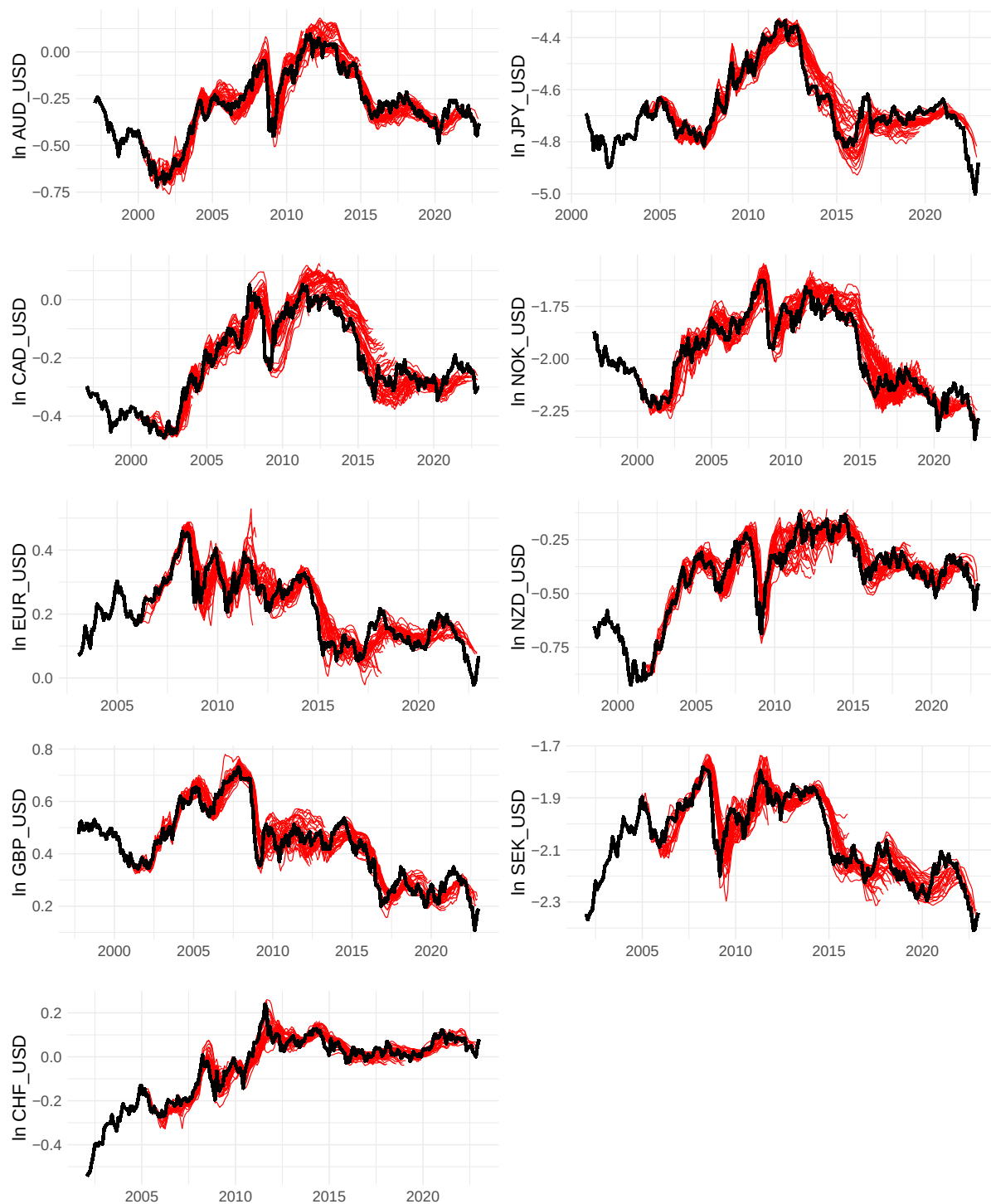
Table A2: Summary Statistics

Variable	Mean	Sd	Max	Min	Rho	ADF	KPSS	N
Nominal Exchange Rate								
AUD/USD	-0.29	0.18	0.10	-0.72	0.98	0.56	0.01	312
CAD/USD	-0.22	0.14	0.05	-0.47	0.98	0.85	0.01	312
CHF/USD	-0.05	0.15	0.24	-0.54	0.96	0.34	0.01	252
EUR/USD	0.21	0.10	0.46	-0.02	0.96	0.03	0.01	240
GBP/USD	0.44	0.13	0.73	0.11	0.98	0.42	0.01	304
JPY/USD	-0.05	0.15	0.24	-0.54	0.96	0.34	0.01	252
NZD/USD	-0.43	0.19	-0.13	-0.93	0.98	0.56	0.01	295
SEK_USD	-2.06	0.15	-1.78	-2.41	0.96	0.17	0.01	253
Credit to Corporation (Money)								
Credit_Australia	6.38	0.44	6.96	5.47	0.99	0.74	0.01	312
Credit_Canada	6.74	0.44	7.58	5.98	0.99	0.84	0.01	312
Credit_Swiss	13.48	0.21	13.91	13.16	0.99	0.20	0.01	252
Credit_EU	8.35	0.15	8.55	8.00	0.98	0.41	0.01	240
Credit_UK	5.80	0.31	6.21	5.12	0.99	0.75	0.01	304
Credit_JPN	14.91	0.10	15.14	14.79	0.98	0.36	0.01	267
Credit_NZL	11.22	0.38	11.80	10.46	0.99	0.69	0.01	295
Credit_Sweden	14.30	0.28	14.91	13.84	0.98	0.91	0.01	253
Credit_US	7.25	0.37	8.03	6.66	0.99	0.44	0.01	312
Price level								
AUD price level	4.46	0.19	4.81	4.12	0.99	0.59	0.01	312
CAD price level	4.51	0.13	4.80	4.28	0.99	0.96	0.01	312
CHF price level	4.61	0.02	4.65	4.55	0.97	0.39	0.01	252
EUR price level	4.57	0.09	4.80	4.39	0.98	0.53	0.01	240
GBP price level	4.50	0.15	4.83	4.25	0.99	0.75	0.01	304
JPN price level	4.59	0.02	4.66	4.56	0.97	0.92	0.01	267
NZD price level	4.51	0.16	4.84	4.24	0.99	0.99	0.01	312
SWE price level	4.59	0.08	4.84	4.45	0.97	0.99	0.01	253
US price level	5.36	0.16	5.70	5.07	0.99	0.92	0.01	312
Unemployment rate								
Australia_un	1.72	0.19	2.27	1.17	0.93	0.50	0.01	312
Canada_un	1.96	0.17	2.62	1.50	0.84	0.02	0.01	312
Swiss_un	1.61	0.36	2.88	0.74	0.91	0.59	0.01	252
EU_un	2.21	0.17	2.55	1.86	0.98	0.96	0.01	240
UK_un	1.72	0.22	2.15	1.28	0.99	0.87	0.01	288
JPN_un	1.38	0.26	1.79	0.79	0.98	0.52	0.01	267
NZL_un	1.59	0.24	2.07	1.16	0.97	0.56	0.01	295
SWE_un	2.00	0.15	2.33	1.67	0.78	0.14	0.03	253
US_un	1.69	0.30	2.67	1.19	0.94	0.58	0.02	312
Interest rate								
Australia_ir	3.85	2.00	7.90	0.01	0.99	0.09	0.01	312
Canada_ir	2.39	1.64	5.92	0.18	0.99	0.38	0.01	312
Swiss_ir	0.59	1.24	3.58	-0.86	0.99	0.29	0.01	312
EU_ir	1.71	1.85	5.11	-0.58	0.99	0.33	0.01	312
UK_ir	0.59	1.24	3.58	-0.86	0.99	0.29	0.01	312
JPN_ir	0.22	0.24	0.89	-0.07	0.99	0.70	0.01	276
NZL_ir	4.33	2.47	9.26	0.26	0.99	0.20	0.01	312
SWE_ir	1.65	1.78	4.49	-0.79	0.99	0.43	0.01	312
US_ir	2.27	2.12	6.73	0.09	0.99	0.55	0.01	312

Note: The table shows the time series which are used in the forecasting exercise. All variables are in log form. Rho denotes the autocorrelation coefficient at first lag. N denotes the number of observations. ADF and KPSS denote p-values of Augmented Dickey-Fuller and Kwiatkowski–Phillips–Schmidt–Shin statistics. The number of lags in ADF and KPSS tests was selected to minimize the AIC.

.2 Additional Figures

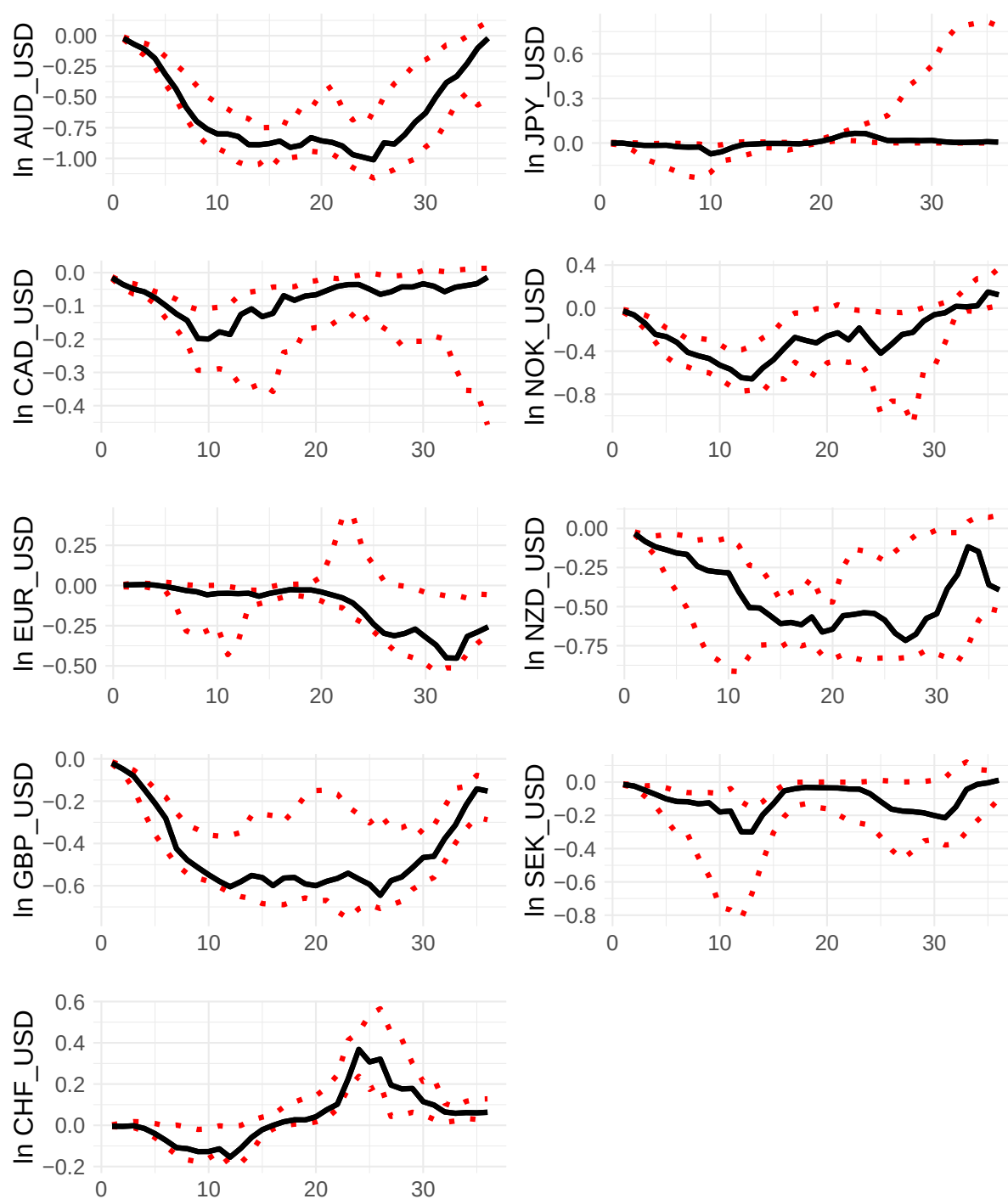
Figure B1: Out of Sample Sequential Forecast



Note: The black line represents realized outcome (log of the nominal exchange rate), while the red lines represent the 36 months ahead forecast calculated at different points in time. DMA framework was used for the estimation.

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

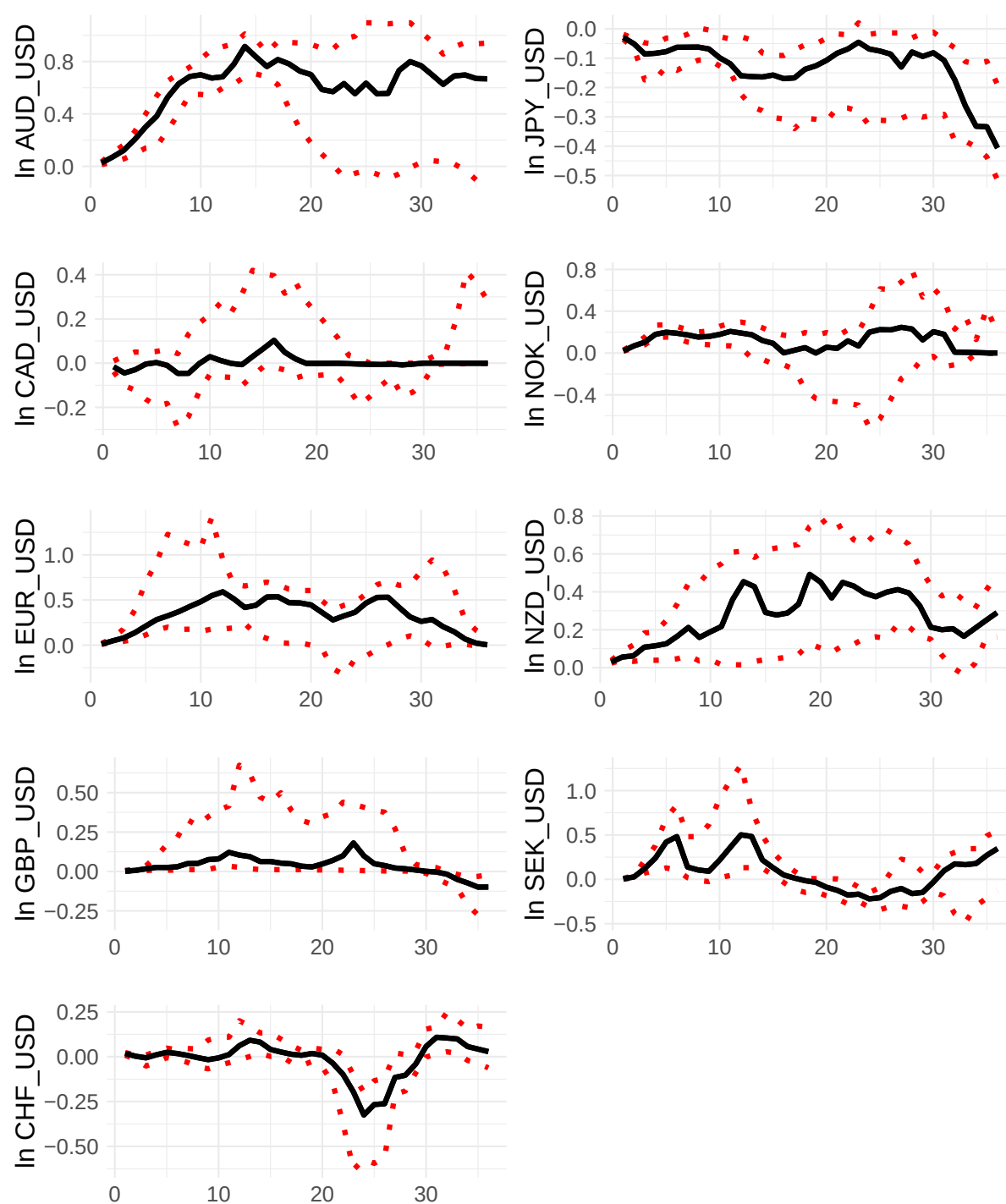
Figure B2: Regression Coefficient - US Money



Note: The graph shows the median of the regression coefficient regarding the US money (loans to corporations) level for different h periods ahead forecast. Dashed lines represent the interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

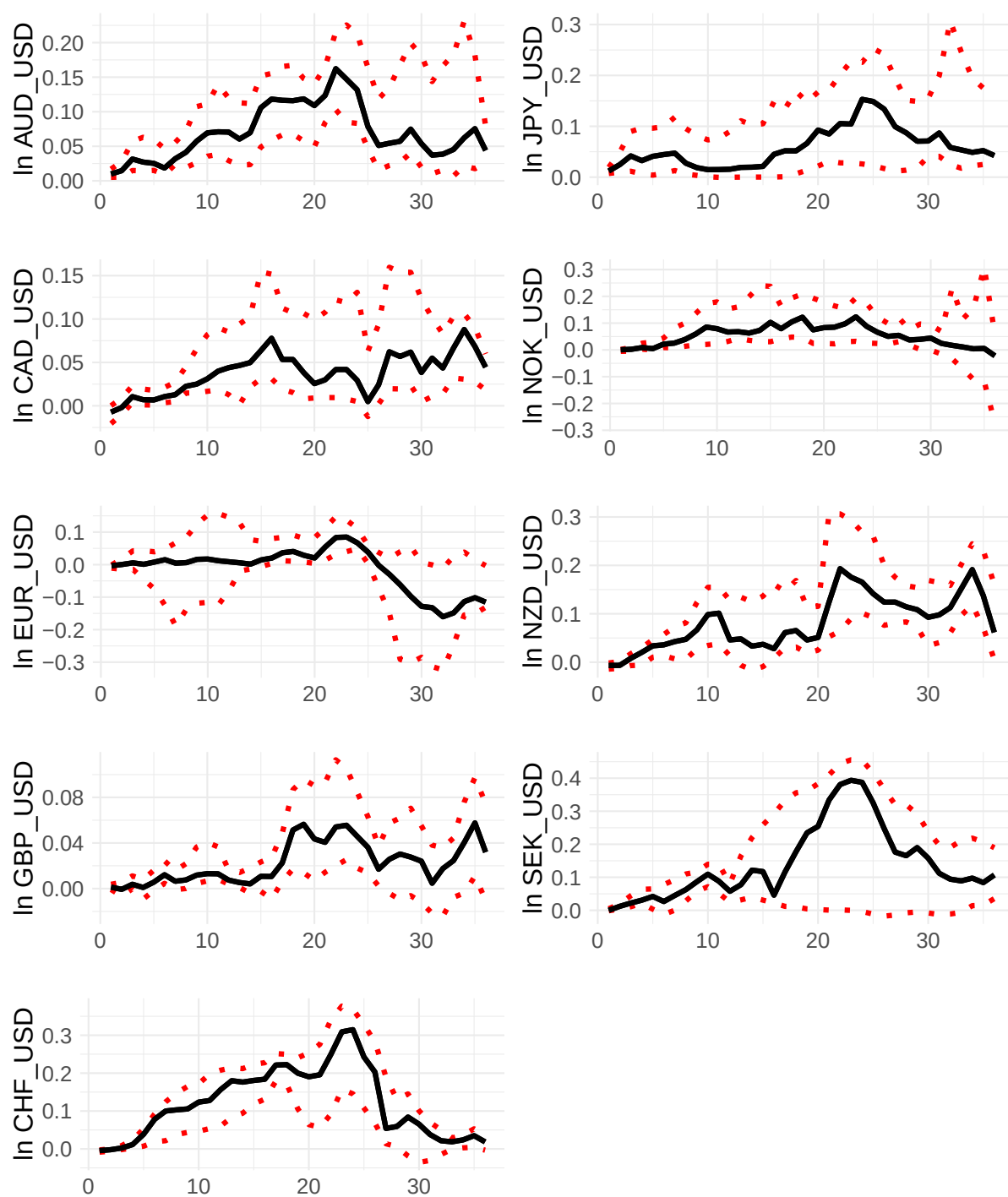
Figure B3: Regression Coefficient - Domestic Country Money



Note: The graph shows the median of the regression coefficient regarding the domestic country money (loans to corporations) level for different h periods ahead forecast. Dashed lines represent the interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

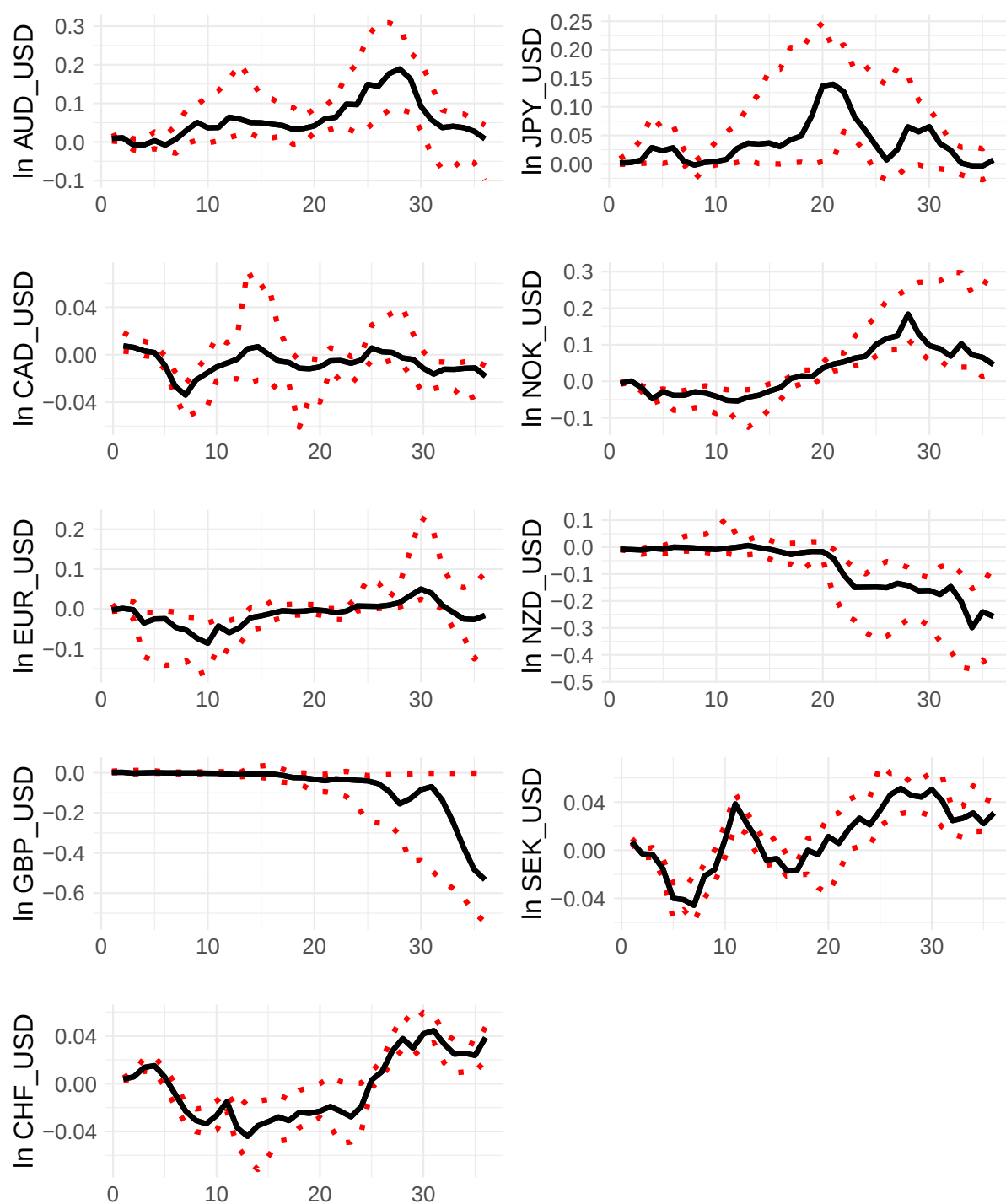
Figure B4: Regression Coefficient - US Unemployment



Note: The graph shows the median of regression coefficient regarding the US unemployment for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

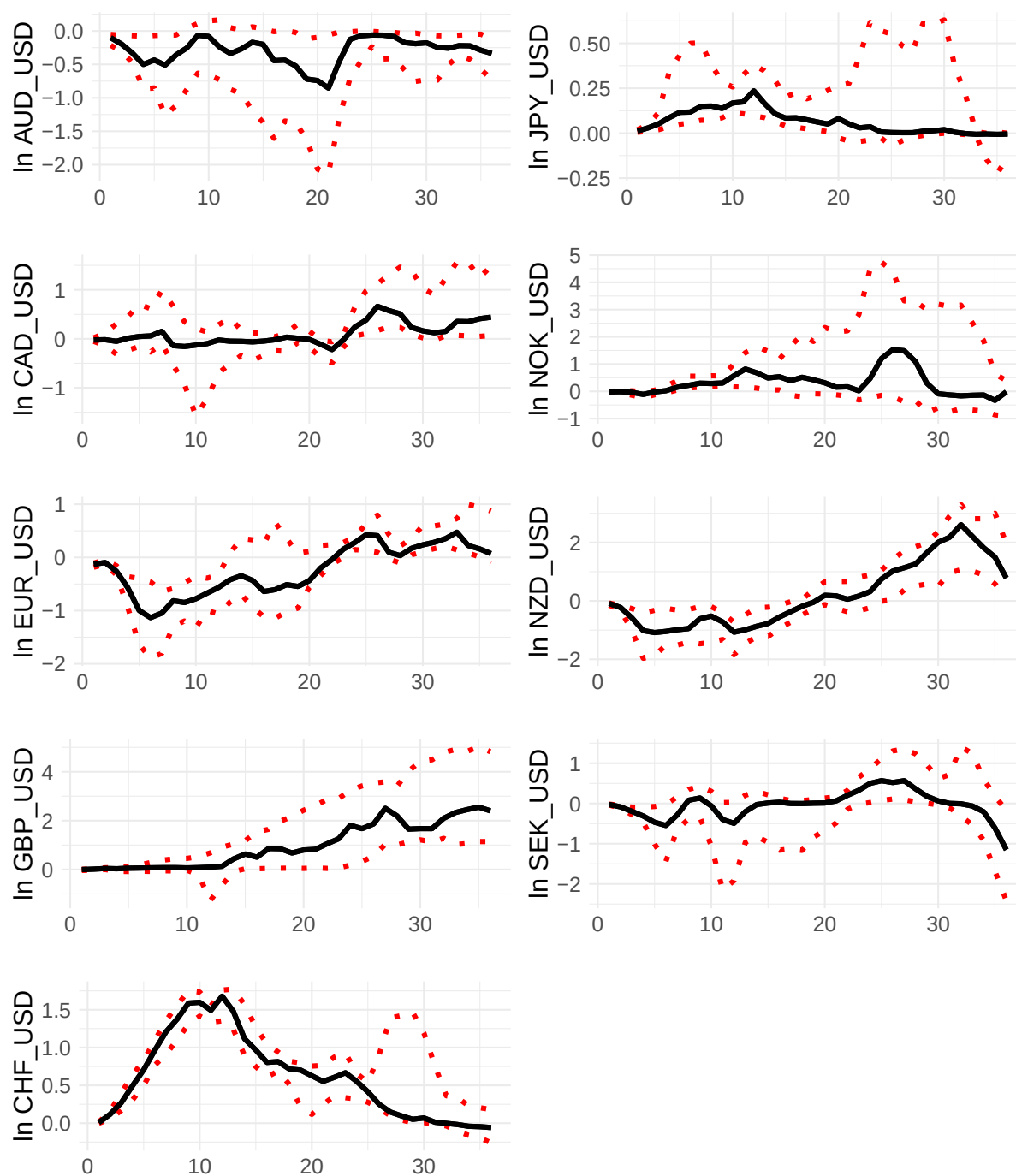
Figure B5: Regression Coefficient - Domestic Country Unemployment



Note: The graph shows the median of regression coefficient regarding the Domestic country unemployment for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

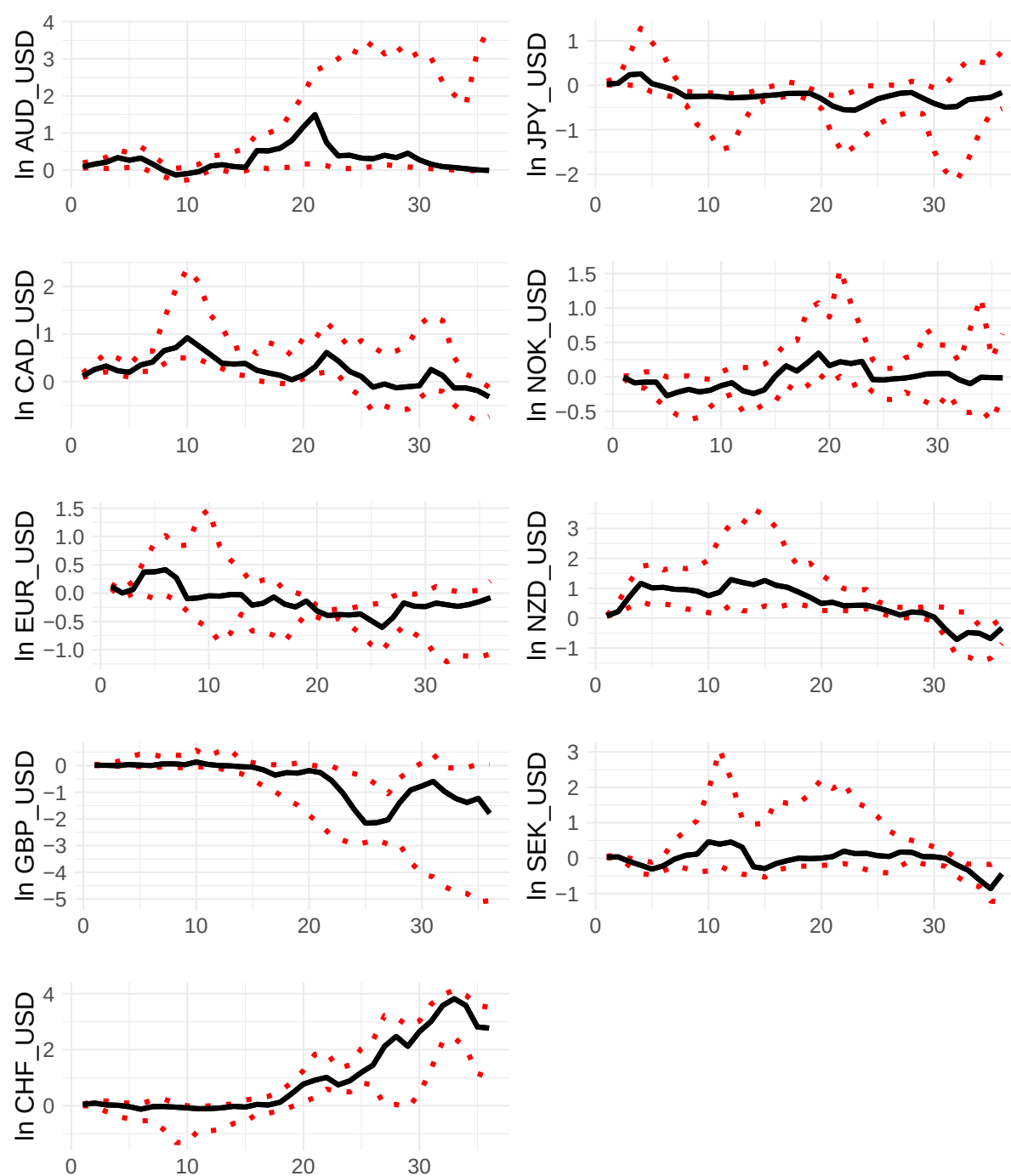
Figure B6: Regression Coefficient - US Price Level



Note: The graph shows the median of regression coefficient regarding the US price level for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

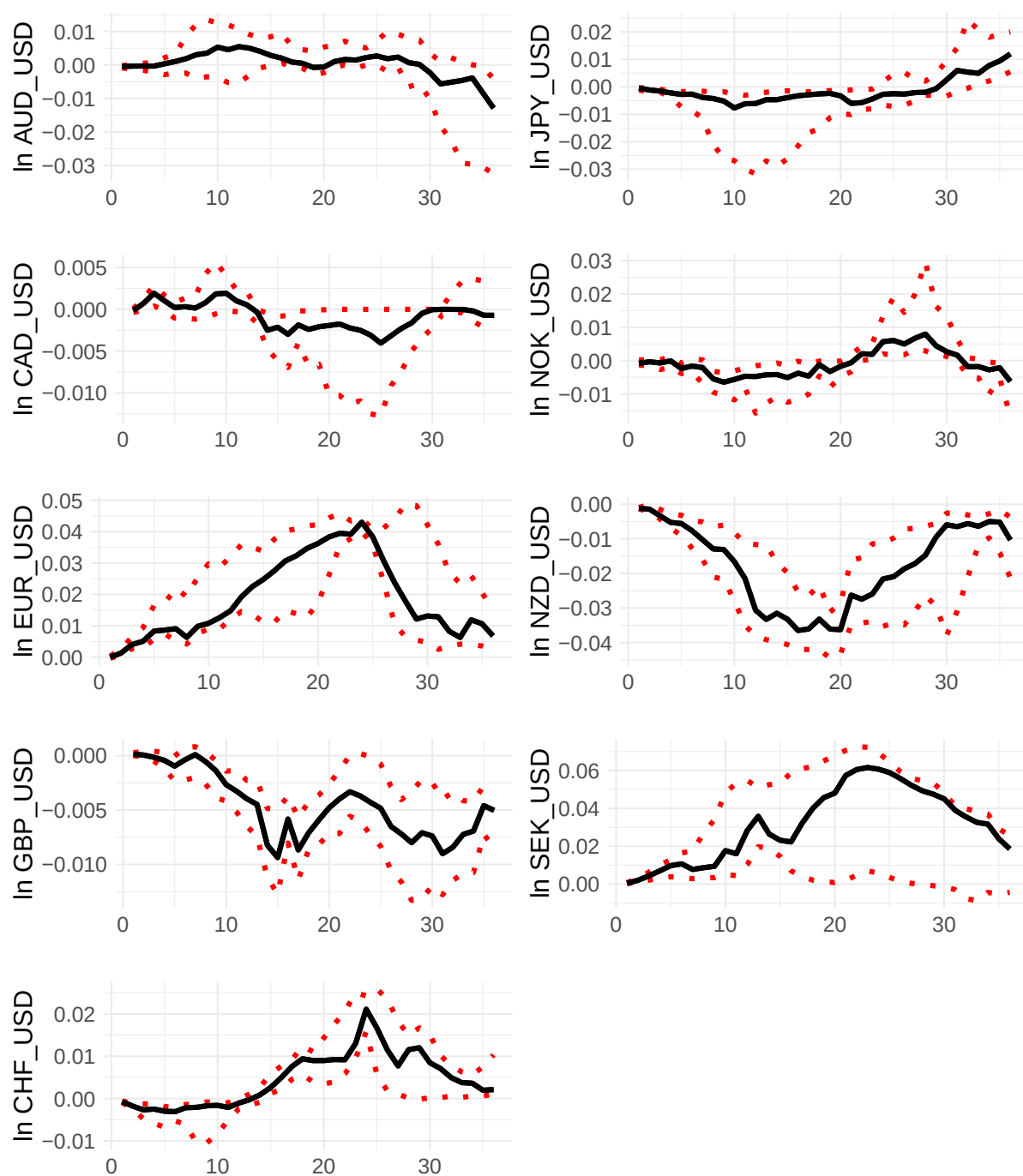
Figure B7: Regression Coefficient - Domestic Country Price Level



Note: The graph shows the median of regression coefficient regarding the domestic country price level for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

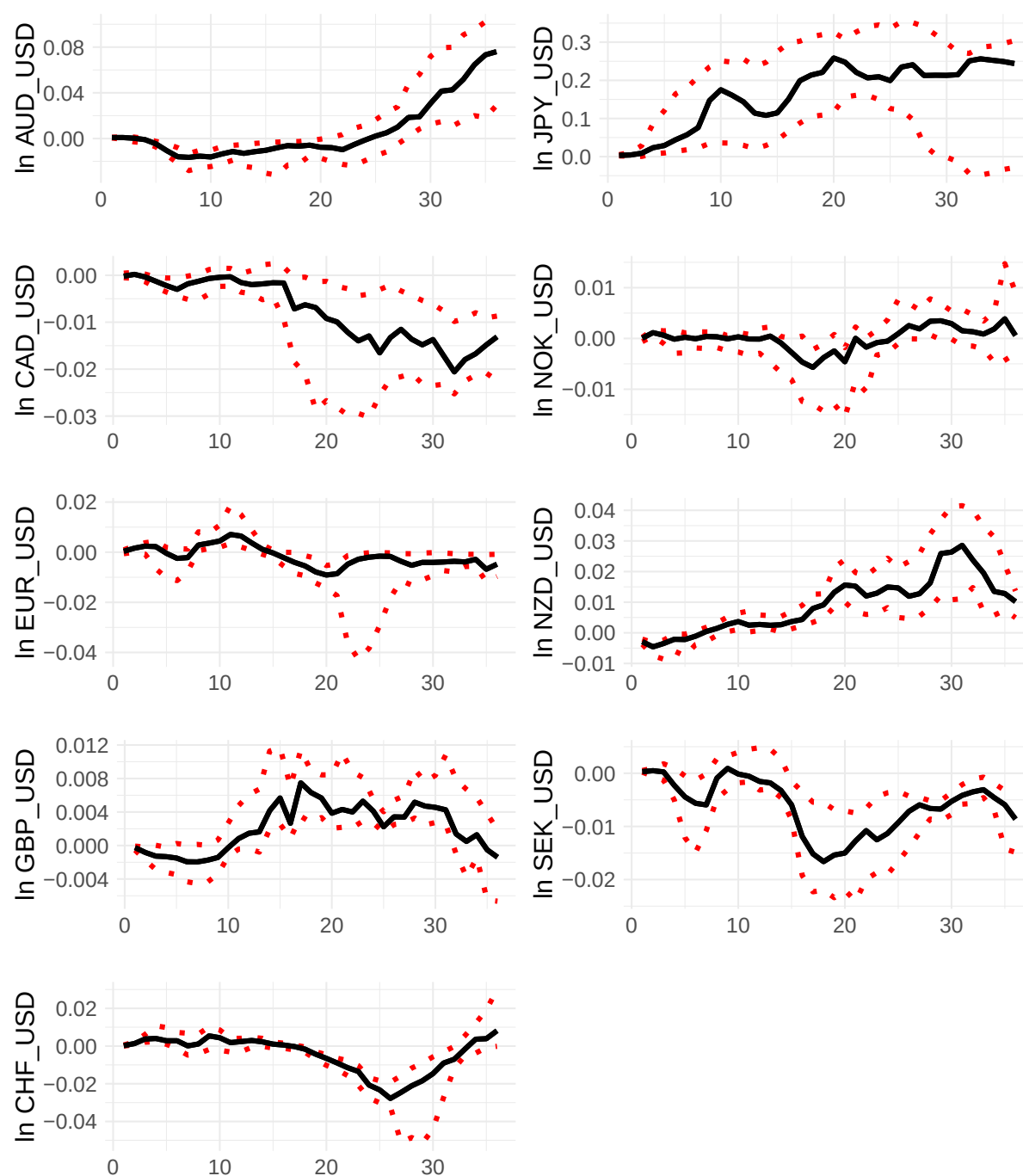
Figure B8: Regression Coefficient - US Interest Rate



Note: The graph shows the median of regression coefficient regarding the US price level for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

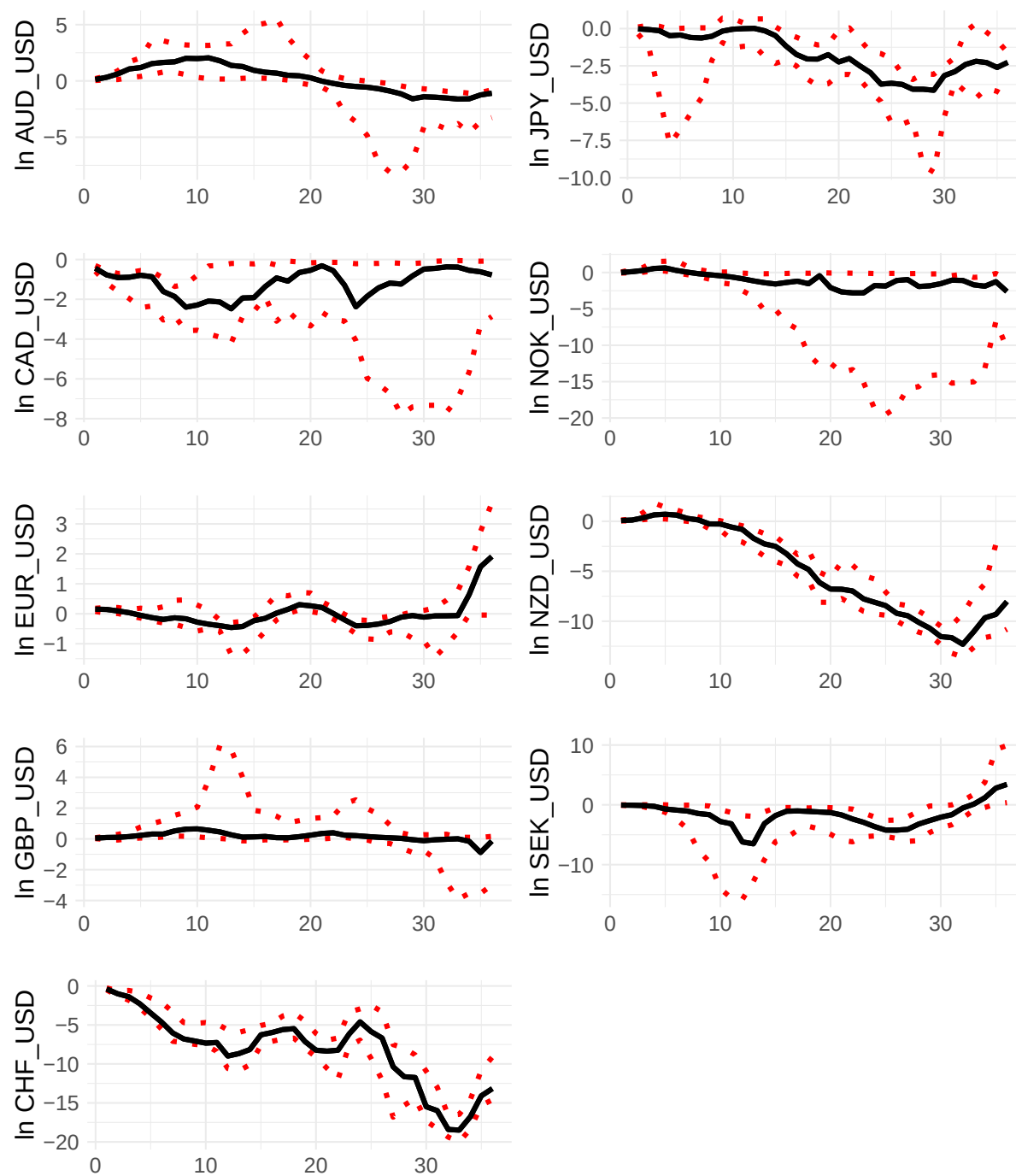
Figure B9: Regression Coefficient - Domestic Country Interest Rate



Note: The graph shows the median of regression coefficient regarding the domestic country price level for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

Figure B10: Regression coefficient - constant term



Note: The graph shows the median of regression coefficient regarding the constant term for different time horizon. Dashed lines represent interquartile range (the 25%–75% quantiles).

Source: Author's calculation, BoC, BoJ, Datastream, ECB DW, FRED, Eurostat, RBA, RBNZ, SCB, SnB

Table B1: Adf Test P-values (Median Values over All DMA Models)

	M1	M3	M6	M9	M12	M24	M36
AUD_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
CAD_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
CHF_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
EUR_USD	0.02	0.01	0.01	0.01	0.01	0.01	0.01
GBP_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
JPY_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
NOK_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.0
NZD_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01
SEK_USD	0.01	0.01	0.01	0.01	0.01	0.01	0.01

Note: ADF tests the null hypothesis that a unit root is present. The number of lags was selected to minimize the AIC. P-value 0.01 means equal or smaller than 0.01. The individual columns (Mx) denote the results for the x month ahead forecast.

Table B2: KPSS Test P-values (Median Values over All DMA Models)

	M1	M3	M6	M9	M12	M24	M36
AUD_USD	0.01	0.01	0.03	0.02	0.02	0.02	0.01
CAD_USD	0.01	0.01	0.01	0.02	0.01	0.01	0.03
CHF_USD	0.10	0.10	0.01	0.01	0.1	0.10	0.02
EUR_USD	0.01	0.01	0.01	0.01	0.02	0.01	0.01
GBP_USD	0.01	0.02	0.01	0.03	0.01	0.05	0.01
JPY_USD	0.01	0.01	0.02	0.02	0.10	0.02	0.01
NOK_USD	0.05	0.01	0.01	0.01	0.01	0.01	0.01
NZD_USD	0.01	0.01	0.01	0.02	0.01	0.01	0.10
SEK_USD	0.01	0.01	0.01	0.02	0.01	0.01	0.01

Note: KPSS tests the null hypothesis that time series are stationary. The number of lags was selected to minimize the AIC. P-value 0.10 means equal or greater than 0.10. The individual columns denote the results for the x month ahead forecast.

Table B3: Out of Sample Forecast Evaluation - Different Benchmark Model 2

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	0.99	0.93	0.86	0.82	0.61	0.50
MAFE_M/MAFE_B	1.00	0.98	0.94	0.90	0.87	0.77	0.67
DMW	0.73	0.45	0.03	0.02	0.02	0.00	0.00
CW	0.04	0.00	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	0.97	0.94	0.87	0.84	0.70	0.65
MAFE_M/MAFE_B	1.00	0.98	0.96	0.92	0.88	0.83	0.80
DMW	0.59	0.19	0.12	0.01	0.01	0.02	0.00
CW	0.01	0.00	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216	204
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.04	1.10	1.06	0.92	0.77	0.56	0.47
MAFE_M/MAFE_B	1.01	1.04	1.00	0.95	0.89	0.76	0.68
DMW	0.97	0.99	0.78	0.20	0.06	0.07	0.13
CW	0.89	0.94	0.19	0.01	0.01	0.01	0.03
N	179	177	174	171	168	156	144
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.95	0.86	0.62	0.55	0.47	0.43	0.55
MAFE_M/MAFE_B	0.97	0.94	0.80	0.73	0.69	0.64	0.70
DMW	0.06	0.07	0.02	0.02	0.01	0.01	0.11
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.01
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.78	0.62	0.45	0.36	0.32	0.21	0.20
MAFE_M/MAFE_B	0.89	0.80	0.68	0.62	0.58	0.46	0.44
DMW	0.00	0.00	0.01	0.01	0.02	0.03	0.02
CW	0.00	0.00	0.00	0.00	0.00	0.01	0.00
N	231	229	226	223	220	208	196
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.02	0.01	0.01	0.01	0.01	0.00	0.00
MAFE_M/MAFE_B	0.13	0.10	0.08	0.07	0.07	0.06	0.05
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	194	192	189	186	183	171	159
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.11	0.06	0.05	0.04	0.03	0.03	0.03
MAFE_M/MAFE_B	0.33	0.24	0.20	0.18	0.18	0.17	0.16
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	239	237	234	231	228	216.00	204
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.97	0.91	0.87	0.79	0.71	0.64	0.63
MAFE_M/MAFE_B	0.97	0.91	0.87	0.79	0.71	0.64	0.63
DMW	0.14	0.13	0.10	0.02	0.02	0.00	0.01
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	222	220	217	214	211	199	187
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.10	0.06	0.04	0.03	0.03	0.02	0.02
MAFE_M/MAFE_B	0.29	0.22	0.17	0.15	0.14	0.12	0.14
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	180	178	175	172	169	150	140

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) represent the results for the x month ahead forecast. RMSFE_M/RMSFE_B denote the ratio of root mean squared forecast error of the DMA framework and OLS benchmark. MAFE_M/MAFE_B represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation. A rolling window approach was used for the estimation of the benchmark model.

Table B4: Out of Sample Forecast Evaluation - Different Benchmark Model 3

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	1.07	1.15	1.16	1.11	1.06	1.16
MAFE_M/MAFE_B	1.01	1.04	1.07	1.06	1.04	0.95	0.99
DMW	0.54	0.82	0.85	0.90	0.82	0.61	0.68
CW	0.00	0.00	0.02	0.04	0.05	0.13	0.27
N	167	165	162	159	156	144	132
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.24	0.14	0.11	0.09	0.09	0.08	0.07
MAFE_M/MAFE_B	0.42	0.33	0.29	0.26	0.26	0.25	0.24
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.98	0.93	0.86	0.82	0.78	0.62	0.50
MAFE_M/MAFE_B	0.99	0.98	0.92	0.89	0.87	0.75	0.67
DMW	0.09	0.04	0.01	0.01	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.94	0.93	0.94	0.87	0.84	0.84	0.67
MAFE_M/MAFE_B	0.97	0.97	0.98	0.92	0.91	0.90	0.81
DMW	0.04	0.21	0.34	0.20	0.19	0.23	0.20
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.98	1.00	1.01	0.98	0.95	0.94	1.10
MAFE_M/MAFE_B	0.99	1.01	0.99	0.99	0.97	0.99	1.03
DMW	0.10	0.50	0.54	0.39	0.32	0.36	0.99
CW	0.03	0.22	0.21	0.07	0.05	0.09	0.21
N	167	165	162	159	156	144	132
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.97	0.93	0.91	0.91	0.91	0.98	1.26
MAFE_M/MAFE_B	0.98	0.97	0.95	0.95	0.94	1.00	1.14
DMW	0.08	0.06	0.16	0.29	0.31	0.47	0.87
CW	0.00	0.00	0.01	0.04	0.06	0.15	0.42
N	167	165	162	159	156	144	132
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.98	0.91	0.77	0.68	0.62	0.52	0.46
MAFE_M/MAFE_B	0.98	0.95	0.88	0.81	0.78	0.72	0.69
DMW	0.13	0.07	0.02	0.02	0.03	0.11	0.12
CW	0.00	0.00	0.00	0.00	0.00	0.02	0.01
N	167	165	162	159	156	144	132
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.67	0.50	0.42	0.38	0.37	0.23	0.19
MAFE_M/MAFE_B	0.78	0.63	0.56	0.53	0.51	0.43	0.38
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.58	0.35	0.22	0.18	0.15	0.10	0.09
MAFE_M/MAFE_B	0.73	0.54	0.42	0.38	0.35	0.28	0.26
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) represent the results for the x month ahead forecast. RMSFE_M/RMSFE_B is the ratio of root mean squared forecast error of the DMA framework and Panel regression benchmark (fixed effects regression with nominal exchange rate as explanatory variable). MAFE_M/MAFE_B represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation. A rolling window approach was used for the estimation of the benchmark model.

Table B5: Out of Sample Forecast Evaluation - Different Benchmark Model 4

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.96	0.88	0.82	0.80	0.65	0.60
MAFE_M/MAFE_B	0.98	0.95	0.89	0.88	0.86	0.80	0.74
DMW	0.28	0.28	0.14	0.12	0.15	0.02	0.03
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.01
N	239	237	234	231	228	216	204
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.98	0.95	0.90	0.88	0.75	0.75
MAFE_M/MAFE_B	0.99	1.00	0.96	0.94	0.92	0.90	0.90
DMW	0.36	0.35	0.28	0.19	0.22	0.14	0.17
CW	0.01	0.00	0.00	0.00	0.00	0.01	0.04
N	239	237	234	231	228	216	204
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.07	1.02	0.88	0.74	0.64	0.65
MAFE_M/MAFE_B	1.00	1.03	0.98	0.91	0.86	0.80	0.76
DMW	0.72	0.89	0.57	0.21	0.08	0.05	0.06
CW	0.11	0.09	0.02	0.00	0.00	0.01	0.00
N	179	177	174	171	168	156	144
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	1.01	0.85	0.81	0.73	0.71	0.85
MAFE_M/MAFE_B	1.00	1.01	0.92	0.90	0.87	0.84	0.88
DMW	0.57	0.54	0.15	0.17	0.13	0.06	0.07
CW	0.12	0.02	0.01	0.01	0.01	0.01	0.03
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.98	0.90	0.81	0.75	0.65	0.64
MAFE_M/MAFE_B	1.00	1.00	0.96	0.92	0.88	0.80	0.79
DMW	0.27	0.36	0.12	0.05	0.05	0.02	0.05
CW	0.02	0.01	0.00	0.00	0.00	0.00	0.00
N	231	229	226	223	220	208	196
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.96	0.84	0.74	0.68	0.52	0.47
MAFE_M/MAFE_B	0.99	0.99	0.93	0.85	0.83	0.74	0.68
DMW	0.35	0.23	0.05	0.03	0.03	0.05	0.02
CW	0.01	0.00	0.00	0.00	0.00	0.01	0.01
N	194	192	189	186	183	171	159
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	1.03	1.00	0.94	0.91	0.77	0.76
MAFE_M/MAFE_B	1.00	1.01	0.98	0.94	0.95	0.90	0.88
DMW	0.56	0.69	0.51	0.37	0.33	0.09	0.11
CW	0.04	0.05	0.01	0.01	0.01	0.00	0.01
N	239	237	234	231	228	216	204
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	1.01	1.01	0.98	0.91	0.88	0.87
MAFE_M/MAFE_B	1.00	1.01	0.99	0.95	0.92	0.87	0.90
DMW	0.56	0.58	0.55	0.45	0.27	0.17	0.03
CW	0.04	0.01	0.00	0.00	0.00	0.00	0.00
N	222	220	217	214	211	199	187
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.04	0.99	0.98	0.90	0.74	0.74
MAFE_M/MAFE_B	1.00	1.02	1.00	0.97	0.95	0.86	0.86
DMW	0.64	0.67	0.45	0.45	0.30	0.20	0.17
CW	0.09	0.07	0.01	0.02	0.01	0.05	0.05
N	180	178	175	172	169	150	140

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) represent the results for the x month ahead forecast. RMSFE_M/RMSFE_B is the ratio of root mean squared forecast error of the DMA framework and OLS benchmark (with real exchange rate). MAFE_M/MAFE_B represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation. A rolling window approach was used for the estimation of the benchmark model.

Table B6: Out of Sample Forecast Evaluation - Different Benchmark Model 5

AUD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.02	1.14	1.32	1.46	1.52	2.01	2.24
MAFE_M/MAFE_B	1.02	1.07	1.12	1.16	1.18	1.30	1.36
DMW	0.87	0.98	0.96	0.97	0.97	0.90	0.85
CW	0.18	0.63	0.78	0.70	0.52	0.71	0.64
N	167	165	162	159	156	144	132
CAD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.62	0.41	0.30	0.26	0.24	0.22	0.19
MAFE_M/MAFE_B	0.73	0.57	0.48	0.44	0.43	0.42	0.38
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
CHF/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.00	0.97	0.93	0.92	0.90	0.80	0.68
MAFE_M/MAFE_B	0.99	0.98	0.94	0.93	0.93	0.86	0.81
DMW	0.41	0.19	0.16	0.17	0.16	0.13	0.11
CW	0.02	0.01	0.01	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
EUR/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.01	1.09	1.16	1.16	1.19	1.32	1.27
MAFE_M/MAFE_B	1.02	1.12	1.29	1.35	1.43	1.78	1.67
DMW	0.85	1.00	1.00	1.00	1.00	1.00	0.97
CW	0.33	0.90	0.99	0.89	0.80	0.98	0.69
N	167	165	162	159	156	144	132
GBP/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.97	0.94	0.93	0.90	0.88	0.80
MAFE_M/MAFE_B	0.97	0.96	0.98	0.95	0.91	0.84	0.80
DMW	0.06	0.20	0.42	0.40	0.35	0.32	0.32
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.01
N	167	165	162	159	156	144	132
JPY/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.99	0.98	1.00	1.05	1.06	1.03	1.13
MAFE_M/MAFE_B	0.99	0.98	1.01	1.02	1.03	1.05	1.12
DMW	0.27	0.35	0.51	0.60	0.59	0.53	0.61
CW	0.00	0.00	0.00	0.01	0.02	0.02	0.00
N	167	165	162	159	156	144	132
NOK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	1.02	1.00	0.85	0.72	0.65	0.53	0.40
MAFE_M/MAFE_B	1.01	0.99	0.90	0.83	0.78	0.70	0.61
DMW	0.71	0.49	0.18	0.06	0.02	0.00	0.01
CW	0.15	0.01	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
NZD/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.88	0.77	0.68	0.62	0.59	0.39	0.30
MAFE_M/MAFE_B	0.92	0.84	0.76	0.70	0.67	0.55	0.48
DMW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132
SEK/USD	M1	M3	M6	M9	M12	M24	M36
RMSFE_M/RMSFE_B	0.87	0.63	0.40	0.32	0.28	0.18	0.15
MAFE_M/MAFE_B	0.92	0.76	0.59	0.52	0.48	0.37	0.34
DMW	0.01	0.00	0.00	0.00	0.00	0.00	0.00
CW	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	167	165	162	159	156	144	132

Note: The dependent variable is the h-step forward difference of the given currency pair. The individual columns (Mx) denote the results for the x month ahead forecast. RMSFE_M/RMSFE_B denote the ratio of root mean squared forecast error of the DMA framework and Panel regression benchmark (fixed effects regression with real exchange rate as explanatory variable).MAFE_M/MAFE_B represents the ratio of the mean absolute forecast error between the two. DMW and CW indicate the p-values for the Diebold-Mariano and Clark-West test statistics, respectively. N refers to the number of estimated forecasts. DMA framework was used for the estimation. A rolling window approach was used for the estimation of the benchmark model.

Concluding Remarks

Asset pricing is an extremely broad topic and generally provides a framework for understanding economic agents' decision-making. This dissertation explores several empirical relationships studied in asset pricing in five different chapters. I use a diverse set of econometric and statistical tools to analyze the predictive power of macroeconomic and financial variables on financial markets. The empirical analysis presented in these articles underscores the variability in the impact of macroeconomic and financial factors on financial markets.

The first chapter of this dissertation empirically investigates the relationship between the development of the exchange rate and exchange rate option implied probabilities. I focus on the application of the Ross recovery theorem in the FOREX market using the CZK/EUR exchange rate. We can also calculate investors' perception of risk based on Ross's recovery theorem.

My findings indicate that both subjective and risk-neutral densities are not unbiased probability estimates of a future exchange rate. However, the results show a significant increase in implied risk aversion during the COVID-19 pandemic. Furthermore, the results presented in this chapter need to be interpreted with caution due to the fact that obtaining risk-neutral and subjective probabilities are two ill-posed problems connected to each other and characterized by numerical instability. The fact that the obtained densities do not correspond to the realization of the processes does not necessarily mean that the market is not able to predict future developments.

The second chapter analyzes stock market development and the role of equity risk premia. In more detail, using the dividend discount model and dividend futures, I present a simple way of extracting the implied equity risk premium. The main contribution of this chapter is the use of dividend futures, which are relatively novel derivative instruments and represent the risk-adjusted expectations of future dividends. This approach allows for the computation of the equity risk premia on a daily basis and, as such, can serve as another indicator for measuring market sentiment, providing an easy-to-understand interpretation. Overall, this approach allows for a better structural interpretation of price movements.

Practically, the second chapter studies the equity risk premium development of the S&P 500 Index and Euro Stoxx 50 Index in the last decade. Using the dividend discount model combined with dividend futures, I show, for example, the development of equity risk premia during the COVID-19 pandemic and Brexit. I illustrate the high variability of the equity risk premium, which is one of the main drivers of price development. I also show the importance of a time-varying structure of interest rates on price development.

The third chapter analyzes the predictive role of inflation and interest rates in forecasting stock returns. I present economically significant in-sample evidence demonstrating that stock returns can be predicted using the cyclical components of inflation and interest rates. The out-of-sample analysis further indicates that these cyclical components provide reasonable out-of-sample predictability for the equity risk premium compared to a historical average benchmark.

The results can be rationalized by the existence of time-varying risk premia and the existence of stochastic trends in both inflation and interest rates. Furthermore, I suggest that properly

accounting for trend inflation changes and the equilibrium real interest rate is crucial for accurate stock return forecasts.

The fourth chapter presents a framework for extracting the supply and demand shocks using the revisions of expectations obtained from a Survey of Professional Forecasters. The main contribution of this article is the identification of these shocks and the subsequent assessment of their impact on the yield curve. The motivation behind this approach is the fact that the practical thinking of many analysts is still dominated by the IS-LM and AD-AS framework and the associated thinking about the business cycle in terms of supply and demand shocks.

The main contribution of this chapter is the identification of supply and demand shocks and the subsequent assessment of their impact on Fed funds rate and bond yields. Specifically, we use structural VAR with sign restrictions to assess the effects of these shocks. This approach allows for the asymmetry in the bond yield movements, where we show that yields react differently to supply and demand shocks. The study empirically shows the different impacts of supply and demand shocks on bond yields. The analysis is performed on the US data ranging from 1969 (q4) to 2022 (q1).

The fifth chapter introduces a dynamic model averaging approach for forecasting nominal exchange rates. This framework incorporates the majority of methods commonly employed in the forecasting literature while also allowing for the examination of parameter and model uncertainty in exchange rate forecasting.

In more detail, the fifth chapter empirically demonstrates the medium and long-term predictability of nine liquid currencies (AUD/USD, CAD/USD, CHF/USD, EUR/USD, GBP/USD, JPY/USD, NOK/USD, NZD/USD, and SEK/USD) over approximately the past two decades, using monthly data. Several theoretical explanations for these findings on predictability are offered. One possible explanation is that by utilizing a large number of variables, the cointegration vector is implicitly identified, suggesting the existence of a fundamental value to which the nominal exchange rate converges. On the other hand, the instability of many parameters within the DMA framework challenges this explanation. A potential reconciliation of these findings may be provided by the 'scapegoat theory of exchange rates.' Additionally, based on the short- and medium-term results, the study posits the existence of mean reversion in exchange rates.

Overall, the findings of this dissertation emphasize the importance of risk aversion and intertemporal preferences in determining the pricing and performance of financial assets. Theoretical models generally provide explanations for asset price fluctuations in response to changes in economic conditions and investors' perceptions of risk. However, this dissertation focuses more on practical applications. In several chapters, I provide an empirical counterpart for values commonly used in theoretical frameworks. Similarly, the forecasting exercises conducted herein allow me to assess whether the theoretical constructs contribute to or justify the observed forecast results. In conclusion, this dissertation examines various standard asset pricing issues, and I believe that, in certain instances, it offers novel insights.

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Appendix A

Responses to Opponents' Reports

I would like to express my gratitude to all the referees for taking the time to review the thesis. Their discussions, suggestions, and points raised certainly have provided me with insight regarding my future work. Below, I summarize my responses to each of their questions and suggestions.

A.0.1 Reviewer: Prof. Evžen Kočenda, Ph.D.

Referee report on the doctoral dissertation:

The doctoral dissertation consists of five chapters that are represented by five individual papers on a unified theme of asset pricing. The titles of papers sufficiently define their content. Individual papers are at various stages of publication (shown in parentheses): 1. Recovery Theorem and the Risk Aversion. Evidence from the Czech Republic (forthcoming in International Journal of Monetary Economics and Finance), 2. Deriving Equity Risk Premium Using Dividend Futures (published in The North American Journal of Economics and Finance), 3. Inflation, Interest Rates and the Predictability of Stock Returns (published in Finance Research Letters), 4. Supply Shocks, Demand Shocks and Yield Curve Dynamics (published in Finance Research Letters). 5. Forecasting Nominal Exchange Rates Using a Dynamic Model Averaging Framework (accepted in Heliyon).

The dissertation is written in a competent manner and shows that the author has mastered his theoretical as well as analytical skills. The work is well structured, and the introduction provides a clear dissertation overview. The references are plentiful, and their use demonstrates that the author is well acquainted with the contemporaneous literature in the field. Some references are missing in the text, but this issue can be corrected at a later stage of the defense process. In the dissertation, standard techniques and novel approaches are applied to analyze a set of issues dealing with asset pricing and financial markets in general. For that, in my further review, I will accentuate more of the dissertation's economic and finance issues when applicable. 1. The first Chapter is forthcoming in the IJMEF and presents an application of the Ross recovery theorem in the forex market. The key contribution rests in a novel approach to extracting risk premium and showing its time-varying nature. The formal framework of the paper

is presented well, and the results of the empirical analysis are reported with sufficient detail. It is a pity that the data ended in the Fall of 2020, given the interesting findings on risk premia related to increased uncertainty due to the covid. In connection with the previous comment, it would be useful to perform the empirical analysis with other currencies whose dynamics and exchange rate arrangement(s) would be suitable with respect to the presented methodology. Larger traded volumes of other currencies might also provide a channel for additional results and provide further comparative perspective.

2. Chapters 2-4 consist of papers that have already been published in respectable academic journals with impact factors. As such, they underwent a thorough refereeing process. All three papers are well-motivated and expertly executed, and my minor comments would lead only to their marginal improvements. For that, I refrain from scrutinizing them further.

3. Chapter 5 presents a dynamic model averaging approach for forecasting nominal exchange rates. This approach is then compared to other ones. An empirical part of the analysis is done with the monthly exchange rate data of major currencies. This is understandable, given the set of other macro variables used in the analysis that are not reported at more granular frequencies. However, while the use of the monthly data is fine from an academic perspective, its use is, to an extent, questionable for practical implications. Further, the set of macroeconomic fundamentals used in the empirical analysis is sensible and grounded in theory. However, market participants exhibit selective attention to the news on numerous variables, including the chosen fundamentals. A dynamic model averaging approach might be a good fit for exploring this angle. The dissertation represents a coherent piece of work with value-added as it brings new evidence related to the topic of asset pricing. My comments do not represent suggestions to rework parts of the analysis, but they are a small set of notes that might help with further research. In my opinion, the dissertation is ready for defense without changes.

Response

Introduction

I would like to sincerely thank the opponent for their careful review of my dissertation and for their constructive and insightful comments. I greatly appreciate the acknowledgment of the theoretical and analytical contributions of my work, as well as the recognition of its structure, coherence, and value-added to the field of asset pricing and financial markets.

Regarding the specific points raised:

Chapter 1 – Ross Recovery Theorem Application in the Forex Market: I acknowledge the opponent’s observation regarding the data ending in the fall of 2020 and the potential for additional insights that could be gained by extending the empirical analysis to other currencies. While such an extension would indeed provide a broader perspective, the scope of the dissertation was defined at the time of writing, and further exploration along these lines could be considered in future research.

Chapters 2-4 – Previously Published Papers: I appreciate the opponent’s recognition that these chapters have already undergone rigorous peer review and that the suggested comments would lead only to marginal improvements. Given their published status, I concur that further modifications are not necessary at this stage.

Chapter 5 – Dynamic Model Averaging for Exchange Rate Forecasting: The discussion on the practical implications of using monthly data is well taken. While the choice of data frequency was guided by theoretical and empirical considerations, future research could indeed explore alternative data frequencies to assess their impact on forecasting accuracy. Additionally, the observation regarding market participants’ selective attention to macroeconomic fundamentals is a valuable point that could serve as a starting point for further studies leveraging dynamic model averaging techniques.

Once again, I am grateful for the opponent’s thoughtful remarks and their endorsement of the dissertation for defense without changes.

A.0.2 Reviewer: Kamil Kladívko, Ph.D.

Review of Martin Casta's dissertation thesis "Five Essays on Asset Pricing"

Introduction

I am pleased to have had the opportunity to review the PhD thesis of Martin Casta. The thesis consists of five chapters, each representing a self-contained paper in the areas of macro-finance, asset pricing, and exchange rate dynamics. The chapters topics are tied together by the thesis' Motivation section. In the following sections, I will briefly summarize each chapter (paper) and provide some comments and questions for consideration. As all five papers have already been published or accepted for publication, these comments and questions are intended for the author's reflection rather than for incorporation into a potential revision. Finally, I will conclude with my overall evaluation of the thesis.

Summary of papers, comments and questions

Please note that my comments and questions specifically refer to the published papers, not to the PhD thesis document.

1. Recovery Theorem and the Risk Aversion

Chapter 1 was accepted to International Journal of Monetary Economics and Finance, see Časta (2024b). This paper applies the Ross Recovery Theorem in the FOREX market, focusing on the CZK/EUR exchange rate. By deriving subjective and risk-neutral densities, it evaluates market perceptions of risk and the implied risk premium, especially during the COVID-19 pandemic. The study reveals significant increases in implied risk aversion and highlights the challenges of applying the recovery theorem in practical contexts.

Comments and questions 1. In my opinion, the mathematical framework and notation would benefit from greater rigor. It would be helpful to include a clear definition of the probability space, the filtrations, and the probability measures on which the problem is formulated. This would enhance the clarity of the exposition.

2. The right-hand side of equation (5) appears inconsistent. If the time horizon (option maturity) is t , the pricing kernel should be m_t , not m_{t+1} . Here m_t would represent the pricing kernel from time 0 to t (Radon-Nikodym derivative multiplied with a deterministic discount factor). Am I misunderstanding this, or is there an alternative interpretation?

3. My understanding is that the FX option pricing used in the paper relies on the Garman-Kolhagen formula, which is adjusted using the Vanna-Volga method. I would appreciate seeing the pricing kernel (change of measure) explicitly stated for the specific pricing model being applied. Sections 2 and 3 appear to be quite general and, in my view, seem to mix continuous and discrete spaces.

4. I am unclear about what is referred to as the "subjective" probability measure. My understanding is as follows: pricing is done under the risk-neutral measure (an equivalent martingale measure, which becomes the unique risk-neutral measure in a complete market setting), while the underlying asset prices (such as the spot FX rate) evolve under the real-world (statistical or physical) measure. The connection between these two measures is determined by

the price of risk, which, in this context, modifies the drift of the Brownian motion. Importantly, the volatility of the Brownian motion remains unchanged under the measure transformation (as stated by Girsanov’s theorem).

It seems, however, that the “subjective” probability measure is being treated as distinct from the real-world probability measure. If this is the case, how exactly does it differ from the real-world measure?

3. Inflation, Interest Rates, and Predictability of Stock Returns

Chapter 3 was published in Finance Research Letters, see Časta (2023). This paper investigates how inflation and interest rates affect stock return predictability. The findings emphasize the importance of cyclical components of these variables in forecasting equity risk premiums, providing evidence for both in-sample and out-of-sample predictive power.

Comments and questions 1. As mentioned in the paper’s introduction, interest rates have been used as stock return predictors repeatedly. My understanding is that the novelty in this paper is the “double” detrending, that is, taking away inflation and a “persistent component” in real rates.

2. The persistent regressor bias is (correctly) admitted in the paper, but as a treatment the Newey-West standard errors are mentioned (second to the last paragraph on page 3. In my view, choosing appropriate standard errors does not cure the Stambaugh bias (persistent regressor bias) – it is a bias; the problem is with the expected value of the estimator and not with its variance. (It may not be a problem in this paper, because the two trends are removed, and thus there should not be persistence. However, this not possible to check in the paper, since no descriptive statistics about the predictors are shown).

4. Supply Shocks, Demand Shocks, and Yield Curve Dynamics Chapter 4 was published in Finance Research Letters, see Časta (2022b). This research examines the asymmetric effects of supply and demand shocks on U.S. bond yields and the federal funds rate. Using data from the Survey of Professional Forecasters, the study identifies these shocks and evaluates their impact on yield curve dynamics.

Comments and questions

1. I would be interested in seeing the time series of the extracted shocks. Additionally, do these shocks exhibit out-of-sample forecasting power within the context of the VAR model used? I realize that the space constraints in Finance Research Letters have precluded the inclusion of this analysis.

2. The SPF and GDP data are observed at a quarterly frequency, which aligns with the frequency of the VAR model. In contrast, interest rate data, including the Nelson-Siegel estimates of the U.S. Treasury yield curve used in the paper, are available at a daily frequency. It might be insightful to examine, by a simple alternative model, the bond market’s immediate or short-term reaction to the shocks.

3. The difference between the Fed funds rate and the 10-year yield can serve as a proxy for the yield curve slope, which appears to align visually with the impulse responses presented in Figures 1 and 2. In other words, the correlation between the two sets of variables is likely quite

high, although I understand why Fed funds rate is of a special importance.

4. I am unclear about the interpretation of $X_t = 0$ in the VAR constraint. Does it indeed represent a strict zero? Additionally, I find it confusing that x_t denotes inflation or GPD, while X_t denotes interest rate variables.

5. Forecasting Nominal Exchange Rates Using Dynamic Model Averaging

Chapter 5 was published in Heliyon, see Časta (2024a). This paper employs a dynamic model averaging (DMA) framework to improve nominal exchange rate forecasts. The study analyzes data from nine major currency pairs and demonstrates significant (both statistically and economically) improvements in medium and long-term forecasting accuracy compared to traditional benchmarks.

1. The DMA method is described as an approximation of linear filtering combined with (nonlinear) Markov switching. The notation and equations presented bear a resemblance to the literature on Kalman filtering (equation (3)) and (latent) Markov switching (equations (8) and (9)). However, Section 4 of Časta (2024a) does not provide sufficient detail for me to implement the DMA approach.

2. Which parameters are specific to the forecast horizon? For instance, are $A_{i,t}, B_{i,t}, C_{i,t}$ in equation (1) dependent on h ? I assume that the parameters α and β in equation (10) are h -dependent. Additionally, should it be $\alpha_{i,t}$ or simply α_i in equation (10)?

3. It is unclear whether the Diebold-Mariano test was applied to the RMSFE or MAFE measure. While the quadratic criterion (RMSFE) is the more commonly used, the MAFE criterion is also a valid option.

4. Figure B.1 in Appendix is interesting but difficult to read. It might be more effective to focus on a few forecast over a shorter period for a single exchange rate. That said, the figure suggests that the DMA model frequently succeeds in predicting abrupt exchange rate changes. For instance, the model appears to have accurately forecasted the sharp decline of the British pound during the 2008 Financial Crisis three years in advance, which is remarkable. How does the model achieve this with its linear structure (equation (1))? Could this be attributed to the Markov switching component?

Response

I would like to express my sincere gratitude to the opponent for their thorough and insightful review of my dissertation. I appreciate the constructive comments, which engage with both the theoretical and empirical aspects of my work. Below, I address the specific points raised in the report.

Regarding the specific points raised:

1) Mathematical Rigor and Notation: I acknowledge that the mathematical framework, particularly the definition of the probability space, filtrations, and probability measures, could have been presented with greater precision. While the exposition was intended to be accessible to a broader audience, I agree that a more rigorous approach would have enhanced clarity.

2) Equation (5) – Consistency of the Pricing Kernel: The opponent is correct in pointing out the issue with equation (5). The pricing kernel should indeed be defined for the correct time horizon, and I acknowledge the presence of a typo in this section. This was an oversight, and I appreciate the clarification.

3) FX Option Pricing and Vanna-Volga Method: The derivation of the Vanna-Volga method is indeed a challenging aspect of the paper. I attempted to provide a formal mathematical derivation but was ultimately unable to do so in a way that satisfied the necessary level of rigor. As a result, the methodology was presented in a more heuristic manner. I recognize that this is a limitation and appreciate the opponent’s suggestion for a more explicit statement of the change of measure. Additionally, the general nature of Sections 2 and 3 was a deliberate choice to maintain a level of abstraction, but I understand that a more structured distinction between continuous and discrete spaces would improve the presentation.

4) Interpretation of the “Subjective” Probability Measure: The opponent raises an important question about the meaning of the “subjective” probability measure. In this context, “subjective” essentially refers to the real-world probability measure. I chose this terminology because it emphasizes that the predictability of (excess) returns can arise both from variations in risk premia and from systematic expectational errors in beliefs. The traditional approach attributes return predictability to time-varying risk premia, but recent research has also explored the role of deviations from full-information rational expectations (FIRE). These deviations introduce an additional source of predictability, as documented (for the Treasury Bonds) in studies such as Bauer & Chernov (2024), Cieslak (2018), and Piazzesi *et al.* (2009). I specifically follow the exposition by Piazzesi *et al.* (2009). The term “subjective” thus captures the possibility that investors’ expectations exhibit (persistent) biases that evolve over time. While the real-world measure is typically associated with the objective statistical distribution of asset returns, the subjective measure reflects how agents perceive and process information, potentially diverging from the true data-generating process.

3. Inflation, Interest Rates, and Predictability of Stock Returns

1) Novelty of the Paper: The opponent is correct in noting that interest rates have been extensively used as stock return predictors. While the “double” detrending approach—removing both inflation and the persistent component of real rates—is an important contribution, the

paper's broader novelty lies in emphasizing the role of inflation detrending in stock return forecasting. While inflation detrending itself is not a new concept in macroeconomic analysis, its application in equity risk premium predictability is relatively unexplored. This aspect distinguishes the paper from existing literature.

2) Persistent Regressor Bias and Statistical Treatment: I acknowledge the opponent's point regarding the Stambaugh bias, which affects the expected value of the estimator rather than just its variance. The use of Newey-West standard errors and bootstrap p-values does not directly eliminate this bias, but it does address issues related to inference, particularly by correcting for size distortions in hypothesis testing. While detrending reduces persistence, a full resolution of the Stambaugh bias would require alternative methods, such as instrumental variable estimation or bias-corrected estimators. Additionally, I note the absence of descriptive statistics on the predictors in the paper, which would have allowed for a clearer assessment of persistence. Due to the letter format of the publication, these results were not included, but I also conducted autocorrelation tests, which provide further insight into the stationarity properties of the predictors.

Below, I provide Autocorrelation Function (ACF) plots for some series used in the model as explanatory variables. (I provide the results regarding monthly data.) The results generally suggest that autocorrelation (persistence) diminishes much more quickly when the series are detrended, which is also suggested by spectrum estimation (not included). Nonetheless, Stambaugh bias is generally more prominent in long-horizon forecasts, where the dependent variable is also highly auto-correlated, which is generally not the case of one- or three-month returns data.

Figure B1: ACF Plots - Part 1

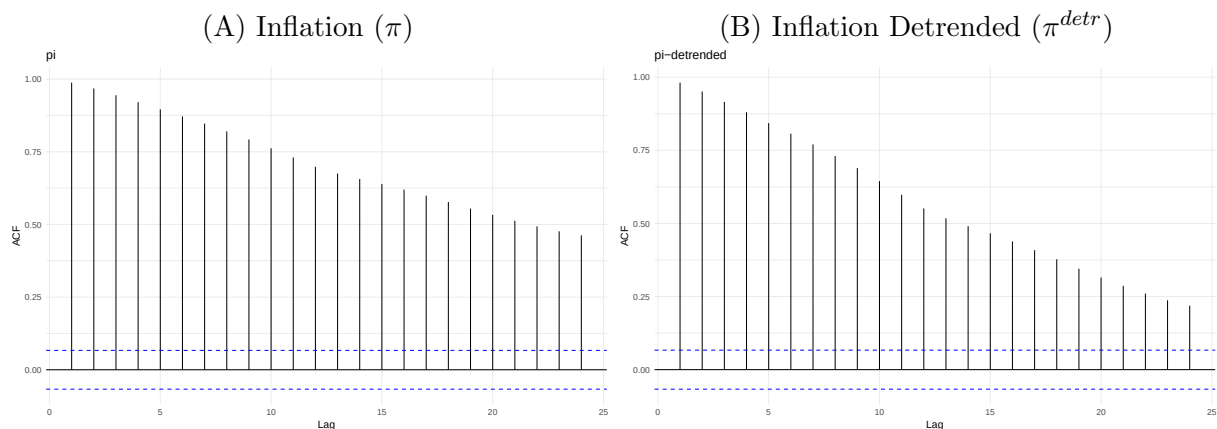
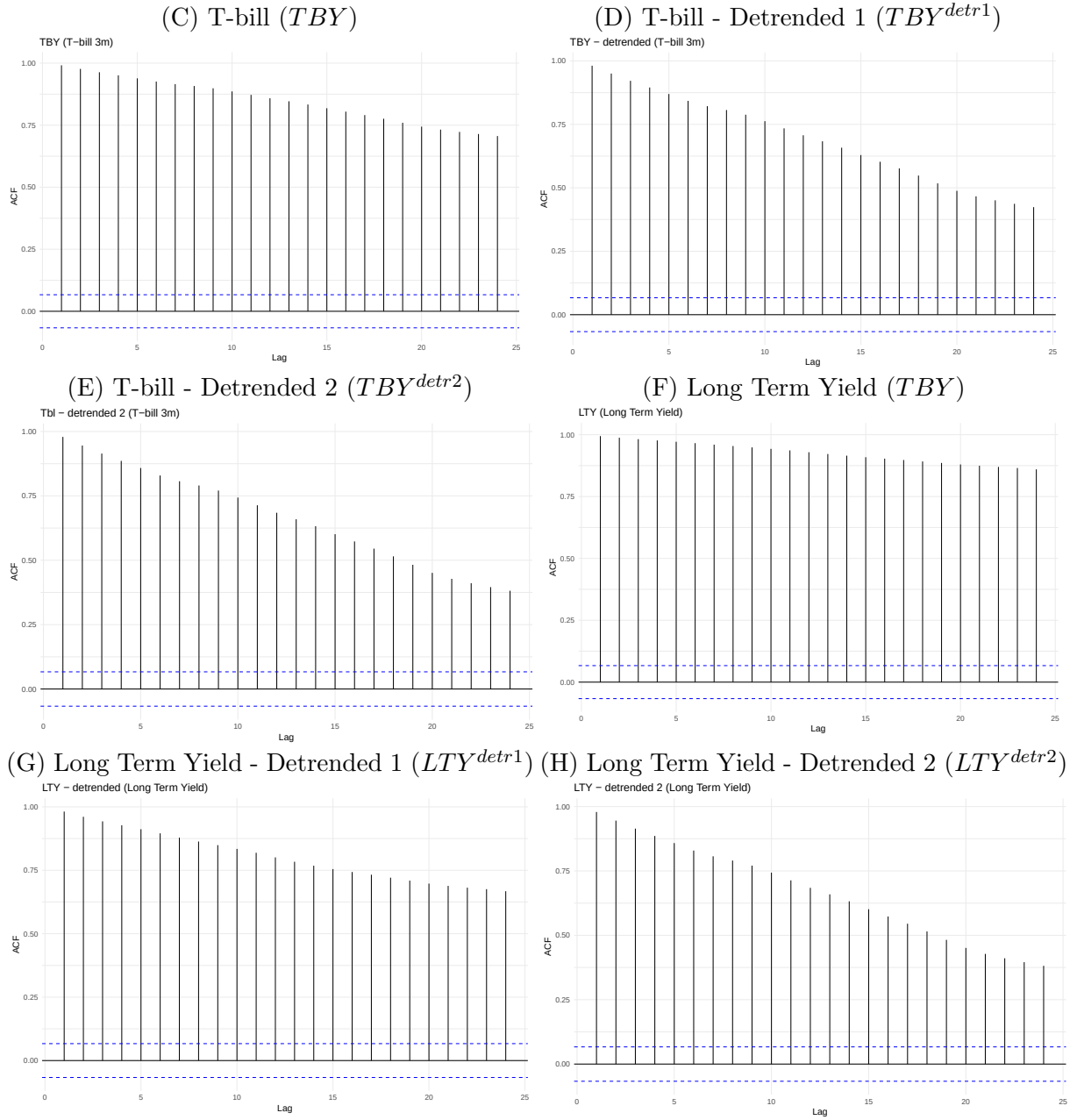


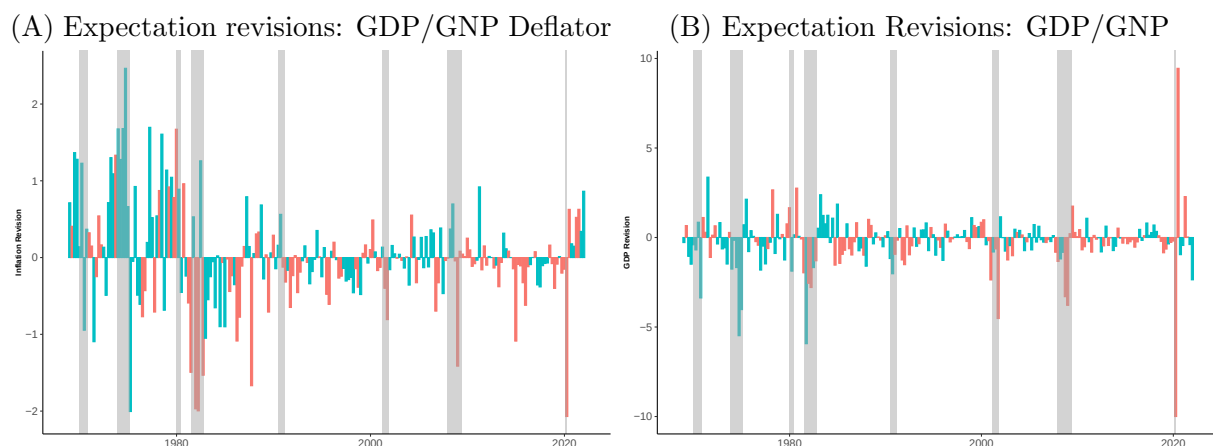
Figure B2: ACF Plots - Part 2



4. Supply Shocks, Demand Shocks, and Yield Curve Dynamics

1) Further Details Regarding Extracted Shocks and Forecasting Power: I appreciate the opponent's interest in examining the extracted shocks and their potential out-of-sample forecasting power within the VAR framework. Due to the Bayesian nature of the VAR model, which relies on multiple draws and a changing rotation matrix under sign restrictions, I am unable to present a single time series of shocks that would be directly comparable across draws. However, to provide further insight into supply and demand dynamics, I can instead examine revisions of expectations. In more detail, I directly divide supply and demand shock based on the sign restrictions, i.e. $shock_{supply(demand),t} = (E_t\pi_t - E_{t-1}\pi_t)$ if $(E_t y_t - E_{t-1}y_t) \times (E_t\pi_t - E_{t-1}\pi_t) < (>)0$, else 0. Figure B3 shows the results. The left panel displays the inflation revision, and the right panel shows the GDP/GNP revision. The revisions were divided based on the supply/demand shock identification described above. Positive co-movement in both series is colored red, e.g., demand shock. Negative co-movement in both series is colored blue, e.g., supply shock. Figure B4 then shows these revisions of expectations summed over the horizon considered, e.g., cumulative expectation revisions. Also, note that this identification mostly corresponds to general intuition. 1970s energy crises are mainly characterized by supply shocks. The recession of 1981–82 (caused by monetary policy) was essentially a demand (shock) recession. The early 2000s recession and the start of the COVID-19 pandemic are also characterized as mostly caused by demand shocks. However, the financial crisis of 2007–2008 is characterized by a series of demand and supply shocks.

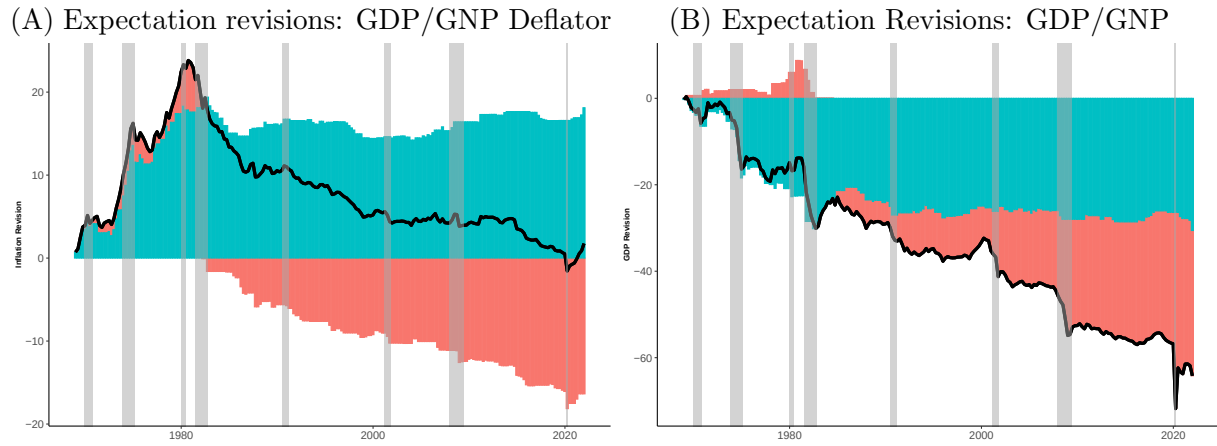
Figure B3: Expectation Revisions



Note: Left panel shows GDP/GNP Deflator revisions. Right panel shows GDP/GNP revisions. GDP revision was winsorized (1 observation) for the graph purpose. Shaded regions denote NBER-dated recessions. Positive co-movement in both series is colored red, e.g., demand shock. Negative co-movement in both series is colored blue, e.g., supply shock.

Furthermore, these shocks can be used as instruments to directly estimate their impact on the variables of interest and to evaluate their out-of-sample predictive power. The impulse response functions, estimated using the local projection approach, remain qualitatively similar. However, the extracted shocks exhibit only weak out-of-sample predictive power, suggesting

Figure B4: Cumulative Expectation Revisions

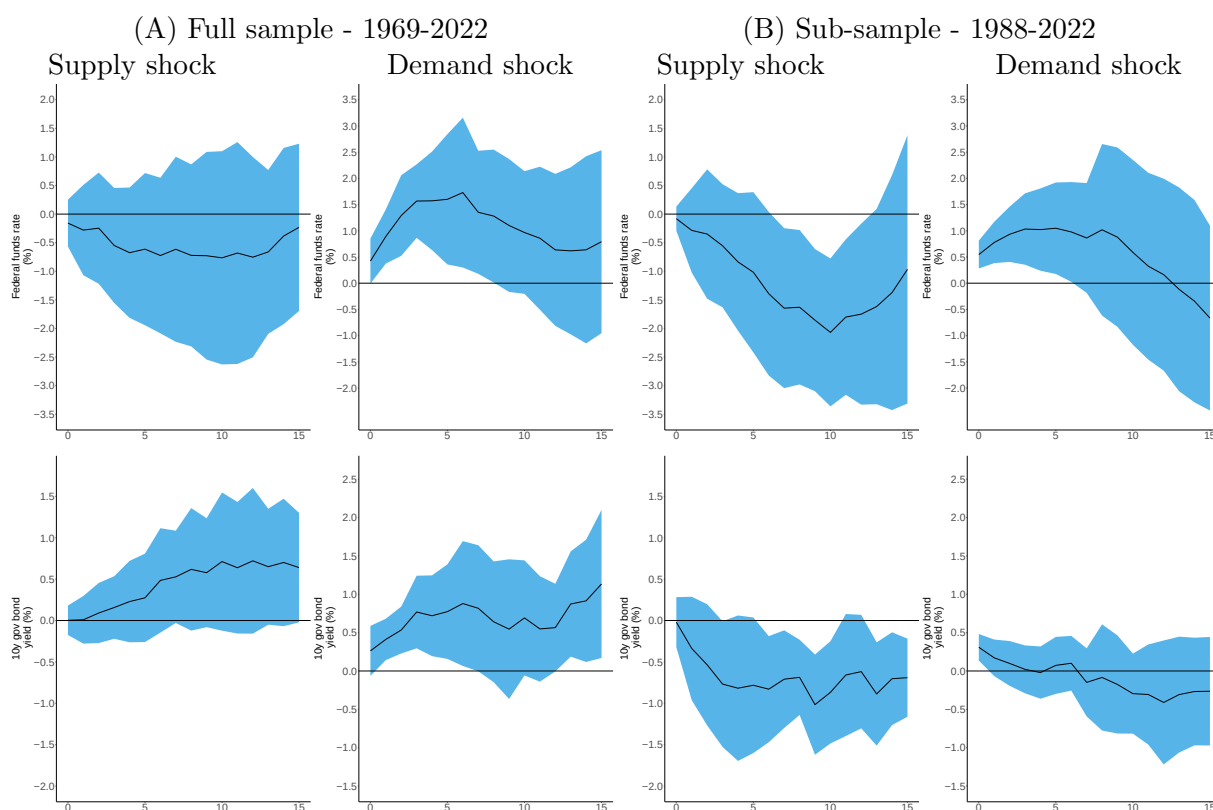


Note: Left panel shows GDP/GNP Deflator revisions. Right panel shows GDP/GNP revisions. GDP revision was winsorized (1 observation) for the graph purpose. Shaded regions denote NBER-dated recessions. Positive co-movement in both series is colored red, e.g., demand shock. Negative co-movement in both series is colored blue, e.g., supply shock.

that while they are useful for understanding historical economic dynamics, their forecasting ability in a real-time setting may be limited.

Figure B5 shows the result from the event study analysis (without any controls). The general finding is that the impact of these shocks on the Fed funds rate and 10-year bond yield is similar to the VAR case. In more detail, the impact of demand shock is clearly identified and is generally very similar to VAR estimation in both full-sample and sub-sample estimation. Thus, the model again confirms that a positive demand shock causes the Fed funds rate to rise, with a peak in the first year. Regarding the 10-year bond yield, the full sample impact seems to be similar to the VAR estimation. The sub-sample estimation shows basically no response in the 10-year yield. In the case of the supply shock, the Fed funds rate again the response is similar to the VAR estimation. However, the 10-year bond yield behaves differently in the full sample case, where we see a rise in the 10-year yield. The sub-sample result regarding the 10-year yield is similar to the VAR estimation.

Figure B5: Event Study - Fed funds rate, 10-year yield



Note: Blue band denote Newey-West robust standard errors (95% confidence intervals).

2) Frequency of Data (Daily Data) and Bond Market Reaction: I appreciate the opponent's suggestion to examine the bond market's immediate or short-term reaction to the identified shocks using an alternative model. The challenge in conducting such an analysis arises from the nature of the data revisions. The cutoff dates for survey responses in the Survey of Professional Forecasters (SPF) have changed over time, as documented in the survey's methodology. However, even the Philadelphia Fed does not provide dates for all periods, making it difficult to precisely align shocks with daily interest rate movements. While this type of analysis would indeed be insightful, the current limitations in data availability prevent a rigorous implementation. If more precise revision timestamps become available in the future, further investigation into the bond market's high-frequency response to shocks could be a valuable extension of this research.

3) Yield Curve Slope and Impulse Responses: I completely agree with the opponent's observation. The difference between the Fed funds rate and the 10-year yield serves as a reasonable proxy for the yield curve slope, and its visual alignment with the impulse responses in Figures 1 and 2 suggests a high correlation between these variables.

This relationship can indeed be interpreted as a form of robustness check, ensuring that the results are not driven by outliers or specific data points. Additionally, my motivation for using the full Nelson-Siegel decomposition was to incorporate all factors, providing a more

comprehensive perspective on yield curve dynamics beyond the simple slope measure.

4) Interpretation of VAR Constraint: Yes, I follow the methodology outlined in Jarcinski's original paper. The approach effectively treats shocks as exogenous variables within a VAR-X framework. This is a fully valid method for computing impulse responses and is asymptotically equivalent to internalizing the shocks within the VAR system. (This was noted, for example, by Karel Mertens in his lecture notes regarding Techniques of Empirical Macroeconomics. See, for example, <https://assets.aeaweb.org/asset-server/files/17943.pdf>)

Regarding notation, I acknowledge that using x_t for inflation or GDP and X_t for interest rate variables may cause some confusion. However, this distinction is in line with the structure of the model and the separation between endogenous and exogenous variables. However, I would now use other notation.

5. Forecasting Nominal Exchange Rates Using Dynamic Model Averaging

1) Methodology: You are correct regarding methodology description. This section was added in response to a reviewer's suggestion and is intended to provide only a brief overview of the method. For implementation details, we recommend referring to the original paper by Raftery *et al.* (2010), which provides a comprehensive discussion of the DMA approach.

2) Parameters: All parameters $A_{i,t}$, $B_{i,t}$, $C_{i,t}$ are dependent on h , but the subscript h suppressed for better readability. Regarding equation (10), the correct notation should be α_i , as this is a simple linear regression. This is a typo in the manuscript.

3) Diebold-Mariano Test Application: The test was applied to the RMSFE (Root Mean Squared Forecast Error) measure, as it is the more commonly used criterion in forecast evaluation. While MAFE (Mean Absolute Forecast Error) is also a valid option, the choice of RMSFE aligns with standard practice in the literature.

4) Diebold-Mariano Test Adjustment for Autocorrelation: Yes, the Diebold-Mariano test statistic was adjusted for autocorrelation. Specifically, I accounted for autocorrelation by using as many lags as the forecasting horizon. This ensures that the test properly accounts for the dependence structure in overlapping forecast errors, leading to more reliable p-values.

4) Figure B.1 and Markov switching component: You are correct regarding the readability of Figure B.1, and it would be useful refining it by focusing on a few forecasts over a shorter period for a single exchange rate. Regarding the model's predictive ability, I believe it results from a combination of factors. First, the presence of mean reversion in exchange rates plays a role—the robustness checks show that a model including lagged exchange rates significantly outperforms a random walk. Second, the Markov switching component is likely crucial, as it allows for shifts in investor attention, aligning with the "scapegoat theory of exchange rates." Notably, the model outperforms a robustness-check model that includes only the exchange rate, whereas a simple linear regression with the same variables does not yield similar improvements.

Conclusion

Once again, I am grateful for the opponent's careful review and valuable comments. While I acknowledge the areas that could have been improved, the articles comprising the dissertation have already been published, and substantive modifications are not feasible at this stage.

Nevertheless, I appreciate these insights, which will be helpful for future research directions.